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MUTANT PORES

Abstract

The invention relates to mutant forms of CsgG. The invention also relates to analyte detection and characterisation using CsgG.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/159,718, filed Jan. 26, 2023, entitled “MUTANT PORES,” which is a continuation of U.S. application Ser. No. 17/215,506, filed Mar. 29, 2021, entitled “MUTANT PORES.” which is a division of U.S. application Ser. No. 16/081,891, filed Aug. 31, 2018, entitled “MUTANT PORES”, which is a national stage filing under 35 U.S.C. 371 of International Patent Application Serial No. PCT/GB2017/050571, filed Mar. 2, 2017, entitled “MUTANT PORES”, which claims foreign priority under 35 U.S.C. § 119(a)-(d) or 35 U.S.C. § 365(b) to British application number GB 1603656.8, filed Mar. 2, 2016, British application number GB 1603657.6, filed Mar. 2, 2016, and British application number GB 1603658.4, filed Mar. 2, 2016. The contents of these applications are incorporated herein by reference in their entireties.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0002] The contents of the electronic sequence listing (O036670063US03-SEQ-LJG.xml; Size: 91,741 bytes; and Date of Creation: Jan. 7, 2025) are herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] The invention relates to mutant forms of CsgG. The invention also relates to analyte detection and characterisation using CsgG.

BACKGROUND OF THE INVENTION

[0004] Nanopore sensing is an approach to sensing that relies on the observation of individual binding or interaction events between analyte molecules and a receptor. Nanopore sensors can be created by placing a single pore of nanometer dimensions in an insulating membrane and measuring voltage-driven ionic transport through the pore in the presence of analyte molecules. The identity of an analyte is revealed through its distinctive current signature, notably the duration and extent of current block and the variance of current levels. Such nanopore sensors are commercially available, such as the MinION™ device sold by Oxford Nanopore Technologies Ltd, comprising an array of nanopores integrated with an electronic chip.

[0005] There is currently a need for rapid and cheap nucleic acid (e.g. DNA or RNA) sequencing technologies across a wide range of applications. Existing technologies are slow and expensive mainly because they rely on amplification techniques to produce large volumes of nucleic acid and require a high quantity of specialist fluorescent chemicals for signal detection. Nanopore sensing has the potential to provide rapid and cheap nucleic acid sequencing by reducing the quantity of nucleotide and reagents required.

[0006] Two of the essential components of sequencing nucleic acids using nanopore sensing are (1) the control of nucleic acid movement through the pore and (2) the discrimination of nucleotides as the nucleic acid polymer is moved through the pore. In the past, to achieve nucleotide discrimination the nucleic acid has been passed through a mutant of hemolysin. This has provided current signatures that have been shown to be sequence dependent. It has also been shown that a

large number of nucleotides contribute to the observed current when a hemolysin pore is used, making a direct relationship between observed current and polynucleotide challenging.

[0007] While the current range for nucleotide discrimination has been improved through mutation of the hemolysin pore, a sequencing system would have higher performance if the current differences between nucleotides could be improved further. In addition, it has been observed that when the nucleic acids are moved through a pore, some current states show high variance. It has also been shown that some mutant hemolysin pores exhibit higher variance than others. While the variance of these states may contain sequence specific information, it is desirable to produce pores that have low variance to simplify the system. It is also desirable to reduce the number of nucleotides that contribute to the observed current.

SUMMARY OF THE INVENTION

[0008] The inventors have surprisingly demonstrated that CsgG and novel mutants thereof may be used to characterise analytes, such as polynucleotides. The invention concerns mutant CsgG monomers. The inventors have surprisingly demonstrated that pores comprising the novel mutant monomers have an enhanced ability to estimate the characteristics of analytes, such as the sequence of polynucleotides. The inventors have made mutant pores that surprisingly provide more consistent movement of a target polynucleotide with respect to, such as through, the pores. The inventors have made mutant pores surprisingly display improved characterisation accuracy. In particular, the mutant pores made by the inventors surprisingly display an increased current range, which makes it easier to discriminate between different nucleotides, and a reduced variance of states, which increases the signal-to-noise ratio. In addition, the mutant pores surprisingly capture nucleotides and polynucleotides more easily.

[0009] All amino-acid substitutions, deletions and/or additions disclosed herein are with reference to a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2, unless stated to the contrary.

[0010] Reference to a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2 encompasses mutant CsgG monomers comprising variants of sequences as set out in the further SEQ ID NOS as disclosed below. Amino-acid substitutions, deletions and/or additions may be made to CsgG monomers comprising a variant of the sequence other than shown in SEQ ID NO:2 that are equivalent to those substitutions, deletions and/or additions disclosed herein with reference to a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO:2.

[0011] A mutant monomer may be considered as an isolated monomer.

[0012] In particular, the invention concerns mutant CsgG monomers in which the arginine (R) at position 192 has been substituted with aspartic acid (D), glutamine (Q), phenylalanine (F), serine (S) or threonine (T). The inventors have surprisingly demonstrated that such monomers, and in particular a monomer comprising a R192D substitution, are much easier to express than monomers without a substitution at position 192. An increased yield is obtained when the CsgG monomer comprises this mutation.

[0013] The mutant csgG monomer may be: [0014] a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2 which comprises R97W; [0015] a mutant CsgG monomer which comprises (a) R192D; (b) R97W/Y and/or R93W/Y, preferably R97W, R93W or R93Y and R97Y; (c) K94Q/N; (d) G103K/R and/or T104KR; and/or (e) F191T, deletion of V105, A106 and I107 and/or deletion of F193, I194, D195, Y196, Q197, R198 and L199.

[0016] The mutant CsgG monomer preferably further comprises Y51A and F56Q.

[0017] Particular mutant CsgG monomers comprise variants of the sequence shown in SEQ ID NO: 2 that comprise the following mutations: [0018] (1) Y51A, F56Q and R192D; [0019] (2) Y51A, F56Q and R97W. [0020] (3) Y51A, F56Q, R192D and R97W; [0021] (4) Y51A, F56Q, R192D and R93W; [0022] (5) Y51A, F56Q, R192D, R93Y and R97Y, or [0023] (6) Y51A, F56Q, R192D and R93W. [0024] (7) the mutations of any one of (1)-(6) and: [0025] (a) deletion of V105,

A106 and I107. [0026] (b) K94Q or K94N; [0027] (c) deletion of D195, Y196, Q197, R198 and L199 or deletion of F193, I194, D195, Y196, Q197, R198 and L199; and/or [0028] (d) F191T [0029] (8) the mutations of any one of (1)-(6) and: [0030] (i) K94Q and deletion of V105, A106 and I107; [0031] (ii) K94N and deletion of V105, A106 and I107; [0032] (iii) F191 T and deletion of V105, A106 and I107; [0033] (iv) K94Q and F191T; [0034] (v) K94N and F191T; [0035] (vi) K94Q, F191T and deletion of V105, A106 and I107; or [0036] (vii) K94N, F191T and deletion of V105, A106 and I107. [0037] (9) the mutations of any one of (1)-(8) and: [0038] T104K or T104R; [0039] L90R; [0040] N91R; [0041] I95R; [0042] A99R; [0043] E101K, E101N, E101Q, E101T or E101H; [0044] E44N or E44Q; and/or [0045] Q42K.

[0046] The invention also provides: [0047] a construct comprising two or more covalently attached CsgG monomers, wherein at least one of the monomers is a mutant monomer of the invention; [0048] a polynucleotide which encodes a mutant monomer of the invention or a construct of the invention; [0049] a homo-oligomeric pore derived from CsgG comprising identical mutant monomers of the invention or identical constructs of the invention; [0050] a hetero-oligomeric pore derived from CsgG comprising at least one mutant monomer of the invention or at least one construct of the invention; [0051] a method for determining the presence, absence or one or more characteristics of a target analyte, comprising: [0052] (a) contacting the target analyte with a pore of the invention such that the target analyte moves with respect to the pore; and [0053] (b) taking one or more measurements as the analyte moves with respect to the pore and thereby determining the presence, absence or one or more characteristics of the analyte; [0054] a method of forming a sensor for characterising a target polynucleotide, comprising forming a complex between a pore of the invention and a polynucleotide binding protein and thereby forming a sensor for characterising the target polynucleotide; [0055] a sensor for characterising a target polynucleotide, comprising a complex between a pore of the invention and a polynucleotide binding protein, [0056] use of a pore of the invention to determine the presence, absence or one or more characteristics of a target analyte; [0057] a kit for characterising a target analyte comprising (a) a pore of the invention and (b) the components of a membrane; [0058] an apparatus for characterising target analytes in a sample, comprising (a) a plurality of a pores of the invention and (b) a plurality of membranes; [0059] a method of characterising a target polynucleotide, comprising: [0060] a) contacting the polynucleotide with a pore of the invention, a polymerase and labelled nucleotides such that phosphate labelled species are sequentially added to the target polynucleotide by the polymerase, wherein the phosphate species contain a label specific for each nucleotide, and [0061] b) detecting the phosphate labelled species using the pore and thereby characterising the polynucleotide; and [0062] a method of producing a mutant monomer of the invention or a construct of the invention, comprising expressing a polynucleotide of the invention in a suitable host cell and thereby producing a mutant monomer of the invention or a construct.

Description

DESCRIPTION OF THE FIGURES

[0063] FIG. 1: Illustrates CsgG from *E. coli*.

[0064] FIG. 2: Illustrates the dimensions of CsgG.

[0065] FIG. 3: Illustrates single G translocation at 10 Å/ns. There is a large barrier for entry of guanine into F56 ring in CsgG-Eco. * = G enters F56 ring. A = G stops interacting with 56 ring. B = G stops interacting with 55 ring. C = G stops interacting with 51 ring.

[0066] FIG. 4: Illustrates ssDNA translocation at 100 Å/ns. A larger force is required to pull the DNA through the constriction for CsgG-Eco.

[0067] FIG. 5: Illustrates ss DNA translocation at 10 Å/ns. CsgG-F56A-N55S and CsG-F56A-N55S-Y51A mutants both have a lower barrier for ssDNA translocation.

[0068] FIGS. **6A-6C**: Mutant pores showing increased range compared with wild-type (WT).

[0069] FIGS. **7A-7C**: Mutant pores showing increased range compared with wild-type (WT).

[0070] FIGS. **8A-8C**: Mutant pores showing increased range compared with wild-type (WT).

[0071] FIGS. **9A-9C**: Mutant pores showing increased throughput compared with wild-type (WT).

[0072] FIGS. **10A-10B**: Mutant pores showing increased throughput compared with wild-type (WT).

[0073] FIGS. **11** and **12**: Mutant pore showing increased insertion compared with wild-type (WT).

[0074] FIG. **13**: shows the DNA construct X used in Example 2. The region labelled 1 corresponded to 30 SpC3 spacers. The region labelled 2 corresponded to SEQ ID NO: 42. The region labelled 3 corresponded to four iSp18 spacers. The region labelled 4 corresponded to SEQ ID NO: 43. The section labelled 5 corresponded to four 5-nitroindoles. The region labelled 6 corresponded to SEQ ID NO: 44. The region labelled 7 corresponded to SEQ ID NO: 45. The region labelled 8 corresponded to SEQ ID NO: 46 which had four iSp18 spacers (the region labelled 9) attached at the 3' end of SEQ ID NO: 46. At the opposite end of the iSp18 spacers was a 3' cholesterol tether (labelled 10). The region labelled 11 corresponded to four SpC3 spacers.

[0075] FIG. **14**: shows an example chromatography trace of Strep trap (GE Healthcare) purification of CsgG protein (x-axis label=elution volume (mL), Y-axis label=Absorbance (mAu)). The sample was loaded in 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM and eluted with 10 mM desthiobiotin. The elution peak in which CsgG protein eluted is labelled as E1.

[0076] FIG. **15**: shows an example of a typical SDS-PAGE visualisation of CsgG protein after the initial strep purification. A 4-20% TGX Gel (Bio Rad) was run at 300 V for 22 minutes in 1×TGS buffer. The gel was stained with Sypro Ruby stain. Lanes 1-3 show the main elution peak (labelled E1 in FIG. **14**) which contained CsgG protein as indicated by the arrow. Lanes 4-6 corresponded to elution fractions of the tail of the main elution peak (labelled E1 in FIG. **14**) which contained contaminants. M shows the molecular weight marker used which was a Novex Sharp Unstained (unit=kD).

[0077] FIG. **16**: Shows an example of a size exclusion chromatogram (SEC) of CsgG protein (120 mL S200 GE healthcare, x-axis label=elution volume (mL), y-axis label=absorbance (mAu)). The SEC was carried out after strep purification and heating the protein sample. The running buffer for SEC was 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS, pH 8.0 and the column was run at 1 mL/minute rate. The trace labelled X shows absorbance at 220 nm and the trace labelled Y shows absorbance at 280 nm. The peak labelled with a star was collected.

[0078] FIG. **17**: shows an example of a typical SDS-PAGE visualisation of CsgG protein after SEC. A 4-20% TGX Gel (Bio Rad) was run at 300V for 22 minutes in 1×TGS buffer and the gel was stained with Sypro Ruby stain. Lane 1 shows CsgG protein sample after strep purification and heating but before SEC. Lanes 2-8 show fractions collected across the peak running approximately 48 mL-60 mL of FIG. **16** (mid peak=55 mL) and labelled with a star in FIG. **16**. M shows the molecular weight marker used which was a Novex Sharp Unstained (unit=kD). The bar corresponding to the CsgG-Eco pore is indicated by an arrow.

[0079] FIGS. **18** to **24**: Mutant pores showing increased range compared with wild-type (WT).

[0080] FIGS. **25** to **30**: Mutant pores showing increased range compared with wild-type (WT).

[0081] FIG. **31** shows snap shots of the enzyme (T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (Δ M1)G1G2)) on top of the pore (CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51 T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20)) taken at 0 and 20 ns during the simulations (Runs 1 to 3).

[0082] FIG. **32** shows snap shots of the enzyme (T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (Δ M1)G1G2)) on top of the pore (CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20)) taken at 30

and 40 ns during the simulations (Runs 1 to 3).

[0083] FIG. **33** shows a snap shot of the enzyme (T4 Dda- (E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1)G1G2) on top of the pore CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 26) taken during the simulations described in Example 5.

[0084] FIG. **34** shows two ten second screen shots of current traces showing translocation of DNA (SEQ ID NO: 51) through MspA mutant x=MspA-((Del-L74/G75/D118/L119)D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K)8 (SEQ ID NO: 50 with mutations D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K and deletion of the amino acids L74/G75/D118/L119) without the control of an enzyme.

[0085] FIG. **35** shows two ten second screen shots of current traces showing translocation of DNA (SEQ ID NO: 51) through CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) without the control of an enzyme.

[0086] FIG. **36** shows two ten second screen shots of current traces showing translocation of DNA (SEQ ID NO: 51) through CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/E101 S/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) without the control of an enzyme.

[0087] FIG. **37** shows an overlay of two gel filtration chromatograms (120 ml S200 column) of the CsgG mutants pores A) CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) and B) CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus). Absorbance at A280 for CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 is labelled A and for CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 is labelled B. Both constructs were grown in 500 ml cultures Expression and purification of both proteins were carried out using exactly the same protocol and same volumes were loaded onto the column. Running Buffer was 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS pH8. The fractional delay with CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 pore was due to different connection configuration used on AKTA Purifier 10. The difference in the absorbance values indicated the amount of proteins expressed with higher absorbance values indicating higher amounts of expressed protein.

[0088] FIG. **38** shows SDS-PAGE analysis of CsgG nanopores. Lanes A-C contained CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9, lanes D-F contained CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 and lane M contained the molecular weight marker. The two pores were expressed and purified using exactly the same protocol. The pores were subjected to electrophoresis on a 4-20% TGX gel (Bio rad cat #5671093) in TGS buffer at 300 V for 22 minutes. The gel was visualised with Sypro Ruby stain (Life Technologies cat #S1200). The same volumes from each pore sample were loaded on the gel to compare the amount of proteins obtained after purification—lanes A and D contained 5 uL, lanes B and E contained 10 uL and lanes C and F contained 15 uL.

[0089] FIG. **39** shows the basecall accuracy of eight CsgC mutant pores compared to the basecall accuracy of a baseline pore, mutant 28 (CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9). The deletion of D195-L199 (Mutant A), F193-L199 (Mutant B) or V105-I107 (Mutant D), or the substitution of F191 T (Mutant C) results in a further improvement in accuracy in addition to the improvement in accuracy resulting from the R97W and R 192D substitutions in mutant 28. The effect on basecall accuracy of deleting V105-I107 was also tested in a mutant pore containing an additional K94Q mutation (Mutant E) and an improvement in accuracy compared to baseline mutant 28 was still observed. Introducing a R93W mutation (Mutant F) or both R93Y and R97W mutations (Mutant H) instead of a R97W mutation (baseline mutant 28) increased the basecall

accuracy. Deleting D195-L199 in addition to R93W (Mutant G) resulted in an enhancement of basecall accuracy.

[0090] FIGS. **40A-40B** show the template speed distribution (FIG. **40A**) and the template accuracy distribution (FIG. **40B**) of the baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9 and Mutant D which comprises an additional deletion of V105-I107. A template DNA was prepared and passed through the mutant pores as described in the Examples. The template speed and accuracy were determined as described in the Examples. FIG. **40A** shows that the speed distribution was tighter when Mutant D was used compared to the baseline mutant. FIG. **40B** shows that mutant D has a tighter distribution of template accuracy compared to the baseline mutant.

[0091] FIG. **41** displays an example “squiggle” that shows the “noisy” pore error mode exhibited by baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9. The top panel of FIG. **41** shows the difference in flow of current through the pore during the “good” and “noisy” pore states. The bottom panel of FIG. **41** shows an expanded view of the transition from “good” state to “noisy” state.

[0092] FIG. **42** shows the reduction in noisy pore state of mutant pores having the same sequence as the baseline mutant 28, which contains Y51A/F56Q/R97W/R192D mutations, with an additional K94N mutation (Mutant I) or an additional K94Q mutation (Mutant J) when compared to baseline mutant 28, averaged over at least 5 runs.

[0093] FIG. **43** illustrates the structure of the template strand having an adapter ligated to each end thereof. The adapter has a T4 Dda helicase enzyme prebound thereto. The sequences of the various parts of the adaptor used in the Examples are shown in SEQ ID NOs: 52 to 55.

[0094] FIG. **44** shows the median time between Thrombin Binding Aptamer (TBA) events for mutant CsgG nanopores comprising one of the following substitutions: Q42K (Mutant K), E44N (Mutant L), E44Q (Mutant M), L90R (Mutant N), N91R (Mutant O), I95R (Mutant P), A99R (Mutant Q), E101H (Mutant R), E101K (Mutant S), E101N (Mutant T) E101Q (Mutant U), E101 T (Mutant V) and Q114K (Mutant W). The median time was significantly reduced compared to the baseline pore comprising the mutations Y51A/F56Q/K94Q/R97W/R192D-del(V105-I107) (Baseline mutant E), all of which are also included in each of the 13 mutants tested. FIG. **44** shows that each of the Q42K, E44N, E44Q, L90R, N91R, I95R, A99R, E101H, E101K, E101N, E101Q, E101T and Q114K substitutions increase template DNA capture rates.

[0095] FIGS. **45A-45C** show sequence alignments of the 21 CsgG homologues corresponding to SEQ ID Nos 2, 5, 6, 7, 27, 28, 29, 30, 32, 36, 3, 35, 31, 40, 33, 34, 37, 39, 38, 41 and 4

[0096] FIGS. **46A-46C** show the same relative sequence alignments as FIG. **45** with predicted alpha helical secondary structure regions additionally shaded.

[0097] FIGS. **47A-47C** show the same relative sequence alignments as FIG. **45** with predicted beta sheet secondary structure regions additionally shaded.

[0098] FIG. **48** shows two examples of raw electrical data for poreAQ and pore97W.

DESCRIPTION OF THE SEQUENCE LISTING

[0099] SEQ ID NO: 1 shows the codon optimised polynucleotide sequence encoding the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. This monomer lacks the signal sequence.

[0100] SEQ ID NO: 2 shows the amino acid sequence of the mature form of the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. This monomer lacks the signal sequence. The abbreviation used for this CsgC=CsgG-Eco.

[0101] SEQ ID NO: 3 shows the amino acid sequence of YP_001453594.1: 1-248 of hypothetical protein CKO_02032 [*Citrobacter koseri* ATCC BAA-895], which is 99% identical to SEQ ID NO: 2.

[0102] SEQ ID NO: 4 shows the amino acid sequence of WP_001787128.1: 16-238 of curli production assembly/transport component CsgG, partial [*Salmonella enterica*], which is 98% to

SEQ ID NO: 2.

[0103] SEQ ID NO: 5 shows the amino acid sequence of KEY44978.1: 16-277 of curli production assembly/transport protein CsgG [*Citrobacter amalonaticus*] which is 98% identical to SEQ ID NO: 2.

[0104] SEQ ID NO: 6 shows the amino acid sequence of YP_003364699.1: 16-277 of curli production assembly/transport component [*Citrobacter rodentium* ICC168], which is 97% identical to SEQ ID NO: 2.

[0105] SEQ ID NO: 7 shows the amino acid sequence of YP_004828099.1: 16-277 of curli production assembly/transport component CsgG [*Enterobacter asburiae* LF7a], which is 94% identical to SEQ ID NO: 2.

[0106] SEQ ID NO: 8 shows the polynucleotide sequence encoding the Phi29 DNA polymerase.

[0107] SEQ ID NO: 9 shows the amino acid sequence of the Phi29 DNA polymerase.

[0108] SEQ ID NO: 10 shows the codon optimised polynucleotide sequence derived from the sbcB gene from *E. coli*. It encodes the exonuclease I enzyme (EcoExo I) from *E. coli*.

[0109] SEQ ID NO: 11 shows the amino acid sequence of exonuclease I enzyme (EcoExo I) from *E. coli*.

[0110] SEQ ID NO: 12 shows the codon optimised polynucleotide sequence derived from the xthA gene from *E. coli*. It encodes the exonuclease III enzyme from *E. coli*.

[0111] SEQ ID NO: 13 shows the amino acid sequence of the exonuclease III enzyme from *E. coli*. This enzyme performs distributive digestion of 5' monophosphate nucleosides from one strand of double stranded DNA (dsDNA) in a 3'-5' direction. Enzyme initiation on a strand requires a 5' overhang of approximately 4 nucleotides.

[0112] SEQ ID NO: 14 shows the codon optimised polynucleotide sequence derived from the recJ gene from *T. thermophilus*. It encodes the RecJ enzyme from *T. thermophilus* (TthRecJ-cd).

[0113] SEQ ID NO: 15 shows the amino acid sequence of the RecJ enzyme from *T. thermophilus* (TthRecJ-cd). This enzyme performs processive digestion of 5' monophosphate nucleosides from ssDNA in a 5'-3' direction. Enzyme initiation on a strand requires at least 4 nucleotides.

[0114] SEQ ID NO: 16 shows the codon optimised polynucleotide sequence derived from the bacteriophage lambda exo (recX) gene. It encodes the bacteriophage lambda exonuclease. [0115]

SEQ ID NO: 17 shows the amino acid sequence of the bacteriophage lambda exonuclease. The sequence is one of three identical subunits that assemble into a trimer. The enzyme performs highly processive digestion of nucleotides from one strand of dsDNA, in a 5'-3' direction

(<http://www.neb.com/nebecomm/products/productM0262.asp>). Enzyme initiation on a strand preferentially requires a 5' overhang of approximately 4 nucleotides with a 5' phosphate.

[0116] SEQ ID NO: 18 shows the amino acid sequence of Hel308 Mbu.

[0117] SEQ ID NO: 19 shows the amino acid sequence of Hel308 Csy.

[0118] SEQ ID NO: 20 shows the amino acid sequence of Hel308 Tga.

[0119] SEQ ID NO: 21 shows the amino acid sequence of Hel308 Mhu.

[0120] SEQ ID NO: 22 shows the amino acid sequence of TraI Eco.

[0121] SEQ ID NO: 23 shows the amino acid sequence of XPD Mbu.

[0122] SEQ ID NO: 24 shows the amino acid sequence of Dda 1993.

[0123] SEQ ID NO: 25 shows the amino acid sequence of Trwc Cba.

[0124] SEQ ID NO: 26 shows the amino acid sequence of WP_006819418.1: 19-280 of transporter [*Yokenella regensburgei*], which is 91% identical to SEQ ID NO: 2.

[0125] SEQ ID NO: 27 shows the amino acid sequence of WP_024556654.1: 16-277 of Curli production assembly/transport protein CsgG [*Cronobacter pulveris*], which is 89% identical to SEQ ID NO: 2.

[0126] SEQ ID NO: 28 shows the amino acid sequence of YP_005400916.1: 16-277 of curli production assembly/transport protein CsgG [*Rahnella aquatilis* HX2], which is 84% identical to SEQ ID NO 2.

[0127] SEQ ID NO: 29 shows the amino acid sequence of KFC99297.1: 20-278 of CsgG family curli production assembly/transport component [*Kluyvera ascorbata* ATCC 33433], which is 82% identical to SEQ ID NO: 2.

[0128] SEQ ID NO: 30 shows the amino acid sequence of KFC86716.1:16-274 of CsgG family curli production assembly/transport component [*Hafnia alvei* ATCC 13337], which is 81% identical to SEQ ID NO: 2.

[0129] SEQ ID NO: 31 shows the amino acid sequence of YP_007340845.1:16-270 of uncharacterised protein involved in formation of curli polymers [*Enterobacteriaceae bacterium* strain FGI 57], which is 76% identical to SEQ ID NO: 2.

[0130] SEQ ID NO: 32 shows the amino acid sequence of WP_010861740.1: 17-274 of curli production assembly/transport protein CsgG [*Plesiomonas shigelloides*], which is 70% identical to SEQ ID NO: 2.

[0131] SEQ ID NO: 33 shows the amino acid sequence of YP_205788.1: 23-270 of curli production assembly/transport outer membrane lipoprotein component CsgG [*Vibrio fischeri* ES114], which is 60% identical to SEQ ID NO: 2.

[0132] SEQ ID NO: 34 shows the amino acid sequence of WP_017023479.1: 23-270 of curli production assembly protein CsgG [*Aliivibrio logei*], which is 59% identical to SEQ ID NO: 2.

[0133] SEQ ID NO: 35 shows the amino acid sequence of WP_007470398.1: 22-275 of Curli production assembly/transport component CsgG [*Photobacterium* sp, AK 15], which is 57% identical to SEQ ID NO: 2.

[0134] SEQ ID NO: 36 shows the amino acid sequence of WP_021231638.1: 17-277 of curli production assembly protein CsgG [*Aeromonas veronii*], which is 56% identical to SEQ ID NO: 2.

[0135] SEQ ID NO: 37 shows the amino acid sequence of WP_033538267.1: 27-265 of curli production assembly/transport protein CsgG [*Shewanella* sp. ECSMB14101], which is 56% identical to SEQ ID NO: 2.

[0136] SEQ ID NO: 38 shows the amino acid sequence of WP_003247972.1: 30-262 of curli production assembly protein CsgG [*Pseudomonas putida*], which is 54% identical to SEQ ID NO: 2.

[0137] SEQ ID NO: 39 shows the amino acid sequence of YP_003557438.1: 1-234 of curli production assembly/transport component CsgG [*Shewanella violacea* DSS12], which is 53% identical to SEQ ID NO: 2.

[0138] SEQ ID NO: 40 shows the amino acid sequence of WP_027859066.1: 36-280 of curli production assembly/transport protein CsgG [*Marinobacterium jannaschii*], which is 53% identical to SEQ ID NO: 2.

[0139] SEQ ID NO: 41 shows the amino acid sequence of CEJ702.sup.22.1: 29-262 of Curli production assembly/transport component CsgG [*Chryseobacterium oranimense* G311], which is 50% identical to SEQ ID NO: 2.

[0140] SEQ ID NO: 42 shows a polynucleotide sequence used in Example 2.

[0141] SEQ ID NO: 43 shows a polynucleotide sequence used in Example 2.

[0142] SEQ ID NO: 44 shows a polynucleotide sequence used in Example 2.

[0143] SEQ ID NO: 45 shows a polynucleotide sequence used in Example 2.

[0144] SEQ ID NO: 46 shows a polynucleotide sequence used in Example 2. Attached to the 3' end of SEQ ID NO: 46 is six iSp18 spacers which are attached at the opposite end to two thymines and a 3' cholesterol TEG.

[0145] SEQ ID NO: 47 shows the polynucleotide sequence of StrepII(C).

[0146] SEQ ID NO: 48 shows the polynucleotide sequence of Pro.

[0147] SEQ ID NO: 49 shows the codon optimised polynucleotide sequence encoding the wild-type MspA monomer. This mutant lacks the signal sequence.

[0148] SEQ ID NO: 50 shows the amino acid sequence of the mature form of the wild-type MspA monomer. This mutant lacks the signal sequence.

[0149] SEQ ID NO: 51 shows the polynucleotide sequence of Thrombin Binding Aptamer used in Examples 7 and 11.

[0150] SEQ ID NO: 52 shows the polynucleotide sequence of a Y-adaptor top strand.

[0151] SEQ ID NO: 53 shows the polynucleotide sequence of a Y-adaptor blocker strand.

[0152] SEQ ID NO: 54 shows the polynucleotide sequence of a Y-adaptor cholesterol tether strand.

[0153] SEQ ID NO: 55 shows the polynucleotide sequence of a Y-adaptor bottom strand. SEQ ID NO. 56 shows the polynucleotide sequence of a 3.6 kb double stranded DNA target sequence used in the Examples.

DETAILED DESCRIPTION OF THE INVENTION

[0154] It is to be understood that different applications of the disclosed products and methods may be tailored to the specific needs in the art. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments of the invention only, and is not intended to be limiting.

[0155] In addition as used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to “a polynucleotide” includes two or more polynucleotides, reference to “a polynucleotide binding protein” includes two or more such proteins, reference to “a helicase” includes two or more helicases, reference to “a monomer” refers to two or more monomers, reference to “a pore” includes two or more pores and the like.

[0156] All publications, patents and patent applications cited herein, whether supra or infra, are hereby incorporated by reference in their entirety.

Mutant CsgG Monomers

[0157] An aspect of the invention provides mutant CsgG monomers. The mutant CsgG monomers may be used to form the pores of the invention. A mutant CsgG monomer is a monomer whose sequence varies from that of a wild-type CsgG monomer and which retains the ability to form a pore. Methods for confirming the ability of mutant monomers to form pores are well-known in the art and are discussed in more detail below.

[0158] Pores constructed from the CsgG monomers of some embodiments of the invention comprising the modification R97W display an increased accuracy as compared to otherwise identical pores without the modification at 97 when characterizing (or sequencing) target polynucleotides. An increased accuracy is also seen when instead of R97W the CsgG monomers of the invention comprise the modification R93W or the modifications R93Y and R97Y. Accordingly, pores may be constructed from one or more mutant CsgG monomers that comprise a modification at R97 or R93 of SEQ ID NO: 2 such that the modification increases the hydrophobicity of the amino acid. For example, such modification may include an amino acid substitution with any amino acid containing a hydrophobic side chain, including, e.g., but not limited to W and Y.

[0159] The CsgG monomers of the invention that comprise R192D/Q/F/S/T are easier to express than monomers which do not have a substitution at position 192 which may be due to the reduction of positive charge. Accordingly position 192 may be substituted with an amino-acid which reduces the positive charge. The monomers of the invention that comprise R192D/Q/F/S/T may also comprise additional modifications which improve the ability of mutant pores formed from the monomers to interact with and characterise analytes, such as polynucleotides.

[0160] The CsgG monomers of some embodiments of the invention that comprise a deletion of V105, A106 and I107, a deletion of F193, I194, D195, Y196, Q197, R198 and L199 or a deletion of D195, Y196, Q197, R198 and L199, and/or F191 T display an increased accuracy when characterizing (or sequencing) target polynucleotides.

[0161] Pores comprising the CsgG monomers of some embodiments of the invention that comprise K94Q or K94N show a reduction in the number of noisy pores (namely those pores that give rise to an increased signal:noise ratio) as compared to identical pores without the mutation at 94 when characterizing (or sequencing) target polynucleotides. Position 94 is found within the vestibule of

the pore and was found to be a particularly sensitive position in relation to the noise of the current signal. Pores comprising the CsgG monomers of some embodiments of the invention that comprise T104K or T104R, N91R, E101K/N/Q/T/H, E44N/Q, Q114K, A99R, I95R, N91R, L90R, E44Q/N and/or Q42K all demonstrate an improved ability to capture target polynucleotides when used to characterize (or sequence) target polynucleotides as compared to identical pores without substitutions at these positions.

[0162] Characterisation, such as sequencing, of a polynucleotide using a transmembrane pore may be carried out such as disclosed in International Application No. PCT/GB2012/052343 (published as WO 2013/041878). As the target polynucleotide moves with respect to, or through the pore, the analyte may be characterised from the distinctive ion current signature produced, typically by measuring the ion current flow through the pore. The level of current measured at any particular time is typically dependent on a group of k polymer (for example nucleotide) units where k is a positive integer and the typical current signature may be represented as a series of current levels indicative of a particular k -mer. The movement of the polynucleotide with respect to, such as through, the pore can be viewed as movement from one k -mer to another or from k -mer to k -mer. Analytical techniques to characterise the polynucleotide may for example involve the use of an HMM, a neural network and for example a Forwards Backwards algorithm or Viterbi algorithm to determine the likelihood of the series of measurements corresponding to a particular sequence. Alternatively the polynucleotide may be characterised by determining a feature vector and comparing the feature vector to another feature vector, which may be known, such as disclosed in International Application No. PCT/GB2013/050381 (published as WO 2013/121224). However, the analytical techniques used to characterise the polynucleotide are not necessarily restricted to the above examples.

[0163] When a monomer of the invention forms a transmembrane pore and is used with a polynucleotide binding protein to characterise a target polynucleotide, some of the modified positions interact with the polynucleotide binding protein. For example, when the monomer forms a transmembrane pore and is used with a polynucleotide binding protein to characterise a target polynucleotide, R97W interacts with the polynucleotide binding protein. Modifying the CsgG monomer in accordance with the invention typically provides more consistent movement of the target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. The modification(s) typically provide more consistent movement from one k -mer to another or from k -mer to k -mer as the target polynucleotide moves with respect to, such as through, the pore. The modification(s) typically allow the target polynucleotide to move with respect to, such as through, the transmembrane pore more smoothly. The modification(s) typically provide more regular or less irregular movement of the target polynucleotide with respect to, such as through, the transmembrane pore.

[0164] Modifying the CsgG monomer in accordance with the invention (e.g. R97W) typically reduces the amount of slipping forward associated with the movement of the target polynucleotide with respect to, such as through, a pore comprising the monomer. Some helicases including the Dda helicase used in the Example move along the polynucleotide in a 5' to 3' direction. When the 5' end of the polynucleotide (the end away from which the helicase moves) is captured by the pore, the helicase works with the direction of the field resulting from the applied potential and moves the threaded polynucleotide into the pore and into the trans chamber. Slipping forward involves the DNA moving forwards relative to the pore (i.e. towards its 3' and away from its 5' end) at least 4 consecutive nucleotides and typically more than 10 consecutive nucleotides. Slipping forward may involve movement forward of 100 consecutive nucleotides or more and this may happen more than once in each strand.

[0165] Modifying the CsgG monomer in accordance with the invention (e.g. R97W) typically reduces the noise associated with the movement of the target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. Unwanted movement of the target

polynucleotide in any dimension as the signal is being analysed typically results in noise in the current signature or level for the k-mer. The modification may reduce this noise by reducing unwanted movement associated with one or more k-mers, such as each k-mer, in the target polynucleotide. The modification may reduce the noise associated with the current level or signature for one or more k-mers, such as each k-mer; in the target polynucleotide.

[0166] The enzyme motors employed for moving the polynucleotide have multiple sub-steps in the full catalytic cycle where ATP is hydrolysed to move the polynucleotide forward one base (eg. binding ATP.Mg, hydrolysing to produce ADP.P.Mg, moving the polynucleotide one base forward, and releasing the ADP/P/Mg by-products). Each sub-step process has a characteristic dwell time distribution determined by the kinetics of the process. If any of these sub-steps of the catalytic cycle move the position of the polynucleotide in the reader (e.g. by moving the polynucleotide relative to the enzyme, or by changing the position of the enzyme on the top of the pore) then this may be observed as a change in current through the pore, as long the change lasts sufficiently long to be detected by the acquisition electronics. If the sub-step processes result in no change of conformation or shift in polynucleotide, or occur too quickly to observe, then in an ideal system the full catalytic cycle will result in only one step change in current for the polynucleotide moving one integer base forward.

[0167] For pores that do not contain R97W (eg Pro-CP1-Eco-(WT-Y51A/F56Q-StrepII(C))9), we observe long dwell time levels where predicted by the model, with an approximately exponential dwell distribution that is dependent on ATP.Mg concentration. For poreAQ we also short-lived substeps current levels in between the major levels, as marked in FIG. 48. Because the sub-step current levels are short-lived, they are most easily observed in the gap between two widely separated current levels. The sub-steps levels correspond to an intermediate approximately 0.5 base movement of the polynucleotide, and under these conditions have an ATP.Mg independent dwell time of approximately 3 milliseconds.

[0168] Pores containing R97W (e.g. Pro-CP1-Eco-(WT-Y51A/F56Q/R97W-StrepII(C))9) shows similar longer lived main levels with ATP.Mg dependent dwell times, but shows no signs of distinct intermediate sub-step current levels under these conditions or at this acquisition frequency (possible explanations being that they do not occur, occur too quickly to be observed, or that the substeps do occur and are slow enough in principle to be observed but that in practice they are not observed due to for example the way in which the enzyme interacts with the pore).

[0169] The raw data traces (FIG. 48) show the ionic current (y-axis, pA) vs. time (x-axis, seconds) trace of an enzyme controlled DNA strand translocation through a nanopore for the pores Pro-CP1-Eco-(WT-Y51A/F56Q/R97W-StrepII(C))9 (Pore 97W) and Pro-CP1-Eco-(WT-Y51A/F56Q-StrepII(C))9 (Pore AQ). Each current level is the result of the sequence held in the nanopore reader altering the flow of ions, and step-wise changes in current are observed when the polynucleotide changes position in the nanopore, for example when the enzyme moves the entire strand forward one base. In this case the DNA strand contains in part a repeating sequence (GGTT)_n. The data was acquired by loading a Dda enzyme onto synthetic DNA polynucleotides and running on a MinION recording raw data output (C is buffer: 500 mM KCl, 25 mM HEPES, pH8, 0.6 mM MgCl₂, 0.6 mM ATP, 140 mV, 37 deg C., 5 kHz acquisition frequency). Pore97W only shows the main current levels from integer step-wise movements of the polynucleotide, with no significant data density between the levels. In comparison, PoreAQ has significant intermediate sub-step levels, as marked by the arrows in FIG. 48.

[0170] The mutant monomers preferably have improved polynucleotide reading properties i.e. display improved polynucleotide capture and nucleotide discrimination. In particular, pores constructed from the mutant monomers preferably capture nucleotides and polynucleotides more easily than the wild type. In addition, pores constructed from the mutant monomers preferably display an increased current range, which makes it easier to discriminate between different nucleotides, and a reduced variance of states, which increases the signal-to-noise ratio.

[0171] In addition, the number of nucleotides contributing to the current as the polynucleotide moves through pores constructed from the mutants is preferably decreased. This makes it easier to identify a direct relationship between the observed current as the polynucleotide moves through the pore and the polynucleotide sequence. In addition, pores constructed from the mutant monomers may display an increased throughput, i.e. are more likely to interact with an analyte, such as a polynucleotide. This makes it easier to characterise analytes using the pores. Pores constructed from the mutant monomers may insert into a membrane more easily. A mutant monomer of the invention comprises a variant of the sequence shown in SEQ ID NO: 2. SEQ ID NO: 2 is the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. A variant of SEQ ID NO: 2 is a polypeptide that has an amino acid sequence which varies from that of SEQ ID NO: 2 and which retains its ability to form a pore. The ability of a variant to form a pore can be assayed using any method known in the art. For instance, the variant may be inserted into an amphiphilic layer along with other appropriate subunits and its ability to oligomerise to form a pore may be determined. Methods are known in the art for inserting subunits into membranes, such as amphiphilic layers. For example, subunits may be suspended in a purified form in a solution containing a triblock copolymer membrane such that it diffuses to the membrane and is inserted by binding to the membrane and assembling into a functional state.

[0172] In all of the discussion herein, the standard one letter codes for amino acids are used. These are as follows: alanine (A), arginine (R), asparagine (N), aspartic acid (D), cysteine (C), glutamic acid (E), glutamine (Q), glycine (G), histidine (H), isoleucine (I), leucine (L), lysine (K), methionine (M), phenylalanine (F), proline (P), serine (S), threonine (T), tryptophan (W), tyrosine (Y) and valine (V). Standard substitution notation is also used, i.e. Q42R means that Q at position 42 is replaced with R.

[0173] In one embodiment of the mutant monomers of the invention, the variant of SEQ ID NO: 2 comprises (a) one or more mutations at the following positions (i.e. mutations at one or more of the following positions) I41, R93, A98, Q100, G103, T104, A106, I107, N108, L113, S115, T117, Y130, K135, E170, S208, D233, D238 and E244 and/or (b) one or more of D43S, E44S, F48S/N/Q/Y/W/I/V/H/R/K, Q87N/R/K, N91K/R, K94R/F/Y/W/L/S/N, R97F/Y/W/V/I/K/S/Q/H, E101I/L/A/H, N102K/Q/L/T/V/S/S1, R110F/G/N, Q114R/K R142Q/S, T150Y/A/V/L/S/Q/N, R192D/Q/F/S/T and D248S/N/Q/K/R. The variant may comprise (a); (b); or (a) and (b).

[0174] In some embodiments of the invention, the variant of SEQ ID NO: 2 comprises R97W.

[0175] In some embodiments of the invention, the variant of SEQ ID NO: 2 comprises R192D/Q/F/S/T, preferably R192D/Q, more preferably R192D. In (a), the variant may comprise modifications at any number and combination of the positions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 or 19 of the positions. In (a), the variant preferably comprises one or more of I41N, R93F/Y/W/L/I/V/N/Q/S, A98K/R, Q100K/R, G103F/W/S/N/K/R, T104R/K, A106R/K, I107R/K/W/F/Y/L/V, N108R/K, L113K/R, S115R/K, T117R/K, Y130W/F/H/O/N, K135L/V/N/Q/S, E170S/N/Q/K/R, S208V/I/F/W/Y/L/T, D233S/N/Q/K/R, D238S/N/Q/K/R and E244S/N/Q/K/R.

[0176] In (a), the variant preferably comprises one or more modifications which provide more consistent movement of a target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. In particular, in (a), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) R93, G103 and I107. The variant may comprise R93; G103; I107; R93 and G103; R93 and I107; G103 and I107; or R93, G103 and I107. The variant preferably comprises one or more of R93F/Y/V/L/I/V/N/Q/S, G103F/W/S/N/K/R and I107R/K/W/F/Y/L/V. These may be present in any combination shown for the positions R93, G103 and I107.

[0177] In (a), the variant preferably comprises one or more modifications which allow pores constructed from the mutant monomers preferably capture nucleotides and polynucleotides more easily. In particular, in (a), the variant preferably comprises one or more mutations at the following positions

(i.e. mutations at one or more of the following positions) I41, T104, A106, N108, L113, S115, T117, E170, D233, D238 and E244. The variant may comprise modifications at any number and combination of the positions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or 11 of the positions. The variant preferably comprises one or more of I41N, T104R/K, A106R/K, N108R/K, L113K/R, S115R/K, T117R/K, E170S/N/Q/K/R, D233S/N/Q/K/R, D238S/N/Q/K/R and E244S/N/Q/K/R. Additionally or alternatively the variant may comprise (c) Q42K/R, E44N/Q, L90R/K, N91R/K, I95R/K, A99R/K, E101H/K/N/Q/T and/or Q114K/R.

[0178] In (a), the variant preferably comprises one or more modifications which provide more consistent movement and increase capture. In particular; in (a) the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) (i) A98, (ii) Q100, (iii) G103 and (iv) I107. The variant preferably comprises one or more of (i) A98R/K, (ii) Q100K/R, (iii) G103K/R and (iv) I107R/K. The variant may comprise {i}; {ii}; {iii}; {iv}; {i,ii}; {i,iii}; {i,iv}; {ii,iii}; {ii,iv}; {iii,iv}; {i,ii,iii}; {i,ii,iv}; {i,iii,iv}; {ii,iii,iv}; or {i,ii,iii,iv}.

[0179] Particularly preferred mutant monomers which provide for increased capture of analytes, such as a polynucleotides include a mutation at one or more of positions Q42, E44, E44, L90, N91, I95, A99, E101 and Q114, which mutation removes the negative charge and/or increases the positive charge at the mutated positions. In particular, the following mutations may be included in a mutant monomer of the invention to produce a CsgG pore that has an improved ability to capture an analyte, preferably a polynucleotide: Q42K, E44N, E44Q, L90R, N91R, I95R, A99R, E101H, E101K, E101N, E101Q, E101T and Q114K. Examples of particular mutant monomers which comprise one of these mutations in combination with other beneficial mutations are described in Example 11.

[0180] In (a), the variant preferably comprises one or more modifications which provide increased characterisation accuracy. In particular, in (a), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) Y130, K135 and S208, such as Y130; K135; S208; Y130 and K135; Y130 and S208; K135 and S208; or Y130, K135 and S208. The variant preferably comprises one or more of Y130W/F/H/Q/N, K135L/V/N/Q/S and R142Q/S. These substitutions may be present in any number and combination as set out for Y130, K135 and S208.

[0181] In (b), the variant may comprise any number and combination of the substitutions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12 of the substitutions. In (b), the variant preferably comprises one or more modifications which provide more consistent movement of a target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. In particular, in (b), the variant preferably comprises one or more one or more of (i) Q87N/R/K, (ii) K94R/F/Y/W/L/S/N, (iii) R97F/Y/W/V/I/K/S/Q/H, (iv) N102K/Q/L/I/V/S/H and (v) R110F/G/N. More preferably, the variant comprises K94D or K94Q and/or R97W or R97Y.

[0182] The variant may comprise {i}; {ii}, {iii}; {iv}, {v}; {i,ii}; {i,iii}; {i,iv}, {i,v}; {ii,iii}; {ii,iv}; {ii,v}; {iii,iv}; {iii,v}; {iv,v}; {i,ii,iii}; {i,ii,iv}; {i,ii,v}; {i,iii,iv}; {i,iii,v}; {i,iv,v}; {ii,iii,iv}; {ii,iii,v}; {ii,iv,v}; {iii,iv,v}; {i,ii,iii,iv}; {i,ii,iii,v}; {i,ii,iv,v}; {i,iii,iv,v}; {ii,iii,iv,v}; or {i,ii,iii,iv,v}. Other preferred variants that are modified to provide more consistent movement of a target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer include (vi) R93W and R93Y. A preferred variant may comprise R93W and R97W, R93Y and R97W, R93W and R97W, or more preferably R93Y and R97Y. The variant may comprise {vi}; {i,vi}; {ii,vi}; {iii,vi}; {iv,vi}; {v,vi}; {i,ii,vi}; {i,iii,vi}; {i,iv,vi}; {i,v,vi}; {ii,iii,vi}; {ii,iv,vi}; {ii,v,vi}; {iii,iv,vi}; {iii,v,vi}; {iv,v,vi}, {i,ii,iii,vi}; {i,ii,iv,vi}; {i,ii,v,vi}, {i,iii,iv,vi}; {i,iii,v,vi}; {i,iv,v,vi}; {ii,iii,iv,vi}; {ii,iii,v,vi}; {ii,iv,v,vi}; {iii,iv,v,vi}, {i,ii,iii,iv,vi}, {i,ii,iii,v,vi}; {i,ii,iv,v,vi}; {i,iii,iv,v,vi}; {ii,iii,iv,v,vi}; or {i,ii,iii,iv,v,vi}.

[0183] In (b), the variant preferably comprises one or modifications which allow pores constructed from the mutant monomers preferably capture nucleotides and polynucleotides more easily. In

particular, in (b), the variant preferably comprises one or more of (i) D43S, (ii) E44S, (iii) N91K/R, (iv) Q114R/K and (v) D248S/N/Q/K/R. The variant may comprise {i}; {ii}; {iii}; {iv}; {v}; {i,ii}; {i,iii}; {i,iv}; {i,v}; {ii,iii}; {ii,iv}; {ii,v}; {iii,iv}; {iii,v}; {iv,v}; {i,ii,iii}; {i,ii,iv}; {i,ii,v}; {i,iii,iv}; {i,iii,v}; {i,iv,v}; {ii,iii,iv}; {ii,iii,v}; {ii,iv,v}; {iii,iv,v}; {i,ii,iii,iv}; {i,ii,iii,v}; {i,ii,iv,v}; {i,iii,iv,v}; {ii,iii,iv,v}; or {i,ii,iii,iv,v}.

[0184] In (b), the variant preferably comprises one or more modifications which provide more consistent movement and increase capture. In particular, in (b), the variant preferably comprises one or more of Q87R/K, E101I/L/A/H and N102K, such as Q87R/K; E101I/L/A/H; N102K; Q87R/K and E101I/L/A/H; Q87R/K and N102K; E101I/L/A/H and N102K; or Q87R/K, E101I/L/A/H and N102K.

[0185] In (b), the variant preferably comprises one or more modifications which provide increased characterisation accuracy. In particular, in (a), the variant preferably comprises F48S/N/Q/Y/W/I/V.

[0186] In (b), the variant preferably comprises one or more modifications which provide increased characterisation accuracy and increased capture. In particular, in (a), the variant preferably comprises F48H/R/K.

[0187] The variant may comprise modifications in both (a) and (b) which provide more consistent movement. The variant may comprise modifications in both (a) and (b) which provide increased capture.

[0188] The invention provides variants of SEQ ID NO: 2 which provide an increased throughput of an assay for characterising an analyte, such as a polynucleotide, using a pore comprising the variant. Such variants may comprise a mutation at K94, preferably K94Q or K94N, more preferably K94Q. Examples of particular mutant monomers which comprise a K94Q or K94N mutation in combination with other beneficial mutations are described in Examples 10 and 11.

[0189] The invention provides variants of SEQ ID NO: 2 which provide increased characterisation accuracy in an assay for characterising an analyte, such as a polynucleotide, using a pore comprising the variant. Such variants include variants that comprise: a mutation at F191, preferably F191T; deletion of V105-I107; deletion of F193-L199 or of D195-L199; and/or a mutation at R93 and/or R97, preferably R93Y, R97Y, or more preferably, R97W, R93W or both R97Y and R97Y. Examples of particular mutant monomers which comprise one or more of these mutations in combination with other beneficial mutations are described in Example 9.

[0190] In another embodiment of the mutant monomers of the invention, the variant of SEQ ID NO: 2 comprises (A) deletion of one or more positions R192, F193, I194, D195, Y196, Q197, R198, L199, L200 and E201 and/or (B) deletion of one or more of V139/G140/D149/T150/V186/Q187/V204/G205 (called band 1 herein), G137/G138/Q151/Y152/Y184/E185/Y206/T207 (called band 2 herein) and A141/R142/G147/A148/A188/G189/G202/E203 (called band 3 herein).

[0191] In (A), the variant may comprise deletion of any number and combination of the positions, such as 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 of the positions. In (A), the variant preferably comprises deletion of [0192] D195, Y196, Q197, R198 and L199; [0193] R192, F193, I194, D195, Y196, Q197, R198, L199 and L200; [0194] Q197, R198, L199 and L200; [0195] I194, D195, Y196, Q197, R198 and L199; [0196] D195, Y196, Q197, R198, L199 and L200; [0197] Y196, Q197, R198, L199, L200 and E201; [0198] Q197, R198, L199, L200 and E201; [0199] Q197, R198, L199; or [0200] F193, I194, D195, Y196, Q197, R198 and L199.

More preferably, the variant comprises deletion of D195, Y196, Q197, R198 and L199 or F193, I194, D195, Y196, Q197, R198 and L199. In (B), any number and combination of bands 1 to 3 may be deleted, such as band 1; band 2; band 3; bands 1 and 2; bands 1 and 3; bands 2 and 3; or bands 1, 2 and 3.

[0201] The variant may comprise deletions according to (A); (B); or (A) and (B).

[0202] The variants comprising deletion of one or more positions according to (A) and/or (B) above may further comprise any of the modifications or substitutions discussed above and below. If

the modifications or substitutions are made at one or more positions which appear after the deletion positions in SEQ ID NO: 2, the numbering of the one or more positions of the modifications or substitutions must be adjusted accordingly. For instance, if L199 is deleted, E244 becomes E243. Similarly, if band 1 is deleted, R192 becomes R186.

[0203] In another embodiment of the mutant monomers of the invention, the variant of SEQ ID NO: 2 comprises (C) deletion of one or more positions V105, A106 and I107. The deletions in accordance with (C) may be made in addition to deletions according to (A) and/or (B).

[0204] The above-described deletions typically reduce the noise associated with the movement of the target polynucleotide with respect to, such as through, a transmembrane pore comprising the monomer. As a result the target polynucleotide can be characterised more accurately.

[0205] In the paragraphs above where different amino acids at a specific position are separated by the/symbol, the/symbol means “or”. For instance, Q87R/K means Q87R or Q87K.

[0206] The invention provides variants of SEQ ID NO: 2 which provide increased capture of an analyte, such as a polynucleotide. Such variants may comprise a mutation at T104, preferably T104R or T104K, a mutation at N91, preferably N91R, a mutation at E101, preferably E101K/N/Q/T/H, a mutation at position E44, preferably E44N or E44Q and/or a mutation at position Q42, preferably Q42K.

[0207] The mutations at different positions in SEQ ID NO: 2 may be combined in any possible way. In particular, a monomer of the invention may comprise one or more mutation that improves accuracy, one or more mutation that reduces noise and/or one or more mutation that enhances capture of an analyte.

[0208] In the mutant monomers of the invention, the variant of SEQ ID NO: 2 preferably comprises one or more of the following (1) one or more mutations at the following positions (i.e. mutations at one or more of the following positions) N40, D43, E44, S54, S57, Q62, R97, E101, E124, E131, R142, T150 and R192, such as one or more mutations at the following positions (i.e. mutations at one or more of the following positions) N40, D43, E44, S54, S57, Q62, E101, E131 and T150 or N40, D43, E44, E101 and E131; (ii) mutations at Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56; (iii) Q42R or Q42K; (iv) K49R; (v) N102R, N102F, N102Y or N102W; (vi) D149N, D149Q or D149R; (vii) E185N, E185Q or E185R; (viii) D195N, D195Q or D195R; (ix) E201N, E201Q or E201R; (x) E203N, E203Q or E203R; and (xi) deletion of one or more of the following positions F48, K49, P50, Y51, P52, A53, S54, N55, F56 and S57. The variant may comprise any combination of (i) to (xi). In particular, the variant may comprise: (where each variant in parentheses { } separated by a space represents an optional variant from the list of variants, namely {i} or {ii} or {iii} or {iv} or {v} and so on) {i} {ii} {iii} {iv} {v} {vi} {vii} {viii} {ix} {x} {xi} {i,ii} {i,iii} {i,iv} {i,v} {i,vi} {i,vii} {i,viii} {i,ix} {i,x} {i,xi} {ii,iii} {ii,iv} {ii,v} {ii,vi} {ii,vii} {ii,viii} {ii,ix} {ii,x} {ii,xi} {iii,iv} {iii,v} {iii,vi} {iii,vii} {iii,viii} {iii,ix} {iii,x} {iii,xi} {iv,v} {iv,vi} {iv,vii} {iv,viii} {iv,ix} {iv,x} {iv,xi} {v,vi} {v,vii} {v,viii} {v,ix} {v,x} {v,xi} {vi,vii} {vi,viii} {vi,ix} {vi,x} {vi,xi} {vii,viii} {vii,ix} {vii,x} {vii,xi} {viii,ix} {viii,x} {viii,xi} {ix,x} {ix,xi} {x,xi} {i,ii,iii} {i,ii,iv} {i,ii,v} {i,ii,vi} {i,ii,vii} {i,ii,viii} {i,ii,ix} {i,ii,x} {i,ii,xi} {i,iii,iv} {i,iii,v} {i,iii,vi} {i,iii,vii} {i,iii,viii} {i,iii,ix} {i,iii,x} {i,iii,xi} {i,iv,v} {i,iv,vi} {i,iv,vii} {i,iv,viii} {i,iv,ix} {i,iv,x} {i,iv,xi} {i,v,vi} {i,v,vii} {i,v,viii} {i,v,ix} {i,v,x} {i,v,xi} {i,vi,vii} {i,vi,viii} {i,vi,ix} {i,vi,x} {i,vi,xi} {i,vii,viii} {i,vii,ix} {i,vii,x} {i,vii,xi} {i,viii,ix} {i,viii,x} {i,viii,xi} {i,ix,x} {i,ix,xi} {i,x,xi} {ii,iii,iv} {ii,iii,v} {ii,iii,vi} {ii,iii,vii} {ii,iii,viii} {ii,iii,ix} {ii,iii,x} {ii,iii,xi} {ii,iv,v} {ii,iv,vi} {ii,iv,vii} {ii,iv,viii} {ii,iv,ix} {ii,iv,x} {ii,iv,xi} {ii,v,vi} {ii,v,vii} {ii,v,viii} {ii,v,ix} {ii,v,x} {ii,v,xi} {ii,vi,vii} {ii,vi,viii} {ii,vi,ix} {ii,vi,x} {ii,vi,xi} {ii,vii,viii} {ii,vii,ix} {ii,vii,x} {ii,vii,xi} {ii,viii,ix} {ii,viii,x} {ii,viii,xi} {ii,ix,x} {ii,ix,xi} {ii,x,xi} {iii,iv,v} {iii,iv,vi} {iii,iv,vii} {iii,iv,viii} {iii,iv,ix} {iii,iv,x} {iii,iv,xi} {iii,v,vi} {iii,v,vii} {iii,v,viii} {iii,v,ix} {iii,v,x} {iii,v,xi} {iii,vi,vii} {iii,vi,viii} {iii,vi,ix} {iii,vi,x} {iii,vi,xi} {iii,vii,viii} {iii,vii,ix} {iii,vii,x} {iii,vii,xi} {iii,viii,ix} {iii,viii,x} {i,ii,iii,iv} {i,ii,iii,v} {i,ii,iii,vi} {i,ii,iii,vii} {i,ii,iii,viii} {i,ii,iii,ix} {i,ii,iii,x} {i,ii,iii,xi} {i,ii,iv,v} {i,ii,iv,vi} {i,ii,iv,vii} {i,ii,iv,viii} {i,ii,iv,ix}

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[0209] If the variant comprises any one of (i) and (iii) to (xi), it may further comprise a mutation at one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0210] In (i), the variant may comprises mutations at any number and combination of N40, D43, E44, S54, S57, Q62, R97, E101, E124, E131, R142, T150 and R192. In (i), the variant preferably comprises one or more mutations at at the following positions (i.e. mutations at one or more of the following positions) N40, D43, E44, S54, S57, Q62, E101, E131 and T150. In (i), the variant preferably comprises one or more mutations at the following positions (i.e. mutations at one or more of the following positions) N40, D43, E44, E101 and E131. In (i), the variant preferably comprises a mutation at S54 and/or S57. In (i), the variant more preferably comprises a mutation at (a) S54 and/or S57 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. If S54 and/or S57 are deleted in (xi), it/they cannot be mutated in (i) and vice versa. In (i), the variant preferably comprises a mutation at T150, such as T150I. Alternatively the variant preferably comprises a mutation at (a) T150 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. In (i), the variant preferably comprises a mutation at Q62, such as Q62R or Q62K. Alternatively the variant preferably comprises a mutation at (a) Q62 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. The variant may comprise a mutation at D43, E44, Q62 or any combination thereof, such as D43, E44, Q62, D43/E44, D43/Q62, E44/Q62 or D43/E44/Q62. Alternatively the variant preferably comprises a mutation at

(a) D43, E44, Q62, D43/Q62, E44/Q62 or D43/E44/Q62 and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0211] In (ii) and elsewhere in this application where different positions are separated by the/symbol, the/symbol means “and” such that Y51/N55 is Y51 and N55. In (ii), the variant preferably comprises mutations at Y51/N55. It has been proposed that the constriction in CsgG is composed of three stacked concentric rings formed by the side chains of residues Y51, N55 and F56 (Goyal et al, 2014, Nature, 516, 250-253). Mutation of these residues in (ii) may therefore decrease the number of nucleotides contributing to the current as the polynucleotide moves through the pore and thereby make it easier to identify a direct relationship between the observed current (as the polynucleotide moves through the pore) and the polynucleotide. F56 may be mutated in any of the ways discussed below with reference to variants and pores useful in the method of the invention.

[0212] In (v), the variant may comprise N102R, N102F, N102Y or N102W. The variant preferably comprises (a) N102R, N102F, N102Y or N102W and (b) a mutation at one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0213] In (xi), any number and combination of K49, P50, Y51, P52, A53, S54, N55, F56 and S57 may be deleted. Preferably one or more of K49, P50, Y51, P52, A53, S54, N55 and S57 may be deleted. If any of Y51, N55 and F56 are deleted in (xi), it/they cannot be mutated in (ii) and vice versa.

[0214] In (i), the variant preferably comprises one or more of the following substitutions N40R, N40K, D43N, D43Q, D43R, D43K, E44N, E44Q, E44R, E44K, S54P, S57P, Q62R, Q62K, R97N, R97G, R97L, E101N, E101Q, E101R, E101K, E101F, E101Y, E101W, E124N, E124Q, E124R, E124K, E124F, E124Y, E124W, E131D, R142E, R142N, T150I, R192E and R192N, such as one or more of N40R, N40K, D43N, D43Q, D43R, D43K, E44N, E44Q, E44R, E44K, S54P, S57P, Q62R, Q62K, E101N, E101Q, E101R, E101K, E101F, E101Y, E101W, E131D and T150I, or one or more of N40R, N40K, D43N, D43Q, D43R, D43K, E44N, E44Q, E44R, E44K, E101N, E101Q, E101R, E101K, E101F, E101Y, E101W and E131D. The variant may comprise any number and combination of these substitutions. In (i), the variant preferably comprises S54P and/or S57P. In (i), the variant preferably comprises (a) S54P and/or S57P and (b) a mutation at one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. The mutations at one or more of Y51, N55 and F56 may be any of those discussed below. In (i), the variant preferably comprises F56A/S57P or S54P/F56A. The variant preferably comprises T150I. Alternatively the variant preferably comprises a mutation at (a) T150I and (b) one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0215] In (i), the variant preferably comprises Q62R or Q62K. Alternatively the variant preferably comprises (a) Q62R or Q62K and (b) a mutation at one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56. The variant may comprise D43N, E44N, Q62R or Q62K or any combination thereof, such as D43N, E44N, Q62R, Q62K, D43N/E44N, D43N/Q62R, D43N/Q62K, E44N/Q62R, E44N/Q62K, D43N/E44N/Q62R or D43N/E44N/Q62K. Alternatively the variant preferably comprises (a) D43N, E44N, Q62R, Q62K, D43N/E44N, D43N/Q62R, D43N/Q62K, E44N/Q62R, E44N/Q62K, D43N/E44N/Q62R or D43N/E44N/Q62K and (b) a mutation at one or more of Y51, N55 and F56, such as at Y51, N55, F56, Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56.

[0216] In (i), the variant preferably comprises D43N.

[0217] In (i), the variant preferably comprises E101R, E101S, E101F or E101N.

[0218] In (i), the variant preferably comprises E124N, E124Q, E124R, E124K, E124F, E124Y, E124W or E124D, such as E124N.

[0219] In (i), the variant preferably comprises R142E and R142N.

[0220] In (i), the variant preferably comprises R97N, R97G or R97L.

[0221] In (i), the variant preferably comprises R192E and R192N.

[0222] In (ii), the variant preferably comprises F56N/N55Q, F56N/N55K, F56N/N55S, F56N/N55G, F56N/N55A, F56N/N55T, F56Q/N55Q, F56Q/N55R, F56Q/N55K, F56Q/N55S, F56Q/N55G, F56Q/N55A, F56Q/N55T, F56R/N55Q, F56R/N55R, F56R/N55K, F56R/N55S, F56R/N55G, F56R/N55A, F56R/N55T, F56S/N55Q, F56S/N55R, F56S/N55K, F56S/N55S, F56S/N55G, F56S/N55A, F56S/N55T, F56G/N55Q, F56G/N55R, F56G/N55K, F56G/N55S, F56G/N55G, F56G/N55A, F56G/N55T, F56A/N55Q, F56A/N55R, F56A/N55K, F56A/N55S, F56A/N55G, F56A/N55A, F56A/N55T, F56K/N55Q, F56K/N55R, F56K/N55K, F56K/N55S, F56K/N55G, F56K/N55A, F56K/N55T, F56N/Y51L, F56N/Y51V, F56N/Y51A, F56N/Y51N, F56N/Y51Q, F56N/Y51S, F56N/Y51G, F56Q/Y51L, F56Q/Y51V, F56Q/Y51A, F56Q/Y51N, F56Q/Y51Q, F56Q/Y51S, F56Q/Y51G, F56R/Y51L, F56R/Y51V, F56R/Y51A, F56R/Y51N, F56R/Y51Q, F56R/Y51S, F56R/Y51G, F56S/Y51L, F56S/Y51V, F56S/Y51A, F56S/Y51N, F56S/Y51Q, F56S/Y51S, F56S/Y51G, F56G/Y51L, F56G/Y51V, F56G/Y51A, F56G/Y51N, F56G/Y51Q, F56G/Y51S, F56G/Y51G, F56A/Y51L, F56A/Y51V, F56A/Y51A, F56A/Y51N, F56A/Y51Q, F56A/Y51S, F56A/Y51G, F56K/Y51L, F56K/Y51V, F56K/Y51A, F56K/Y51N, F56K/Y51Q, F56K/Y51S, F56K/Y51G, N55Q/Y51L, N55Q/Y51V, N55Q/Y51A, N55Q/Y51N, N55Q/Y51Q, N55Q/Y51S, N55Q/Y51G, N55R/Y51L, N55R/Y51V, N55R/Y51A, N55R/Y51N, N55R/Y51Q, N55R/Y51S, N55R/Y51G, N55K/Y51L, N55K/Y51V, N55K/Y51A, N55K/Y51N, N55K/Y51Q, N55K/Y51S, N55K/Y51G, N55S/Y51L, N55S/Y51V, N55S/Y51A, N55S/Y51N, N55S/Y51Q, N55S/Y51S, N55S/Y51G, N55G/Y51L, N55G/Y51V, N55G/Y51A, N55G/Y51N, N55G/Y51Q, N55G/Y51S, N55G/Y51G, N55A/Y51L, N55A/Y51V, N55A/Y51A, N55A/Y51N, N55A/Y51Q, N55A/Y51S, N55A/Y51G, N55T/Y51L, N55T/Y51V, N55T/Y51A, N55T/Y51N, N55T/Y51Q, N55T/Y51S, N55T/Y51G, F56N/N55Q/Y51L, F56N/N55Q/Y51V, F56N/N55Q/Y51A, F56N/N55Q/Y51N, F56N/N55Q/Y51Q, F56N/N55Q/Y51S, F56N/N55Q/Y51G, F56N/N55R/Y51L, F56N/N55R/Y51V, F56N/N55R/Y51A, F56N/N55R/Y51N, F56N/N55R/Y51Q, F56N/N55R/Y51S, F56N/N55R/Y51G, F56N/N55K/Y51L, F56N/N55K/Y51V, F56N/N55K/Y51A, F56N/N55K/Y51N, F56N/N55K/Y51Q, F56N/N55K/Y51S, F56N/N55K/Y51G, F56N/N55S/Y51L, F56N/N55S/Y51V, F56N/N55S/Y51A, F56N/N55S/Y51N, F56N/N55S/Y51Q, F56N/N55S/Y51S, F56N/N55S/Y51G, F56N/N55G/Y51L, F56N/N55G/Y51V, F56N/N55G/Y51A, F56N/N55G/Y51N, F56N/N55G/Y51Q, F56N/N55G/Y51S, F56N/N55G/Y51G, F56N/N55A/Y51L, F56N/N55A/Y51V, F56N/N55A/Y51A, F56N/N55A/Y51N, F56N/N55A/Y51Q, F56N/N55A/Y51S, F56N/N55A/Y51G, F56N/N55T/Y51L, F56N/N55T/Y51V, F56N/N55T/Y51A, F56N/N55T/Y51N, F56N/N55T/Y51Q, F56N/N55T/Y51S, F56N/N55T/Y51G, F56Q/N55Q/Y51L, F56Q/N55Q/Y51V, F56Q/N55Q/Y51A, F56Q/N55Q/Y51N, F56Q/N55Q/Y51Q, F56Q/N55Q/Y51S, F56Q/N55Q/Y51G, F56Q/N55R/Y51L, F56Q/N55R/Y51V, F56Q/N55R/Y51A, F56Q/N55R/Y51N, F56Q/N55R/Y51Q, F56Q/N55R/Y51S, F56Q/N55R/Y51G, F56Q/N55K/Y51L, F56Q/N55K/Y51V, F56Q/N55K/Y51A, F56Q/N55K/Y51N, F56Q/N55K/Y51Q, F56Q/N55K/Y51S, F56Q/N55K/Y51G, F56Q/N55S/Y51L, F56Q/N55S/Y51V, F56Q/N55S/Y51A, F56Q/N55S/Y51N, F56Q/N55S/Y51Q, F56Q/N55S/Y51S, F56Q/N55S/Y51G, F56Q/N55G/Y51L, F56Q/N55G/Y51V, F56Q/N55G/Y51A, F56Q/N55G/Y51N, F56Q/N55G/Y51Q, F56Q/N55G/Y51S, F56Q/N55G/Y51G, F56Q/N55A/Y51L, F56Q/N55A/Y51V, F56Q/N55A/Y51A, F56Q/N55A/Y51N, F56Q/N55A/Y51Q, F56Q/N55A/Y51S, F56Q/N55A/Y51G, F56Q/N55T/Y51L, F56Q/N55T/Y51V, F56Q/N55T/Y51A, F56Q/N55T/Y51N, F56Q/N55T/Y51Q, F56Q/N55T/Y51S, F56Q/N55T/Y51G, F56R/N55Q/Y51L, F56R/N55Q/Y51V, F56R/N55Q/Y51A, F56R/N55Q/Y51N, F56R/N55Q/Y51Q, F56R/N55Q/Y51S, F56R/N55Q/Y51G, F56R/N55R/Y51L, F56R/N55R/Y51V, F56R/N55R/Y51A, F56R/N55R/Y51N, F56R/N55R/Y51Q, F56R/N55R/Y51S, F56R/N55R/Y51G, F56R/N55K/Y51L, F56R/N55K/Y51V, F56R/N55K/Y51A, F56R/N55K/Y51N,

[illegible]

F56K/N55A/Q, F56K/N55A/Y51G, F56K/N55A/Y51L, F56K/N55T/Y51V, F56K/N55T/Y51A, F56K/N55T/Y51N, F56K/N55T/Y51Q, F56K/N55T/Y51S, F56K/N55T/Y51G, F56E/N55R, F56E/N55K, F56D/N55R, F56D/N55K, F56R/N55E, F56R/N55D, F56K/N55E or F56K/N55D.

[0223] In (ii), the variant preferably comprises Y51R/F56Q, Y51N/F56N, Y51M/F56Q, Y51L/F56Q, Y51I/F56Q, Y51V/F56Q, Y51A/F56Q, Y51P/F56Q, Y51G/F56Q, Y51C/F56Q, Y51Q/F56Q, Y51N/F56Q, Y51S/F56Q, Y51E/F56Q, Y51D/F56Q, Y51K/F56Q or Y51H/F56Q.

[0224] In (ii), the variant preferably comprises Y51T/F56Q, Y51Q/F56Q or Y51A/F56Q.

[0225] In (ii), the variant preferably comprises Y51T/F56F, Y51T/F56M, Y51T/F56L, Y51T/F56I, Y51T/F56V, Y51T/F56A, Y51T/F56P, Y51T/F56G, Y51T/F56C, Y51T/F56Q, Y51T/F56N, Y51T/F56T, Y51T/F56S, Y51T/F56E, Y51T/F56D, Y51T/F56K, Y51T/F56H or Y51T/F56R.

[0226] In (ii), the variant preferably comprises Y51T/N56Q, Y51T/N55S or Y51T/N55A.

[0227] In (ii), the variant preferably comprises Y51A/F56F, Y51A/F56L, Y51A/F56I, Y51A/F56V, Y51A/F56A, Y51A/F56P, Y51A/F56G, Y51A/F56C, Y51A/F56Q, Y51A/F56N, Y51A/F56T, Y51A/F56S, Y51A/F56E, Y51A/F56D, Y51A/F56K, Y51A/F56H or Y51A/F56R.

[0228] In (ii), the variant preferably comprises Y51C/F56A, Y51E/F56A, Y51D/F56A, Y51K/F56A, Y51H/F56A, Y51Q/F56A, Y51N/F56A, Y51S/F56A, Y51P/F56A or Y51V/F56A.

[0229] In (xi), the variant preferably comprises deletion of Y51/P52, Y51/P52/A53, P50 to P52, P50 to A53, K49 to Y51, K49 to A53 and replacement with a single proline (P), K49 to S54 and replacement with a single P, Y51 to A53, Y51 to S54, N55/F56, N55 to S57, N55/F56 and replacement with a single P, N55/F56 and replacement with a single glycine (G), N55/F56 and replacement with a single alanine (A), N55/F56 and replacement with a single P and Y51N, N55/F56 and replacement with a single P and Y51Q, N55/F56 and replacement with a single P and Y51S, N55/F56 and replacement with a single G and Y51N, N55/F56 and replacement with a single G and Y51Q, N55/F56 and replacement with a single G and Y51S, N55/F56 and replacement with a single A and Y51N, N55/F56 and replacement with a single A/Y51Q or N55/F56 and replacement with a single A and Y51S.

[0230] The variant more preferably comprises D195N/E203N, D195Q/E203N, D195N/E203Q, D195Q/E203Q, E201N/E203N, E201Q/E203N, E201N/E203Q, E201Q/E203Q, E185N/E203Q, E185Q/E203Q, E185N/E203N, E185Q/E203N, D195N/E201N/E203N, D195Q/E201N/E203N, D195N/E201Q/E203N, D195N/E201N/E203Q, D195Q/E201Q/E203N, D195Q/E201N/E203Q, D195N/E201Q/E203Q, D195Q/E201Q/E203Q, D149N/E201N, D149Q/E201N, D149N/E201Q, D149Q/E201Q, D149N/E201N/D195N, D149Q/E201N/D195N, D149N/E201Q/D195N, D149N/E201N/D195Q, D149Q/E201Q/D195N, D149Q/E201N/D195Q, D149N/E201Q/D195Q, D149Q/E201Q/D195Q, D149N/E203N, D149Q/E203N, D149N/E203Q, D149Q/E203Q, D149N/E185N/E201N, D149Q/E185N/E201N, D149N/E185Q/E201N, D149N/E185N/E201Q, D149Q/E185Q/E201N, D149Q/E185N/E201Q, D149N/E185Q/E201Q, D149Q/E185Q/E201Q, D149N/E185N/E203N, D149Q/E185N/E203N, D149N/E185Q/E203N, D149N/E185N/E203Q, D149Q/E185Q/E203N, D149Q/E185N/E203Q, D149N/E185Q/E203Q, D149Q/E185Q/E203Q, D149N/E185N/E201N/E203N, D149Q/E185N/E201N/E203N, D149N/E185Q/E201N/E203N, D149N/E185N/E201Q/E203N, D149N/E185N/E201N/E203Q, D149Q/E185Q/E201N/E203N, D149Q/E185N/E201Q/E203N, D149N/E185Q/E201Q/E203N, D149N/E185Q/E201N/E203Q, D149Q/E185Q/E201Q/E203Q, D149Q/E185Q/E201N/E203Q, D149Q/E185N/E201Q/E203Q, D149N/E185Q/E201Q/E203Q, D149Q/E185Q/E201Q/E203N, D149N/E185N/D195N/E201N/E203N, D149Q/E185N/D195N/E201N/E203N, D149N/E185Q/D195N/E201N/E203N, D149N/E185N/D195Q/E201N/E203N, D149N/E185N/D195N/E201Q/E203N, D149N/E185N/D195N/E201N/E203Q, D149Q/E185Q/D195N/E201N/E203N, D149Q/E185N/D195Q/E201N/E203N, D149Q/B185N/D195N/E201Q/E203N, D149Q/E185N/D195N/E201N/E203Q, D149N/E185Q/D195Q/E201N/E203N,

D149N/E185Q/D195N/E201Q/E203N, D149N/E185Q/D195N/E201N/E203Q,
D149N/E185N/D195Q/E201Q/E203N, D149N/E185N/D195Q/E201N/E203Q,
D149N/E185N/D195N/E201Q/E203Q, D149Q/E185Q/D195Q/201N/E203N,
D149Q/E185Q/D195N/E201Q/E203N, D149Q/E185Q/D195N/E201N/E203Q,
D149Q/E185N/D195Q/E201Q/E203N, D149Q/E185N/D195Q/E201N/E203Q,
D149Q/E185N/D195N/E201Q/E203Q, D149N/E185Q/D195Q/E201Q/E203N,
D149N/E185Q/D195Q/E201N/E203Q, D149N/E185Q/D195N/E201Q/E203Q,
D149N/E185N/D195Q/E201Q/E203Q, D149Q/E185Q/D195Q/E201Q/E203N,
D149Q/E185Q/D195Q/E201N/E203Q, D149Q/E185Q/D195N/E201Q/E203Q,
D149Q/E185N/D195Q/E201Q/E203Q, D149N/E185Q/D195Q/E201Q/E203Q,
D149Q/E185Q/D195Q/E201Q/E203Q, D149N/E185R/E201N/E203N,
D149Q/E185R/E201N/E203N, D149N/E185R/E201Q/E203N, D149N/E185R/E201N/E203Q,
D149Q/E185R/E201Q/E203N, D149Q/E185R/E201N/E203Q, D149N/B185R/E201Q/E203Q,
D149Q/E185R/E201Q/E203Q, D149R/E185N/E201N/E203N, D149R/E185Q/E201N/E203N,
D149R/E185N/E201Q/E203N, D149R/E185N/E201N/E203Q, D149R/E185Q/E201Q/E203N,
D149R/E185Q/E201N/E203Q, D149R/E185N/E201Q/E203Q, D149R/E185Q/E201Q/E203Q,
D149R/E185N/D195N/E201N/E203N, D149R/E185Q/D195N/E201N/E203N,
D149R/E185N/D195Q/E201N/E203N, D149R/E185N/D195N/E201Q/E203N,
D149R/E185Q/D195N/E201N/E203Q, D149R/E185Q/D195Q/E201N/E203N,
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D149R/E185N/D195Q/E201Q/E203Q, D149R/E185Q/D195Q/E201Q/E203Q,
D149N/E185R/D195N/E201N/E203N, D149Q/E185R/D195N/E201N/E203N,
D149N/E185R/D195Q/E201N/E203N, D149N/E185R/D195N/E201Q/E203N,
D149N/E185R/D195N/E201N/E203Q, D149Q/E185R/D195Q/E201N/E203N,
D149Q/E185R/D195N/E201Q/E203N, D149Q/E185R/D195N/E20IN/E203Q,
D149N/E185R/D195Q/E201Q/E203N, D149N/E185R/D195Q/E201N/E203Q,
D149N/E185R/D195N/E201Q/E203Q, D149Q/E185R/D195Q/E201Q/E203N,
D149Q/E185R/D195Q/E201N/E203Q, D149Q/E185R/D195N/E201Q/E203Q,
D149N/E185R/D195Q/E201Q/E203Q, D149Q/E185R/D195Q/E201Q/E203Q,
D149N/E185R/D195N/E201R/E203N, D149Q/E185R/D195N/E201R/E203N,
D149N/E185R/D195Q/E201R/E203N, D149N/E185R/D195N/E201R/E203Q,
D149Q/E185R/D195Q/E201R/E203N, D149Q/E185R/D195N/E201R/E203Q,
D149N/E185R/D195Q/E201R/E203Q, D149Q/E185R/D195Q/E201R/E203Q, E131D/K49R,
E101N/N102F, E101N/N102Y, E101N/N102W, E101F/N102F, E101F/N102Y, E101F/N102W,
E101Y/N102F, E101Y/N102Y, E101Y/N102W, E101W/N102F, E101W/N102Y, E101W/N102W,
E101N/N102R, E101F/N102R, E101Y/N102R or E101W/N102F.

[0231] Preferred variants of the invention which form pores in which fewer nucleotides contribute to the current as the polynucleotide moves through the pore comprise Y51A/F56A, Y51A/F56N, Y511/F56A, Y51L/F56A, Y51T/F56A, Y511/F56N, Y51L/F56N or Y51T/F56N or more preferably Y511/F56A, Y51L/F56A or Y51T/F56A. As discussed above, this makes it easier to identify a direct relationship between the observed current (as the polynucleotide moves through the pore) and the polynucleotide.

[0232] Preferred variants which form pores displaying an increased range comprise mutations at the following positions: [0233] Y51, F56, D149, E185, E201 and E203; [0234] N55 and F56; [0235] Y51 and F56; [0236] Y51, N55 and F56; or [0237] F56 and N102.

[0238] Preferred variants which form pores displaying an increased range comprise: [0239] Y51N, F56A, D149N, E185R, E20IN and E203N; [0240] N55S and F56Q; [0241] Y51A and F56A;

[0242] Y51A and F56N; [0243] Y51I and F56A; [0244] Y51L and F56A; [0245] Y51T and F56A; [0246] Y51I and F56N; [0247] Y51L and F56N; [0248] Y51T and F56N; [0249] Y51T and F56Q; [0250] Y51A, N55S and F56A; [0251] Y51A, N55S and F56N, [0252] Y51T, N55S and F56Q; or [0253] F56Q and N102R.

[0254] Preferred variants which form pores in which fewer nucleotides contribute to the current as the polynucleotide moves through the pore comprise mutations at the following positions: [0255] N55 and F56, such as N55X and F56Q, wherein X is any amino acid; or [0256] Y51 and F56, such as Y51X and F56Q, wherein X is any amino acid.

[0257] Particularly preferred variants comprise Y51A and F56Q.

[0258] Preferred variants which form pores displaying an increased throughput comprise mutations at the following positions: [0259] D149, E185 and E203; [0260] D149, E185, E201 and E203; or [0261] D149, E185, D195, E201 and E203.

[0262] Preferred variants which form pores displaying an increased throughput comprise: [0263] D149N, E185N and E203N; [0264] D149N, E185N, E201N and E203N; [0265] D149N, E185R, D195N, E201N and E203N; or [0266] D149N, E185R, D195N, E201R and E203N.

[0267] Preferred variants which form pores in which capture of the polynucleotide is increased comprise the following mutations: [0268] D43N/Y51T/F56Q; [0269] E44N/Y51T/F56Q; [0270] D43N/E44N/Y51T/F56Q; [0271] Y51T/F56Q/Q62R; [0272] D43N/Y51T/F56Q/Q62R; [0273] E44N/Y51T/F56Q/Q62R; or [0274] D43N/E44N/Y51T/F56Q/Q62R.

[0275] Preferred variants comprise the following mutations: [0276] D149R/E185R/E201R/E203R or Y51T/F56Q/D149R/E185R/E201R/E203R; [0277] D149N/E185N/E201N/E203N or Y51T/F56Q/D149N/E185N/E201N/E203N; [0278] E201R/E203R or Y51T/F56Q/E201R/E203R [0279] E201N/E203R or Y51T/F56Q/E201N/E203R; [0280] E203R or Y51T/F56Q/E203R; [0281] E203N or Y51T/F56Q/E203N; [0282] E201R or Y51T/F56Q/E201R; [0283] E201N or Y51T/F56Q/E201N, [0284] E185R or Y51T/F56Q/E185R; [0285] E185N or Y51T/F56Q/E185N; [0286] D149R or Y51T/F56Q/D149R; [0287] D149N or Y51T/F56Q/D149N; [0288] R142E or Y51T/F56Q/R142E; [0289] R142N or Y51T/F56Q/R142N; [0290] R192E or Y51T/F56Q/R192E; or [0291] R192N or Y51T/F56Q/R192N.

[0292] Preferred variants comprise the following mutations: [0293] Y51A/F56Q/E101N/N102R; [0294] Y51A/F56Q/R97N/N102G; [0295] Y51A/F56Q/R97N/N102R; [0296] Y51A/F56Q/R97N; [0297] Y51A/F56Q/R97G; [0298] Y51A/F56Q/R97L; [0299] Y51A/F56Q/N102R; [0300] Y51A/F56Q/N102F; [0301] Y51A/F56Q/N102G; [0302] Y51A/F56Q/E101R; [0303] Y51A/F56Q/E101F; [0304] Y51A/F56Q/E101N; or [0305] Y51A/F56Q/E101G

[0306] The variant preferably further comprises a mutation at T150. A preferred variant which forms a pore displaying an increased insertion comprises T150I. A mutation at T150, such as T150I, may be combined with any of the mutations or combinations of mutations discussed above.

[0307] A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) a mutation at Y51 and/or F56. A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b)

Y51R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M and/or F56R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M. A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) Y51L/V/A/N/Q/S/G and/or F56A/Q/N. A preferred variant of SEQ ID NO: 2 comprises (a) R97W and (b) Y51A and/or F56Q. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A and F56Q.

[0308] In the mutant monomers of the invention, the variant of SEQ ID NO: 2 preferably comprises a mutation at R192. The variant preferably comprises R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) a mutation at Y51 and/or F56 and (c) a mutation at R192, such as R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) Y51R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M and/or F56R/H/K/D/E/S/T/N/Q/C/G/P/A/V/I/L/M and (c) a mutation at R192, such as R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) Y51L/V/A/N/Q/S/G and/or F56A/Q/N and (c) a

mutation at R192, such as R192D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises (a) R97W, (b) Y51A and/or F56Q and (c) a mutation at R192, such as R192 D/Q/F/S/T/N/E, R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A, F56Q and R192D/Q/F/S/T or R192D/Q. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A, F56Q and R192D. A preferred variant of SEQ ID NO: 2 comprises R97W, Y51A, F56Q and R192Q. In the paragraphs above where different amino acids at a specific position are separated by the/symbol, the/symbol means “or”. For instance, R192D/Q means R192D or R192Q.

[0309] In the mutant monomers of the invention, the variant of SEQ ID NO: 2 preferably comprises a mutation at R93. A preferred variant of SEQ ID NO: 2 comprises (a) R93W and (b) a mutation at Y51 and/or F56, preferably Y51A and F56Q. D or R192N.deletion of V105, A106 and I107.

[0310] Any of the above preferred variants of SEQ ID NO: 2 may comprise a K94N/Q mutation. Any of the above preferred variants of SEQ ID NO: 2 may comprise a F191T mutation. The invention also provides a mutant CsgG monomer comprising a variant of the sequence shown in SEQ ID NO: 2 comprising the combination of mutations present in a variant disclosed in the Examples.

[0311] Methods for introducing or substituting naturally-occurring amino acids are well known in the art. For instance, methionine (M) may be substituted with arginine (R) by replacing the codon for methionine (ATG) with a codon for arginine (CGT) at the relevant position in a polynucleotide encoding the mutant monomer. The polynucleotide can then be expressed as discussed below.

[0312] Methods for introducing or substituting non-naturally-occurring amino acids are also well known in the art. For instance, non-naturally-occurring amino acids may be introduced by including synthetic aminoacyl-tRNAs in the IVTT system used to express the mutant monomer. Alternatively, they may be introduced by expressing the mutant monomer in *E. coli* that are auxotrophic for specific amino acids in the presence of synthetic (i.e. non-naturally-occurring) analogues of those specific amino acids. They may also be produced by naked ligation if the mutant monomer is produced using partial peptide synthesis.

Variants

[0313] In addition to the specific mutations discussed above, the variant may include other mutations. Over the entire length of the amino acid sequence of SEQ ID NO: 2, a variant will preferably be at least 50% homologous to that sequence based on amino acid identity. More preferably, the variant may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 2 over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95%, amino acid identity over a stretch of 100 or more, for example 125, 150, 175 or 200 or more, contiguous amino acids (“hard homology”).

[0314] Standard methods in the art may be used to determine homology. For example the UWGCG Package provides the BESTFIT program which can be used to calculate homology, for example used on its default settings (Devereux et al (1984) *Nucleic Acids Research* 12, p387-395). The PILEUP and BLAST algorithms can be used to calculate homology or line up sequences (such as identifying equivalent residues or corresponding sequences (typically on their default settings)), for example as described in Altschul S. F. (1993) *J Mol Evol* 36:290-300; Altschul, S. F et al (1990) *J Mol Biol* 215:403-10. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (<http://www.ncbi.nlm.nih.gov/>).

[0315] SEQ ID NO: 2 is the wild-type CsgG monomer from *Escherichia coli* Str. K-12 substr. MC4100. The variant of SEQ ID NO: 2 may comprise any of the substitutions present in another CsgG homologue. Preferred CsgG homologues are shown in SEQ ID NOs: 3 to 7 and 26 to 41. The variant may comprise combinations of one or more of the substitutions present in SEQ ID NOs: 3

to 7 and 26 to 41 compared with SEQ ID NO: 2. For example, mutations may be made at any one or more of the positions in SEQ ID NO: 2 that differ between SEQ ID NO: 2 and any one of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41. Such a mutation may be a substitution of an amino acid in SEQ ID NO: 2 with an amino acid from the corresponding position in any one of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41. Alternatively, the mutation at any one of these positions may be a substitution with any amino acid, or may be a deletion or insertion mutation, such as deletion or insertion of 1 to 10 amino acids, such as of 2 to 8 or 3 to 6 amino acids. Other than the mutations disclosed herein, the amino acids that are conserved between SEQ ID NO: 2 and all of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41 are preferably present in a variant of the invention.

However, conservative mutations may be made at any one or more of these positions that are conserved between SEQ ID NO. 2 and all of SEQ ID NOs: 3 to 7 and SEQ ID NOs: 26 to 41.

[0316] The invention provides a pore-forming CsgG mutant monomer that comprises any one or more of the amino acids described herein as being substituted into a specific position of SEQ ID NO: 2 at a position in the structure of the CsgG monomer that corresponds to the specific position in SEQ ID NO: 2. Corresponding positions may be determined by standard techniques in the art. For example, the PILEUP and BLAST algorithms mentioned above can be used to align the sequence of a CsgG monomer with SEQ ID NO: 2 and hence to identify corresponding residues.

[0317] In particular, the invention provides a pore-forming CsgG mutant monomer that comprises any one or more of the following: [0318] a W at a position corresponding to R97 in SEQ ID NO:2; [0319] a W at a position corresponding to R93 in SEQ ID NO:2; [0320] a Y at a position corresponding to R97 in SEQ ID NO: 2; [0321] a Y at a position corresponding to R93 in SEQ ID NO: 2; [0322] a Y at each of the positions corresponding to R93 and R97 in SEQ ID NO: 2; [0323] a D at the position corresponding to R192 in SEQ ID NO:2; [0324] deletion of the residues at the positions corresponding to V105-I107 in SEQ ID NO:2; [0325] deletion of the residues at one or more of the positions corresponding to F193 to L199 in SEQ ID NO: 2; [0326] deletion of the residues the positions corresponding to F195 to L199 in SEQ ID NO: 2; [0327] deletion of the residues the positions corresponding to F193 to L199 in SEQ ID NO: 2; [0328] a T at the position corresponding to F191 in SEQ ID NO: 2; [0329] a Q at the position corresponding to K49 in SEQ ID NO: 2; [0330] a N at the position corresponding to K49 in SEQ ID NO: 2; [0331] a Q at the position corresponding to K42 in SEQ ID NO: 2; [0332] a Q at the position corresponding to E44 in SEQ ID NO: 2; [0333] a N at the position corresponding to E44 in SEQ ID NO: 2; [0334] a R at the position corresponding to L90 in SEQ ID NO: 2; [0335] a R at the position corresponding to L91 in SEQ ID NO: 2; [0336] a R at the position corresponding to 195 in SEQ ID NO: 2; [0337] a R at the position corresponding to A99 in SEQ ID NO: 2; [0338] a H at the position corresponding to E101 in SEQ ID NO: 2; [0339] a K at the position corresponding to E101 in SEQ ID NO: 2; [0340] a N at the position corresponding to E101 in SEQ ID NO: 2; [0341] a Q at the position corresponding to E101 in SEQ ID NO: 2; [0342] a T at the position corresponding to E101 in SEQ ID NO: 2; [0343] a K at the position corresponding to Q114 in SEQ ID NO: 2.

[0344] The CsgG pore-forming monomer of the invention preferably further comprises an A at the position corresponding to Y51 in SEQ ID NO: 2 and/or a Q at the position corresponding to F56 in SEQ ID NO: 2.

[0345] The pore-forming mutant monomer typically retains the ability to form the same 3D structure as the wild-type CsgG monomer, such as the same 3D structure as a CsgG monomer having the sequence of SEQ ID NO: 2. The 3D structure of CsgG is known in the art and is disclosed, for example, in Cao et al (2014) PNAS E5439-E5444. Any number of mutations may be made in the wild-type CsgG sequence in addition to the mutations described herein provided that the CsgG mutant monomer retains the improved properties imparted on it by the mutations of the present invention.

[0346] Typically the CsgG monomer will retain the ability to form a structure comprising three alpha-helices and five beta-sheets. The present inventors have shown in particular that mutations

may be made at least in the region of CsgG which is N-terminal to the first alpha helix (which starts at S63 in SEQ ID NO:2), in the second alpha helix (from G85 to A99 of SEQ ID NO: 2), in the loop between the second alpha helix and the first beta sheet (from Q100 to N120 of SEQ ID NO: 2), in the fourth and fifth beta sheets (S173 to R192 and R198 to T107 of SEQ ID NO: 2, respectively) and in the loop between the fourth and fifth beta sheets (F193 to Q197 of SEQ ID NO: 2) without affecting the ability of the CsgG monomer to form a transmembrane pore, which transmembrane pore is capable of translocating polypeptides. Therefore, it is envisaged that further mutations may be made in any of these regions in any CsgG monomer without affecting the ability of the monomer to form a pore that can translocate polynucleotides. It is also expected that mutations may be made in other regions, such as in any of the alpha helices (S63 to R76, G85 to A99 or V211 to L236 of SEQ ID NO: 2) or in any of the beta sheets (I121 to N133, K135 to R142, 1146 to R162, S173 to R192 or R198 to T107 of SEQ ID NO: 2) without affecting the ability of the monomer to form a pore that can translocate polynucleotides. It is also expected that deletions of one or more amino acids can be made in any of the loop regions linking the alpha helices and beta sheets and/or in the N-terminal and/or C-terminal regions of the CsgG monomer without affecting the ability of the monomer to form a pore that can translocate polynucleotides.

[0347] Amino acid substitutions may be made to the amino acid sequence of SEQ ID NO: 2 in addition to those discussed above, for example up to 1, 2, 3, 4, 5, 10, 20 or 30 substitutions. Conservative substitutions replace amino acids with other amino acids of similar chemical structure, similar chemical properties or similar side-chain volume. The amino acids introduced may have similar polarity, hydrophilicity, hydrophobicity, basicity, acidity, neutrality or charge to the amino acids they replace. Alternatively, the conservative substitution may introduce another amino acid that is aromatic or aliphatic in the place of a pre-existing aromatic or aliphatic amino acid. Conservative amino acid changes are well-known in the art and may be selected in accordance with the properties of the 20 main amino acids as defined in Table 2 below. Where amino acids have similar polarity, this can also be determined by reference to the hydropathy scale for amino acid side chains in Table 3.

TABLE-US-00001 TABLE 2 Chemical properties of amino acids Ala aliphatic, hydrophobic, Met hydrophobic, neutral neutral Cys polar, hydrophobic, Asn polar, hydrophilic, neutral neutral Asp polar, hydrophilic, Pro hydrophobic, neutral charged (−) Glu polar, hydrophilic, Gln polar, hydrophilic, charged (−) neutral Phe aromatic, hydrophobic, Arg polar, hydrophilic, neutral charged (+) Gly aliphatic, neutral Ser polar, hydrophilic, neutral His aromatic, polar, Thr polar, hydrophilic, hydrophilic, neutral charged (+) Ile aliphatic, hydrophobic, Val aliphatic, hydrophobic, neutral neutral Lys polar, hydrophilic, Trp aromatic, hydrophobic, charged(+) neutral Leu aliphatic, hydrophobic, Tyr aromatic, polar, neutral hydrophobic

TABLE-US-00002 TABLE 3 Hydropathy scale Side Chain Hydropathy Ile 4.5 Val 4.2 Leu 3.8 Phe 2.8 Cys 2.5 Met 1.9 Ala 1.8 Gly −0.4 Thr −0.7 Ser −0.8 Trp −0.9 Tyr −1.3 Pro −1.6 His −3.2 Glu −3.5 Gln −3.5 Asp −3.5 Asn −3.5 Lys −3.9 Arg −4.5

[0348] One or more amino acid residues of the amino acid sequence of SEQ ID NO: 2 may additionally be deleted from the polypeptides described above. Up to 1, 2, 3, 4, 5, 10, 20 or 30 or more residues may be deleted.

[0349] Variants may include fragments of SEQ ID NO: 2. Such fragments retain pore forming activity. Fragments may be at least 50, at least 100, at least 150, at least 200 or at least 250 amino acids in length. Such fragments may be used to produce the pores. A fragment preferably comprises the membrane spanning domain of SEQ ID NO: 2, namely K135-Q153 and S183-S208.

[0350] One or more amino acids may be alternatively or additionally added to the polypeptides described above. An extension may be provided at the amino terminal or carboxy terminal of the amino acid sequence of SEQ ID NO: 2 or polypeptide variant or fragment thereof. The extension may be quite short, for example from 1 to 10 amino acids in length. Alternatively, the extension may be longer, for example up to 50 or 100 amino acids. A carrier protein may be fused to an

amino acid sequence according to the invention. Other fusion proteins are discussed in more detail below.

[0351] As discussed above, a variant is a polypeptide that has an amino acid sequence which varies from that of SEQ ID NO: 2 and which retains its ability to form a pore. A variant typically contains the regions of SEQ ID NO: 2 that are responsible for pore formation. The pore forming ability of CsgG, which contains a β -barrel, is provided by β -sheets in each subunit. A variant of SEQ ID NO: 2 typically comprises the regions in SEQ ID NO: 2 that form β -sheets, namely K135-Q153 and S183-S208. One or more modifications can be made to the regions of SEQ ID NO: 2 that form β -sheets as long as the resulting variant retains its ability to form a pore. A variant of SEQ ID NO: 2 preferably includes one or more modifications, such as substitutions, additions or deletions, within its-helices and/or loop regions.

[0352] The monomers derived from CsgG may be modified to assist their identification of purification, for example by the addition of a streptavidin tag or by the addition of a signal sequence to promote their secretion from a cell where the monomer does not naturally contain such a sequence. Other suitable tags are discussed in more detail below. The monomer may be labelled with a revealing label. The revealing label may be any suitable label which allows the monomer to be detected. Suitable labels are described below.

[0353] The monomer derived from CsgG may also be produced using D-amino acids. For instance, the monomer derived from CsgG may comprise a mixture of L-amino acids and D-amino acids. This is conventional in the art for producing such proteins or peptides.

[0354] The monomer derived from CsgG contains one or more specific modifications to facilitate nucleotide discrimination. The monomer derived from CsgG may also contain other non-specific modifications as long as they do not interfere with pore formation. A number of non-specific side chain modifications are known in the art and may be made to the side chains of the monomer derived from CsgG. Such modifications include, for example, reductive alkylation of amino acids by reaction with an aldehyde followed by reduction with NaBH_4 , amidination with methylacetimidate or acylation with acetic anhydride.

[0355] The monomer derived from CsgG can be produced using standard methods known in the art. The monomer derived from CsgG may be made synthetically or by recombinant means. For example, the monomer may be synthesised by in vitro translation and transcription (IVTT). Suitable methods for producing pores and monomers are discussed in International Application Nos. PCT/GB09/001690 (published as WO 2010/004273), PCT/GB09/001679 (published as WO 2010/004265) or PCT/GB10/000133 (published as WO 2010/086603). Methods for inserting pores into membranes are discussed.

[0356] In some embodiments, the mutant monomer is chemically modified. The mutant monomer can be chemically modified in any way and at any site. The mutant monomer is preferably chemically modified by attachment of a molecule to one or more cysteines (cysteine linkage), attachment of a molecule to one or more lysines, attachment of a molecule to one or more non-natural amino acids, enzyme modification of an epitope or modification of a terminus. Suitable methods for carrying out such modifications are well-known in the art. The mutant monomer may be chemically modified by the attachment of any molecule. For instance, the mutant monomer may be chemically modified by attachment of a dye or a fluorophore.

[0357] In some embodiments, the mutant monomer is chemically modified with a molecular adaptor that facilitates the interaction between a pore comprising the monomer and a target nucleotide or target polynucleotide sequence. The presence of the adaptor improves the host-guest chemistry of the pore and the nucleotide or polynucleotide sequence and thereby improves the sequencing ability of pores formed from the mutant monomer. The principles of host-guest chemistry are well-known in the art. The adaptor has an effect on the physical or chemical properties of the pore that improves its interaction with the nucleotide or polynucleotide sequence. The adaptor may alter the charge of the barrel or channel of the pore or specifically interact with or

bind to the nucleotide or polynucleotide sequence thereby facilitating its interaction with the pore.
[0358] The molecular adaptor is preferably a cyclic molecule, a cyclodextrin, a species that is capable of hybridization, a DNA binder or interchelator, a peptide or peptide analogue, a synthetic polymer, an aromatic planar molecule, a small positively-charged molecule or a small molecule capable of hydrogen-bonding.

[0359] The adaptor may be cyclic. A cyclic adaptor preferably has the same symmetry as the pore. The adaptor preferably has eight-fold or nine-fold symmetry since CsgG typically has eight or nine subunits around a central axis. This is discussed in more detail below.

[0360] The adaptor typically interacts with the nucleotide or polynucleotide sequence via host-guest chemistry. The adaptor is typically capable of interacting with the nucleotide or polynucleotide sequence. The adaptor comprises one or more chemical groups that are capable of interacting with the nucleotide or polynucleotide sequence. The one or more chemical groups preferably interact with the nucleotide or polynucleotide sequence by non-covalent interactions, such as hydrophobic interactions, hydrogen bonding, Van der Waal's forces, π -cation interactions and/or electrostatic forces. The one or more chemical groups that are capable of interacting with the nucleotide or polynucleotide sequence are preferably positively charged. The one or more chemical groups that are capable of interacting with the nucleotide or polynucleotide sequence more preferably comprise amino groups. The amino groups can be attached to primary, secondary or tertiary carbon atoms. The adaptor even more preferably comprises a ring of amino groups, such as a ring of 6, 7 or 8 amino groups. The adaptor most preferably comprises a ring of eight amino groups. A ring of protonated amino groups may interact with negatively charged phosphate groups in the nucleotide or polynucleotide sequence.

[0361] The correct positioning of the adaptor within the pore can be facilitated by host-guest chemistry between the adaptor and the pore comprising the mutant monomer. The adaptor preferably comprises one or more chemical groups that are capable of interacting with one or more amino acids in the pore. The adaptor more preferably comprises one or more chemical groups that are capable of interacting with one or more amino acids in the pore via non-covalent interactions, such as hydrophobic interactions, hydrogen bonding, Van der Waal's forces, π -cation interactions and/or electrostatic forces. The chemical groups that are capable of interacting with one or more amino acids in the pore are typically hydroxyls or amines. The hydroxyl groups can be attached to primary, secondary or tertiary carbon atoms. The hydroxyl groups may form hydrogen bonds with uncharged amino acids in the pore. Any adaptor that facilitates the interaction between the pore and the nucleotide or polynucleotide sequence can be used.

[0362] Suitable adaptors include, but are not limited to, cyclodextrins, cyclic peptides and cucurbiturils. The adaptor is preferably a cyclodextrin or a derivative thereof. The cyclodextrin or derivative thereof may be any of those disclosed in Eliseev, A. V., and Schneider, H-J. (1994) *J. Am. Chem. Soc.* 116, 6081-6088. The adaptor is more preferably heptakis-6-amino- β -cyclodextrin (am.sub.7- β CD), 6-monodeoxy-6-monoamino- β -cyclodextrin (am.sub.i- β CD) or heptakis-(6-deoxy-6-guanidino)-cyclodextrin (gu.sub.7- β CD). The guanidino group in gu.sub.7- β CD has a much higher pKa than the primary amines in am.sub.7- β CD and so it is more positively charged. This gu- β CD adaptor may be used to increase the dwell time of the nucleotide in the pore, to increase the accuracy of the residual current measured, as well as to increase the base detection rate at high temperatures or low data acquisition rates.

[0363] If a succinimidyl 3-(2-pyridyldithio) propionate (SPDP) crosslinker is used as discussed in more detail below, the adaptor is preferably heptakis (6-deoxy-6-amino)-6-N-mono(2-pyridyl)dithiopropionyl- β -cyclodextrin (am.sub.6amPDP.sub.1- β CD).

[0364] More suitable adaptors include γ -cyclodextrins, which comprise 9 sugar units (and therefore have nine-fold symmetry). The γ -cyclodextrin may contain a linker molecule or may be modified to comprise all or more of the modified sugar units used in the β -cyclodextrin examples discussed above.

[0365] The molecular adaptor is preferably covalently attached to the mutant monomer. The adaptor can be covalently attached to the pore using any method known in the art. The adaptor is typically attached via chemical linkage. If the molecular adaptor is attached via cysteine linkage, the one or more cysteines have preferably been introduced to the mutant, for instance in the barrel, by substitution. The mutant monomer may be chemically modified by attachment of a molecular adaptor to one or more cysteines in the mutant monomer. The one or more cysteines may be naturally-occurring, i.e. at positions 1 and/or 215 in SEQ ID NO: 2. Alternatively, the mutant monomer may be chemically modified by attachment of a molecule to one or more cysteines introduced at other positions. The cysteine at position 215 may be removed, for instance by substitution, to ensure that the molecular adaptor does not attach to that position rather than the cysteine at position 1 or a cysteine introduced at another position. The reactivity of cysteine residues may be enhanced by modification of the adjacent residues. For instance, the basic groups of flanking arginine, histidine or lysine residues will change the pKa of the cysteines thiol group to that of the more reactive S group. The reactivity of cysteine residues may be protected by thiol protective groups such as dTNB. These may be reacted with one or more cysteine residues of the mutant monomer before a linker is attached. The molecule may be attached directly to the mutant monomer. The molecule is preferably attached to the mutant monomer using a linker, such as a chemical crosslinker or a peptide linker.

[0366] Suitable chemical crosslinkers are well-known in the art. Preferred crosslinkers include 2,5-dioxopyrrolidin-1-yl 3-(pyridin-2-yl)disulfanylpropanoate, 2,5-dioxopyrrolidin-1-yl 4-(pyridin-2-yl)disulfanylbutanoate and 2,5-dioxopyrrolidin-1-yl 8-(pyridin-2-yl)disulfanyloctanoate. The most preferred crosslinker is succinimidyl 3-(2-pyridyldithio)propionate (SPDP). Typically, the molecule is covalently attached to the bifunctional crosslinker before the molecule/crosslinker complex is covalently attached to the mutant monomer but it is also possible to covalently attach the bifunctional crosslinker to the monomer before the bifunctional crosslinker/monomer complex is attached to the molecule.

[0367] The linker is preferably resistant to dithiothreitol (DTT). Suitable linkers include, but are not limited to, iodoacetamide-based and Maleimide-based linkers.

[0368] In other embodiment, the monomer may be attached to a polynucleotide binding protein. This forms a modular sequencing system that may be used in the methods of sequencing of the invention. Polynucleotide binding proteins are discussed below.

[0369] The polynucleotide binding protein is preferably covalently attached to the mutant monomer. The protein can be covalently attached to the monomer using any method known in the art. The monomer and protein may be chemically fused or genetically fused. The monomer and protein are genetically fused if the whole construct is expressed from a single polynucleotide sequence. Genetic fusion of a monomer to a polynucleotide binding protein is discussed in International Application No. PCT/GB09/001679 (published as WO 2010/004265).

[0370] If the polynucleotide binding protein is attached via cysteine linkage, the one or more cysteines have preferably been introduced to the mutant by substitution. The one or more cysteines are preferably introduced into loop regions which have low conservation amongst homologues indicating that mutations or insertions may be tolerated. They are therefore suitable for attaching a polynucleotide binding protein. In such embodiments, the naturally-occurring cysteine at position 251 may be removed. The reactivity of cysteine residues may be enhanced by modification as described above.

[0371] The polynucleotide binding protein may be attached directly to the mutant monomer or via one or more linkers. The molecule may be attached to the mutant monomer using the hybridization linkers described in International Application No. PCT/GB10/000132 (published as WO 2010/086602). Alternatively, peptide linkers may be used. Peptide linkers are amino acid sequences. The length, flexibility and hydrophilicity of the peptide linker are typically designed such that it does not disturb the functions of the monomer and molecule. Preferred flexible

peptide linkers are stretches of 2 to 20, such as 4, 6, 8, 10 or 16, serine and/or glycine amino acids. More preferred flexible linkers include (SG).sub.1, (SG).sub.2, (SG).sub.3, (SG).sub.4, (SG).sub.5 and (SG).sub.8 wherein S is serine and G is glycine. Preferred rigid linkers are stretches of 2 to 30, such as 4, 6, 8, 16 or 24, proline amino acids. More preferred rigid linkers include (P) 12 wherein P is proline.

[0372] The mutant monomer may be chemically modified with a molecular adaptor and a polynucleotide binding protein.

[0373] The molecule (with which the monomer is chemically modified) may be attached directly to the monomer or attached via a linker as disclosed in International Application Nos.

PCT/GB09/001690 (published as WO 2010/004273), PCT/GB09/001679 (published as WO 2010/004265) or PCT/GB10/000133 (published as WO 2010/086603).

[0374] Any of the proteins described herein, such as the mutant monomers and pores of the invention, may be modified to assist their identification or purification, for example by the addition of histidine residues (a his tag), aspartic acid residues (an asp tag), a streptavidin tag, a flag tag, a SUMO tag, a GST tag or a MBP tag, or by the addition of a signal sequence to promote their secretion from a cell where the polypeptide does not naturally contain such a sequence. An alternative to introducing a genetic tag is to chemically react a tag onto a native or engineered position on the protein. An example of this would be to react a gel-shift reagent to a cysteine engineered on the outside of the protein. This has been demonstrated as a method for separating hemolysin hetero-oligomers (Chem Biol. 1997 July; 4 (7): 497-505).

[0375] Any of the proteins described herein, such as the mutant monomers and pores of the invention, may be labelled with a revealing label. The revealing label may be any suitable label which allows the protein to be detected. Suitable labels include, but are not limited to, fluorescent molecules, radioisotopes, e.g. ¹²⁵I, ³⁵S, enzymes, antibodies, antigens, polynucleotides and ligands such as biotin.

[0376] Any of the proteins described herein, such as the monomers or pores of the invention, may be made synthetically or by recombinant means. For example, the protein may be synthesised by in vitro translation and transcription (IVTT). The amino acid sequence of the protein may be modified to include non-naturally occurring amino acids or to increase the stability of the protein. When a protein is produced by synthetic means, such amino acids may be introduced during production. The protein may also be altered following either synthetic or recombinant production.

[0377] Proteins may also be produced using D-amino acids. For instance, the protein may comprise a mixture of L-amino acids and D-amino acids. This is conventional in the art for producing such proteins or peptides.

[0378] The protein may also contain other non-specific modifications as long as they do not interfere with the function of the protein. A number of non-specific side chain modifications are known in the art and may be made to the side chains of the protein(s). Such modifications include, for example, reductive alkylation of amino acids by reaction with an aldehyde followed by reduction with NaBH₄, amidination with methylacetimidate or acylation with acetic anhydride.

[0379] Any of the proteins described herein, including the monomers and pores of the invention, can be produced using standard methods known in the art. Polynucleotide sequences encoding a protein may be derived and replicated using standard methods in the art. Polynucleotide sequences encoding a protein may be expressed in a bacterial host cell using standard techniques in the art. The protein may be produced in a cell by in situ expression of the polypeptide from a recombinant expression vector. The expression vector optionally carries an inducible promoter to control the expression of the polypeptide. These methods are described in Sambrook, J. and Russell, D. (2001). Molecular Cloning: A Laboratory Manual, 3rd Edition. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY.

[0380] Proteins may be produced in large scale following purification by any protein liquid chromatography system from protein producing organisms or after recombinant expression. Typical

protein liquid chromatography systems include FPLC, AKTA systems, the Bio-Cad system, the Bio-Rad BioLogic system and the Gilson HPLC system.

Constructs

[0381] The invention also provides a construct comprising two or more covalently attached CsgG monomers, wherein at least one of the monomers is a mutant monomer of the invention. The construct of the invention retains its ability to form a pore. This may be determined as discussed above. One or more constructs of the invention may be used to form pores for characterising, such as sequencing, polynucleotides. The construct may comprise at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9 or at least 10 monomers. The construct preferably comprises two monomers. The two or more monomers may be the same or different.

[0382] At least one monomer in the construct is a mutant monomer of the invention. 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more or 10 or more monomers in the construct may be mutant monomers of the invention. All of the monomers in the construct are preferably mutant monomers of the invention. The mutant monomers may be the same or different. In a preferred embodiment, the construct comprises two mutant monomers of the invention.

[0383] The mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. The barrels of the mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. Length may be measured in number of amino acids and/or units of length.

[0384] The construct may comprise one or more monomers which are not mutant monomers of the invention. CsgG mutant monomers which are non mutant monomers of the invention include monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 in which none of the amino acids/positions discussed above have been mutated. At least one monomer in the construct may comprise SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of the sequence shown in SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41. A comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 is at least 50% homologous to SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over its entire sequence based on amino acid identity. More preferably, the comparative variant may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over the entire sequence.

[0385] The monomers in the construct are preferably genetically fused. Monomers are genetically fused if the whole construct is expressed from a single polynucleotide sequence. The coding sequences of the monomers may be combined in any way to form a single polynucleotide sequence encoding the construct.

[0386] The monomers may be genetically fused in any configuration. The monomers may be fused via their terminal amino acids. For instance, the amino terminus of the one monomer may be fused to the carboxy terminus of another monomer. The second and subsequent monomers in the construct (in the amino to carboxy direction) may comprise a methionine at their amino terminal ends (each of which is fused to the carboxy terminus of the previous monomer). For instance, if M is a monomer (without an amino terminal methionine) and mM is a monomer with an amino terminal methionine, the construct may comprise the sequence M-mM, M-mM-mM or M-mM-mM-mM. The presences of these methionines typically results from the expression of the start codons (i.e. ATGs) at the 5' end of the polynucleotides encoding the second or subsequent monomers within the polynucleotide encoding entire construct. The first monomer in the construct (in the amino to carboxy direction) may also comprise a methionine (e.g. mM-mM, mM-mM-mM or mM-mM-mM-mM).

[0387] The two or more monomers may be genetically fused directly together. The monomers are preferably genetically fused using a linker. The linker may be designed to constrain the mobility of the monomers. Preferred linkers are amino acid sequences (i.e. peptide linkers). Any of the peptide linkers discussed above may be used.

[0388] In another preferred embodiment, the monomers are chemically fused. Two monomers are chemically fused if the two parts are chemically attached, for instance via a chemical crosslinker. Any of the chemical crosslinkers discussed above may be used. The linker may be attached to one or more cysteine residues introduced into a mutant monomer of the invention. Alternatively, the linker may be attached to a terminus of one of the monomers in the construct.

[0389] If a construct contains different monomers, crosslinkage of monomers to themselves may be prevented by keeping the concentration of linker in a vast excess of the monomers. Alternatively, a “lock and key” arrangement may be used in which two linkers are used. Only one end of each linker may react together to form a longer linker and the other ends of the linker each react with a different monomers. Such linkers are described in International Application No.

PCT/GB10/000132 (published as WO 2010/086602).

Polynucleotides

[0390] The present invention also provides polynucleotide sequences which encode a mutant monomer of the invention. The mutant monomer may be any of those discussed above. The polynucleotide sequence preferably comprises a sequence at least 50%, 60%, 70%, 80%, 90% or 95% homologous based on nucleotide identity to the sequence of SEQ ID NO: 1 over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95% nucleotide identity over a stretch of 300 or more, for example 375, 450, 525 or 600 or more, contiguous nucleotides (“hard homology”). Homology may be calculated as described above. The polynucleotide sequence may comprise a sequence that differs from SEQ ID NO: 1 on the basis of the degeneracy of the genetic code.

[0391] The present invention also provides polynucleotide sequences which encode any of the genetically fused constructs of the invention. The polynucleotide preferably comprises two or more variants of the sequence shown in SEQ ID NO: 1. The polynucleotide sequence preferably comprises two or more sequences having at least 50%, 60%, 70%, 80%, 90% or 95% homology to SEQ ID NO: 1 based on nucleotide identity over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95% nucleotide identity over a stretch of 600 or more, for example 750, 900, 1050 or 1200 or more, contiguous nucleotides (“hard homology”). Homology may be calculated as described above.

[0392] Polynucleotide sequences may be derived and replicated using standard methods in the art. Chromosomal DNA encoding wild-type CsgG may be extracted from a pore producing organism, such as *Escherichia coli*. The gene encoding the pore subunit may be amplified using PCR involving specific primers. The amplified sequence may then undergo site-directed mutagenesis. Suitable methods of site-directed mutagenesis are known in the art and include, for example, combine chain reaction. Polynucleotides encoding a construct of the invention can be made using well-known techniques, such as those described in Sambrook, J. and Russell, D. (2001). Molecular Cloning: A Laboratory Manual, 3rd Edition. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY.

[0393] The resulting polynucleotide sequence may then be incorporated into a recombinant replicable vector such as a cloning vector. The vector may be used to replicate the polynucleotide in a compatible host cell. Thus polynucleotide sequences may be made by introducing a polynucleotide into a replicable vector, introducing the vector into a compatible host cell, and growing the host cell under conditions which bring about replication of the vector. The vector may be recovered from the host cell. Suitable host cells for cloning of polynucleotides are known in the art and described in more detail below.

[0394] The polynucleotide sequence may be cloned into suitable expression vector. In an

expression vector, the polynucleotide sequence is typically operably linked to a control sequence which is capable of providing for the expression of the coding sequence by the host cell. Such expression vectors can be used to express a pore subunit.

[0395] The term “operably linked” refers to a juxtaposition wherein the components described are in a relationship permitting them to function in their intended manner. A control sequence “operably linked” to a coding sequence is ligated in such a way that expression of the coding sequence is achieved under conditions compatible with the control sequences. Multiple copies of the same or different polynucleotide sequences may be introduced into the vector.

[0396] The expression vector may then be introduced into a suitable host cell. Thus, a mutant monomer or construct of the invention can be produced by inserting a polynucleotide sequence into an expression vector, introducing the vector into a compatible bacterial host cell, and growing the host cell under conditions which bring about expression of the polynucleotide sequence. The recombinantly-expressed monomer or construct may self-assemble into a pore in the host cell membrane. Alternatively, the recombinant pore produced in this manner may be removed from the host cell and inserted into another membrane. When producing pores comprising at least two different monomers or constructs, the different monomers or constructs may be expressed separately in different host cells as described above, removed from the host cells and assembled into a pore in a separate membrane, such as a rabbit cell membrane or a synthetic membrane.

[0397] The vectors may be for example, plasmid, virus or phage vectors provided with an origin of replication, optionally a promoter for the expression of the said polynucleotide sequence and optionally a regulator of the promoter. The vectors may contain one or more selectable marker genes, for example a tetracycline resistance gene. Promoters and other expression regulation signals may be selected to be compatible with the host cell for which the expression vector is designed. A T7, trc, lac, ara or λ .sub.L promoter is typically used.

[0398] The host cell typically expresses the monomer or construct at a high level. Host cells transformed with a polynucleotide sequence will be chosen to be compatible with the expression vector used to transform the cell. The host cell is typically bacterial and preferably *Escherichia coli*. Any cell with a λ DE3 lysogen, for example C41 (DE3), BL21 (DE3), JM109 (DE3), B834 (DE3), TUNER, Origami and Origami B, can express a vector comprising the T7 promoter. In addition to the conditions listed above any of the methods cited in Cao et al, 2014, PNAS, Structure of the nonameric bacterial amyloid secretion channel, doi-1411942111 and Goyal et al, 2014, Nature, 516, 250-253 structural and mechanistic insights into the bacterial amyloid secretion channel CsgG may be used to express the CsgG proteins.

[0399] The invention also comprises a method of producing a mutant monomer of the invention or a construct of the invention. The method comprises expressing a polynucleotide of the invention in a suitable host cell. The polynucleotide is preferably part of a vector and is preferably operably linked to a promoter.

Pores

[0400] The invention also provides various pores. The pores of the invention are ideal for characterising, such as sequencing, polynucleotide sequences because they can discriminate between different nucleotides with a high degree of sensitivity. The pores can surprisingly distinguish between the four nucleotides in DNA and RNA. The pores of the invention can even distinguish between methylated and unmethylated nucleotides. The base resolution of pores of the invention is surprisingly high. The pores show almost complete separation of all four DNA nucleotides. The pores further discriminate between deoxycytidine monophosphate (dCMP) and methyl-dCMP based on the dwell time in the pore and the current flowing through the pore.

[0401] The pores of the invention can also discriminate between different nucleotides under a range of conditions. In particular, the pores will discriminate between nucleotides under conditions that are favourable to the characterising, such as sequencing, of nucleic acids. The extent to which the pores of the invention can discriminate between different nucleotides can be controlled by altering

the applied potential, the salt concentration, the buffer, the temperature and the presence of additives, such as urea, betaine and DTT. This allows the function of the pores to be fine-tuned, particularly when sequencing. This is discussed in more detail below. The pores of the invention may also be used to identify polynucleotide polymers from the interaction with one or more monomers rather than on a nucleotide by nucleotide basis.

[0402] A pore of the invention may be isolated, substantially isolated, purified or substantially purified. A pore of the invention is isolated or purified if it is completely free of any other components, such as lipids or other pores. A pore is substantially isolated if it is mixed with carriers or diluents which will not interfere with its intended use. For instance, a pore is substantially isolated or substantially purified if it is present in a form that comprises less than 10%, less than 5%, less than 2% or less than 1% of other components, such as triblock copolymers, lipids or other pores. Alternatively, a pore of the invention may be present in a membrane. Suitable membranes are discussed below.

[0403] A pore of the invention may be present as an individual or single pore. Alternatively, a pore of the invention may be present in a homologous or heterologous population of two or more pores.

Homo-Oligomeric Pores

[0404] The invention also provides a homo-oligomeric pore derived from CsgG comprising identical mutant monomers of the invention. The homo-oligomeric pore may comprise any of the mutants of the invention. The homo-oligomeric pore of the invention is ideal for characterising, such as sequencing, polynucleotides. The homo-oligomeric pore of the invention may have any of the advantages discussed above.

[0405] The homo-oligomeric pore may contain any number of mutant monomers. The pore typically comprises at least 7, at least 8, at least 9 or at least 10 identical mutant monomers, such as 7, 8, 9 or 10 mutant monomers. The pore preferably comprises eight or nine identical mutant monomers. One or more, such as 2, 3, 4, 5, 6, 7, 8, 9 or 10, of the mutant monomers is preferably chemically modified as discussed above.

[0406] Methods for making pores are discussed in more detail below.

Hetero-Oligomeric Pores

[0407] The invention also provides a hetero-oligomeric pore derived from CsgG comprising at least one mutant monomer of the invention. The hetero-oligomeric pore of the invention is ideal for characterising, such as sequencing, polynucleotides. Hetero-oligomeric pores can be made using methods known in the art (e.g. Protein Sci. 2002 July; 11 (7): 1813-24).

[0408] The hetero-oligomeric pore contains sufficient monomers to form the pore. The monomers may be of any type. The pore typically comprises at least 7, at least 8, at least 9 or at least 10 monomers, such as 7, 8, 9 or 10 monomers. The pore preferably comprises eight or nine monomers.

[0409] In a preferred embodiment, all of the monomers (such as 10, 9, 8 or 7 of the monomers) are mutant monomers of the invention and at least one of them differs from the others. In a more preferred embodiment, the pore comprises eight or nine mutant monomers of the invention and at least one of them differs from the others. They may all differ from one another.

[0410] The mutant monomers of the invention in the pore are preferably approximately the same length or are the same length. The barrels of the mutant monomers of the invention in the pore are preferably approximately the same length or are the same length. Length may be measured in number of amino acids and/or units of length.

[0411] In another preferred embodiment, at least one of the mutant monomers is not a mutant monomer of the invention. In this embodiment, the remaining monomers are preferably mutant monomers of the invention. Hence, the pore may comprise 9, 8, 7, 6, 5, 4, 3, 2 or 1 mutant monomers of the invention. Any number of the monomers in the pore may not be a mutant monomer of the invention. The pore preferably comprises seven or eight mutant monomers of the invention and a monomer which is not a monomer of the invention. The mutant monomers of the

invention may be the same or different.

[0412] The mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. The barrels of the mutant monomers of the invention in the construct are preferably approximately the same length or are the same length. Length may be measured in number of amino acids and/or units of length.

[0413] The pore may comprise one or more monomers which are not mutant monomers of the invention. CsgG monomers which are not mutant monomers of the invention include monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 in which none of the amino acids/positions discussed above in relation to the invention have been mutated/substituted. A comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 is typically at least 50% homologous to SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over its entire sequence based on amino acid identity. More preferably, the comparative variant may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 over the entire sequence.

[0414] In all the embodiments discussed above, one or more, such as 2, 3, 4, 5, 6, 7, 8, 9 or 10, of the mutant monomers is preferably chemically modified as discussed above.

[0415] Methods for making pores are discussed in more detail below.

Construct-Containing Pores

[0416] The invention also provides a pore comprising at least one construct of the invention. A construct of the invention comprises two or more covalently attached monomers derived from CsgG wherein at least one of the monomers is a mutant monomer of the invention. In other words, a construct must contain more than one monomer. The pore contains sufficient constructs and, if necessary, monomers to form the pore. For instance, an octameric pore may comprise (a) four constructs each comprising two constructs, (b) two constructs each comprising four monomers or (b) one construct comprising two monomers and six monomers that do not form part of a construct. For instance, an nonameric pore may comprise (a) four constructs each comprising two constructs and one monomer that does not form part of a construct, (b) two constructs each comprising four monomers and a monomer that does not form part of a construct or (b) one construct comprising two monomers and seven monomers that do not form part of a construct. Other combinations of constructs and monomers can be envisaged by the skilled person.

[0417] At least two of the monomers in the pore are in the form of a construct of the invention. The construct, and hence the pore, comprises at least one mutant monomer of the invention. The pore typically comprises at least 7, at least 8, at least 9 or at least 10 monomers, such as 7, 8, 9 or 10 monomers, in total (at least two of which must be in a construct). The pore preferably comprises eight or nine monomers (at least two of which must be in a construct).

[0418] The construct containing pore may be a homo-oligomer (i.e. include identical constructs) or be a hetero-oligomer (i.e. where at least one construct differs from the others).

[0419] A pore typically contains (a) one construct comprising two monomers and (b) 5, 6, 7 or 8 monomers. The construct may be any of those discussed above. The monomers may be any of those discussed above, including mutant monomers of the invention, monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 and mutant monomers comprising a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 as discussed above.

[0420] Another typical pore comprises more than one construct of the invention, such as two, three or four constructs of the invention. If necessary, such pores further comprise sufficient additional monomers or constructs to form the pore. The additional monomer(s) may be any of those

discussed above, including mutant monomers of the invention, monomers comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 and mutant monomers comprising a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 as discussed above. The additional construct(s) may be any of those discussed above or may be a construct comprising two or more covalently attached CsgG monomers each comprising a monomer comprising SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 or a comparative variant of SEQ ID NO: 2, 3, 4, 5, 6, 7, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40 or 41 as discussed above.

[0421] A further pore of the invention comprises only constructs comprising 2 monomers, for example a pore may comprise 4, 5, 6, 7 or 8 constructs comprising 2 monomers. At least one construct is a construct of the invention, i.e. at least one monomer in the at least one construct, and preferably each monomer in the at least one construct, is a mutant monomer of the invention. All of the constructs comprising 2 monomers may be constructs of the invention.

[0422] A specific pore according to the invention comprises four constructs of the invention each comprising two monomers, wherein at least one monomer in each construct, and preferably each monomer in each construct, is a mutant monomer of the invention. The constructs may oligomerise into a pore with a structure such that only one monomer of each construct contributes to the channel of the pore. Typically the other monomers of the construct will be on the outside of the channel of the pore. For example, pores of the invention may comprise 7, 8, 9 or 10 constructs comprising 2 monomers where the channel comprises 7, 8, 9 or 10 monomers.

[0423] Mutations can be introduced into the construct as described above. The mutations may be alternating, i.e. the mutations are different for each monomer within a two monomer construct and the constructs are assembled as a homo-oligomer resulting in alternating modifications. In other words, monomers comprising MutA and MutB are fused and assembled to form an A-B: A-B: A-B: A-B pore. Alternatively, the mutations may be neighbouring, i.e. identical mutations are introduced into two monomers in a construct and this is then oligomerised with different mutant monomers or constructs. In other words, monomers comprising MutA are fused followed by oligomerisation with MutB-containing monomers to form A-A: B: B: B: B: B: B.

[0424] One or more of the monomers of the invention in a construct-containing pore may be chemically-modified as discussed above.

Analyte Characterisation

[0425] The invention provides a method of determining the presence, absence or one or more characteristics of a target analyte. The method involves contacting the target analyte with a pore of the invention such that the target analyte moves with respect to, such as through, the pore and taking one or more measurements as the analyte moves with respect to the pore and thereby determining the presence, absence or one or more characteristics of the analyte. The target analyte may also be called the template analyte or the analyte of interest.

[0426] Steps (a) and (b) are preferably carried out with a potential applied across the pore. As discussed in more detail below, the applied potential typically results in the formation of a complex between the pore and a polynucleotide binding protein. The applied potential may be a voltage potential. Alternatively, the applied potential may be a chemical potential. An example of this is using a salt gradient across an amphiphilic layer. A salt gradient is disclosed in Holden et al., J Am Chem Soc. 2007 Jul. 11; 129 (27): 8650-5.

[0427] The method is for determining the presence, absence or one or more characteristics of a target analyte. The method may be for determining the presence, absence or one or more characteristics of at least one analyte. The method may concern determining the presence, absence or one or more characteristics of two or more analytes. The method may comprise determining the presence, absence or one or more characteristics of any number of analytes, such as 2, 5, 10, 15, 20, 30, 40, 50, 100 or more analytes. Any number of characteristics of the one or more analytes may be determined, such as 1, 2, 3, 4, 5, 10 or more characteristics.

[0428] The target analyte is preferably a metal ion, an inorganic salt, a polymer, an amino acid, a peptide, a polypeptide, a protein, a nucleotide, an oligonucleotide, a polynucleotide, a dye, a bleach, a pharmaceutical, a diagnostic agent, a recreational drug, an explosive or an environmental pollutant. The method may concern determining the presence, absence or one or more characteristics of two or more analytes of the same type, such as two or more proteins, two or more nucleotides or two or more pharmaceuticals. Alternatively, the method may concern determining the presence, absence or one or more characteristics of two or more analytes of different types, such as one or more proteins, one or more nucleotides and one or more pharmaceuticals.

[0429] The target analyte can be secreted from cells. Alternatively, the target analyte can be an analyte that is present inside cells such that the analyte must be extracted from the cells before the invention can be carried out.

[0430] The analyte is preferably an amino acid, a peptide, a polypeptides and/or a protein. The amino acid, peptide, polypeptide or protein can be naturally-occurring or non-naturally-occurring. The polypeptide or protein can include within them synthetic or modified amino acids. A number of different types of modification to amino acids are known in the art. Suitable amino acids and modifications thereof are above. For the purposes of the invention, it is to be understood that the target analyte can be modified by any method available in the art.

[0431] The protein can be an enzyme, an antibody, a hormone, a growth factor or a growth regulatory protein, such as a cytokine. The cytokine may be selected from interleukins, preferably IFN-1, IL-1, IL-2, IL-4, IL-5, IL-6, IL-10, IL-12 and IL-13, interferons, preferably IL- γ , and other cytokines such as TNF- α . The protein may be a bacterial protein, a fungal protein, a virus protein or a parasite-derived protein.

[0432] The target analyte is preferably a nucleotide, an oligonucleotide or a polynucleotide. Nucleotides and polynucleotides are discussed below. Oligonucleotides are short nucleotide polymers which typically have 50 or fewer nucleotides, such 40 or fewer, 30 or fewer, 20 or fewer, 10 or fewer or 5 or fewer nucleotides. The oligonucleotides may comprise any of the nucleotides discussed below, including the abasic and modified nucleotides.

[0433] The target analyte, such as a target polynucleotide, may be present in any of the suitable samples discussed below.

[0434] The pore is typically present in a membrane as discussed below. The target analyte may be coupled or delivered to the membrane using of the methods discussed below.

[0435] Any of the measurements discussed below can be used to determine the presence, absence or one or more characteristics of the target analyte. The method preferably comprises contacting the target analyte with the pore such that the analyte moves with respect to, such as moves through, the pore and measuring the current passing through the pore as the analyte moves with respect to the pore and thereby determining the presence, absence or one or more characteristics of the analyte.

[0436] The target analyte is present if the current flows through the pore in a manner specific for the analyte (i.e. if a distinctive current associated with the analyte is detected flowing through the pore). The analyte is absent if the current does not flow through the pore in a manner specific for the nucleotide. Control experiments can be carried out in the presence of the analyte to determine the way in which it affects the current flowing through the pore.

[0437] The invention can be used to differentiate analytes of similar structure on the basis of the different effects they have on the current passing through a pore. Individual analytes can be identified at the single molecule level from their current amplitude when they interact with the pore. The invention can also be used to determine whether or not a particular analyte is present in a sample. The invention can also be used to measure the concentration of a particular analyte in a sample. Analyte characterisation using pores other than CsgG is known in the art.

Polynucleotide Characterisation

[0438] The invention provides a method of characterising a target polynucleotide, such as sequencing a polynucleotide. There are two main strategies for characterising or sequencing

polynucleotides using nanopores, namely strand characterisation/sequencing and exonuclease characterisation/sequencing. The method of the invention may concern either method.

[0439] In strand sequencing, the DNA is translocated through the nanopore either with or against an applied potential. Exonucleases that act progressively or processively on double stranded DNA can be used on the cis side of the pore to feed the remaining single strand through under an applied potential or the trans side under a reverse potential. Likewise, a helicase that unwinds the double stranded DNA can also be used in a similar manner. A polymerase may also be used. There are also possibilities for sequencing applications that require strand translocation against an applied potential, but the DNA must be first “caught” by the enzyme under a reverse or no potential. With the potential then switched back following binding the strand will pass cis to trans through the pore and be held in an extended conformation by the current flow. The single strand DNA exonucleases or single strand DNA dependent polymerases can act as molecular motors to pull the recently translocated single strand back through the pore in a controlled stepwise manner, trans to cis, against the applied potential.

[0440] In one embodiment, the method of characterising a target polynucleotide involves contacting the target sequence with a pore of the invention and a helicase enzyme. Any helicase may be used in the method. Suitable helicases are discussed below. Helicases may work in two modes with respect to the pore. First, the method is preferably carried out using a helicase such that it controls movement of the target sequence through the pore with the field resulting from the applied voltage. In this mode 5' end of the DNA is first captured in the pore, and the enzyme controls movement of the DNA into the pore such that the target sequence is passed through the pore with the field until it finally translocates through to the trans side of the bilayer. Alternatively, the method is preferably carried out such that a helicase enzyme controls movement of the target sequence through the pore against the field resulting from the applied voltage. In this mode 3' end of the DNA is first captured in the pore, and the enzyme controls movement of the DNA through the pore such that the target sequence is pulled out of the pore against the applied field until finally ejected back to the cis side of the bilayer.

[0441] In exonuclease sequencing, an exonuclease releases individual nucleotides from one end of the target polynucleotide and these individual nucleotides are identified as discussed below. In another embodiment, the method of characterising a target polynucleotide involves contacting the target sequence with a pore and an exonuclease enzyme. Any of the exonuclease enzymes discussed below may be used in the method. The enzyme may be covalently attached to the pore as discussed below.

[0442] Exonucleases are enzymes that typically latch onto one end of a polynucleotide and digest the sequence one nucleotide at a time from that end. The exonuclease can digest the polynucleotide in 5' to 3' direction or 3' to 5' direction. The end of the polynucleotide to which the exonuclease binds is typically determined through the choice of enzyme used and/or using methods known in the art. Hydroxyl groups or cap structures at either end of the polynucleotide may typically be used to prevent or facilitate the binding of the exonuclease to a particular end of the polynucleotide.

[0443] The method involves contacting the polynucleotide with the exonuclease so that the nucleotides are digested from the end of the polynucleotide at a rate that allows characterisation or identification of a proportion of nucleotides as discussed above. Methods for doing this are well known in the art. For example, Edman degradation is used to successively digest single amino acids from the end of polypeptide such that they may be identified using High Performance Liquid Chromatography (HPLC). A homologous method may be used in the present invention.

[0444] The rate at which the exonuclease functions is typically slower than the optimal rate of a wild-type exonuclease. A suitable rate of activity of the exonuclease in the method of the invention involves digestion of from 0.5 to 1000 nucleotides per second, from 0.6 to 500 nucleotides per second, 0.7 to 200 nucleotides per second, from 0.8 to 100 nucleotides per second, from 0.9 to 50 nucleotides per second or 1 to 20 or 10 nucleotides per second. The rate is preferably 1, 10, 100,

500 or 1000 nucleotides per second. A suitable rate of exonuclease activity can be achieved in various ways. For example, variant exonucleases with a reduced optimal rate of activity may be used in accordance with the invention.

[0445] In the strand characterisation embodiment, the method comprises contacting the polynucleotide with a pore of the invention such that the polynucleotide moves with respect to, such as through, the pore and taking one or more measurements as the polynucleotide moves with respect to the pore, wherein the measurements are indicative of one or more characteristics of the polynucleotide, and thereby characterising the target polynucleotide.

[0446] In the exonucleotide characterisation embodiment, the method comprises contacting the polynucleotide with a pore of the invention and an exonuclease such that the exonuclease digests individual nucleotides from one end of the target polynucleotide and the individual nucleotides move with respect to, such as through, the pore and taking one or more measurements as the individual nucleotides move with respect to the pore, wherein the measurements are indicative of one or more characteristics of the individual nucleotides, and thereby characterising the target polynucleotide.

[0447] An individual nucleotide is a single nucleotide. An individual nucleotide is one which is not bound to another nucleotide or polynucleotide by a nucleotide bond. A nucleotide bond involves one of the phosphate groups of a nucleotide being bound to the sugar group of another nucleotide. An individual nucleotide is typically one which is not bound by a nucleotide bond to another polynucleotide of at least 5, at least 10, at least 20, at least 50, at least 100, at least 200, at least 500, at least 1000 or at least 5000 nucleotides. For example, the individual nucleotide has been digested from a target polynucleotide sequence, such as a DNA or RNA strand. The nucleotide can be any of those discussed below.

[0448] The individual nucleotides may interact with the pore in any manner and at any site. The nucleotides preferably reversibly bind to the pore via or in conjunction with an adaptor as discussed above. The nucleotides most preferably reversibly bind to the pore via or in conjunction with the adaptor as they pass through the pore across the membrane. The nucleotides can also reversibly bind to the barrel or channel of the pore via or in conjunction with the adaptor as they pass through the pore across the membrane.

[0449] During the interaction between the individual nucleotide and the pore, the nucleotide typically affects the current flowing through the pore in a manner specific for that nucleotide. For example, a particular nucleotide will reduce the current flowing through the pore for a particular mean time period and to a particular extent. In other words, the current flowing through the pore is distinctive for a particular nucleotide. Control experiments may be carried out to determine the effect a particular nucleotide has on the current flowing through the pore. Results from carrying out the method of the invention on a test sample can then be compared with those derived from such a control experiment in order to identify a particular nucleotide in the sample or determine whether a particular nucleotide is present in the sample. The frequency at which the current flowing through the pore is affected in a manner indicative of a particular nucleotide can be used to determine the concentration of that nucleotide in the sample. The ratio of different nucleotides within a sample can also be calculated. For instance, the ratio of dCMP to methyl-dCMP can be calculated.

[0450] The method involves measuring one or more characteristics of the target polynucleotide. The target polynucleotide may also be called the template polynucleotide or the polynucleotide of interest.

[0451] This embodiment also uses a pore of the invention. Any of the pores and embodiments discussed above with reference to the target analyte may be used.

Polynucleotide

[0452] A polynucleotide, such as a nucleic acid, is a macromolecule comprising two or more nucleotides. The polynucleotide or nucleic acid may comprise any combination of any nucleotides. The nucleotides can be naturally occurring or artificial. One or more nucleotides in the

polynucleotide can be oxidized or methylated. One or more nucleotides in the polynucleotide may be damaged. For instance, the polynucleotide may comprise a pyrimidine dimer. Such dimers are typically associated with damage by ultraviolet light and are the primary cause of skin melanomas. One or more nucleotides in the polynucleotide may be modified, for instance with a label or a tag. Suitable labels are described below. The polynucleotide may comprise one or more spacers.

[0453] A nucleotide typically contains a nucleobase, a sugar and at least one phosphate group. The nucleobase and sugar form a nucleoside.

[0454] The nucleobase is typically heterocyclic. Nucleobases include, but are not limited to, purines and pyrimidines and more specifically adenine (A), guanine (G), thymine (T), uracil (U) and cytosine (C).

[0455] The sugar is typically a pentose sugar. Nucleotide sugars include, but are not limited to, ribose and deoxyribose. The sugar is preferably a deoxyribose.

[0456] The polynucleotide preferably comprises the following nucleosides: deoxyadenosine (dA), deoxyuridine (dU) and/or thymidine (dT), deoxyguanosine (dG) and deoxycytidine (dC).

[0457] The nucleotide is typically a ribonucleotide or deoxyribonucleotide. The nucleotide typically contains a monophosphate, diphosphate or triphosphate. The nucleotide may comprise more than three phosphates, such as 4 or 5 phosphates. Phosphates may be attached on the 5' or 3' side of a nucleotide. Nucleotides include, but are not limited to, adenosine monophosphate (AMP), guanosine monophosphate (GMP), thymidine monophosphate (TMP), uridine monophosphate (UMP), 5-methylcytidine monophosphate, 5-hydroxymethylcytidine monophosphate, cytidine monophosphate (CMP), cyclic adenosine monophosphate (cAMP), cyclic guanosine monophosphate (cGMP), deoxyadenosine monophosphate (dAMP), deoxyguanosine monophosphate (dGMP), deoxythymidine monophosphate (dTMP), deoxyuridine monophosphate (dUMP), deoxycytidine monophosphate (dCMP) and deoxymethylcytidine monophosphate. The nucleotides are preferably selected from AMP, TMP, GMP, CMP, UMP, dAMP, dTMP, dGMP, dCMP and dUMP.

[0458] A nucleotide may be abasic (i.e. lack a nucleobase). A nucleotide may also lack a nucleobase and a sugar (i.e. is a C3 spacer).

[0459] The nucleotides in the polynucleotide may be attached to each other in any manner. The nucleotides are typically attached by their sugar and phosphate groups as in nucleic acids. The nucleotides may be connected via their nucleobases as in pyrimidine dimers.

[0460] The polynucleotide may be single stranded or double stranded. The polynucleotide is preferably single stranded. Single stranded polynucleotide characterization is referred to as ID in the Examples. At least a portion of the polynucleotide may be double stranded.

[0461] The polynucleotide can be a nucleic acid, such as deoxyribonucleic acid (DNA) or ribonucleic acid (RNA). The polynucleotide can comprise one strand of RNA hybridised to one strand of DNA. The polynucleotide may be any synthetic nucleic acid known in the art, such as peptide nucleic acid (PNA), glycerol nucleic acid (GNA), threose nucleic acid (TNA), locked nucleic acid (LNA) or other synthetic polymers with nucleotide side chains. The PNA backbone is composed of repeating N-(2-aminoethyl)-glycine units linked by peptide bonds. The GNA backbone is composed of repeating glycol units linked by phosphodiester bonds. The TNA backbone is composed of repeating threose sugars linked together by phosphodiester bonds. LNA is formed from ribonucleotides as discussed above having an extra bridge connecting the 2' oxygen and 4' carbon in the ribose moiety. Bridged nucleic acids (BNAs) are modified RNA nucleotides. They may also be called constrained or inaccessible RNA. BNA monomers can contain a five-membered, six-membered or even a seven-membered bridged structure with a "fixed" C3'-endo sugar puckering. The bridge is synthetically incorporated at the 2', 4'-position of the ribose to produce a 2', 4'-BNA monomer.

[0462] The polynucleotide is most preferably ribonucleic acid (RNA) or deoxyribonucleic acid (DNA).

[0463] The polynucleotide can be any length. For example, the polynucleotide can be at least 10, at least 50, at least 100, at least 150, at least 200, at least 250, at least 300, at least 400 or at least 500 nucleotides or nucleotide pairs in length. The polynucleotide can be 1000 or more nucleotides or nucleotide pairs, 5000 or more nucleotides or nucleotide pairs in length or 100000 or more nucleotides or nucleotide pairs in length.

[0464] Any number of polynucleotides can be investigated. For instance, the method of the invention may concern characterising 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 50, 100 or more polynucleotides. If two or more polynucleotides are characterised, they may be different polynucleotides or two instances of the same polynucleotide.

[0465] The polynucleotide can be naturally occurring or artificial. For instance, the method may be used to verify the sequence of a manufactured oligonucleotide. The method is typically carried out in vitro.

Sample

[0466] The polynucleotide is typically present in any suitable sample. The invention is typically carried out on a sample that is known to contain or suspected to contain the polynucleotide. Alternatively, the invention may be carried out on a sample to confirm the identity of a polynucleotide whose presence in the sample is known or expected.

[0467] The sample may be a biological sample. The invention may be carried out in vitro using a sample obtained from or extracted from any organism or microorganism. The organism or microorganism is typically archaeal, prokaryotic or eukaryotic and typically belongs to one of the five kingdoms: plantae, animalia, fungi, monera and protista. The invention may be carried out in vitro on a sample obtained from or extracted from any virus. The sample is preferably a fluid sample. The sample typically comprises a body fluid of the patient. The sample may be urine, lymph, saliva, mucus or amniotic fluid but is preferably blood, plasma or serum.

[0468] Typically, the sample is human in origin, but alternatively it may be from another mammal animal such as from commercially farmed animals such as horses, cattle, sheep, fish, chickens or pigs or may alternatively be pets such as cats or dogs. Alternatively, the sample may be of plant origin, such as a sample obtained from a commercial crop, such as a cereal, legume, fruit or vegetable, for example wheat, barley, oats, canola, maize, soya, rice, rhubarb, bananas, apples, tomatoes, potatoes, grapes, tobacco, beans, lentils, sugar cane, cocoa, cotton.

[0469] The sample may be a non-biological sample. The non-biological sample is preferably a fluid sample. Examples of non-biological samples include surgical fluids, water such as drinking water, sea water or river water, and reagents for laboratory tests.

[0470] The sample is typically processed prior to being used in the invention, for example by centrifugation or by passage through a membrane that filters out unwanted molecules or cells, such as red blood cells. The sample may be measured immediately upon being taken. The sample may also be typically stored prior to assay, preferably below -70° C.

Characterisation

[0471] The method may involve measuring two, three, four or five or more characteristics of the polynucleotide. The one or more characteristics are preferably selected from (i) the length of the polynucleotide, (ii) the identity of the polynucleotide, (iii) the sequence of the polynucleotide, (iv) the secondary structure of the polynucleotide and (v) whether or not the polynucleotide is modified. Any combination of (i) to (v) may be measured in accordance with the invention, such as {i}, {ii}, {iii}, {iv}, {v}, {i,ii}, {i,iv}, {i,v}, {ii,iii}, {ii,iv}, {ii,v}, {iii,iv}, {iii,v}, {iv,v}, {i,ii,iii}, {i,ii,iv}, {i,ii,v}, {i,iii,iv}, {i,iii,v}, {i,iv,v}, {ii,iii,iv}, {ii,iii,v}, {ii,iv,v}, {iii,iv,v}, {i,ii,iii,iv}, {i,ii,iii,v}, {i,ii,iv,v}, {i,iii,iv,v}, {ii,iii,iv,v} or {i,ii,iii,iv,v}. Different combinations of (i) to (v) may be measured for the first polynucleotide compared with the second polynucleotide, including any of those combinations listed above.

[0472] For (i), the length of the polynucleotide may be measured for example by determining the number of interactions between the polynucleotide and the pore or the duration of interaction

between the polynucleotide and the pore.

[0473] For (ii), the identity of the polynucleotide may be measured in a number of ways. The identity of the polynucleotide may be measured in conjunction with measurement of the sequence of the polynucleotide or without measurement of the sequence of the polynucleotide. The former is straightforward; the polynucleotide is sequenced and thereby identified. The latter may be done in several ways. For instance, the presence of a particular motif in the polynucleotide may be measured (without measuring the remaining sequence of the polynucleotide). Alternatively, the measurement of a particular electrical and/or optical signal in the method may identify the polynucleotide as coming from a particular source.

[0474] For (iii), the sequence of the polynucleotide can be determined as described previously. Suitable sequencing methods, particularly those using electrical measurements, are described in Stoddart D et al., *Proc Natl Acad Sci*, 12; 106 (19): 7702-7, Lieberman K R et al, *J Am Chem Soc.* 2010,132 (50): 17961-72, and International Application WO 2000/28312.

[0475] For (iv), the secondary structure may be measured in a variety of ways. For instance, if the method involves an electrical measurement, the secondary structure may be measured using a change in dwell time or a change in current flowing through the pore. This allows regions of single-stranded and double-stranded polynucleotide to be distinguished.

[0476] For (v), the presence or absence of any modification may be measured. The method preferably comprises determining whether or not the polynucleotide is modified by methylation, by oxidation, by damage, with one or more proteins or with one or more labels, tags or spacers. Specific modifications will result in specific interactions with the pore which can be measured using the methods described below. For instance, methylcytosine may be distinguished from cytosine on the basis of the current flowing through the pore during its interaction with each nucleotide.

[0477] The target polynucleotide is contacted with a pore of the invention. The pore is typically present in a membrane. Suitable membranes are discussed below. The method may be carried out using any apparatus that is suitable for investigating a membrane/pore system in which a pore is present in a membrane. The method may be carried out using any apparatus that is suitable for transmembrane pore sensing. For example, the apparatus comprises a chamber comprising an aqueous solution and a barrier that separates the chamber into two sections. The barrier typically has an aperture in which the membrane containing the pore is formed. Alternatively the barrier forms the membrane in which the pore is present.

[0478] The method may be carried out using the apparatus described in International Application No. PCT/GB08/000562 (WO 2008/102120).

[0479] A variety of different types of measurements may be made. This includes without limitation: electrical measurements and optical measurements. Possible electrical measurements include: current measurements, impedance measurements, tunnelling measurements (Ivanov A P et al., *Nano Lett.* 2011 Jan. 12; 11 (1): 279-85), and FET measurements (International Application WO 2005/124888). Optical measurements may be combined with electrical measurements (Soni G V et al., *Rev Sci Instrum.* 2010 January; 81 (1): 014301). The measurement may be a transmembrane current measurement such as measurement of ionic current flowing through the pore.

[0480] Electrical measurements may be made using standard single channel recording equipment as describe in Stoddart D et al., *Proc Natl Acad Sci*, 12; 106 (19): 7702-7, Lieberman K R et al, *J Am Chem Soc.* 2010; 132 (50): 17961-72, and International Application WO 2000/28312. Alternatively, electrical measurements may be made using a multi-channel system, for example as described in International Application WO 2009/077734 and International Application WO 2011/067559.

[0481] The method is preferably carried out with a potential applied across the membrane. The applied potential may be a voltage potential. Alternatively, the applied potential may be a chemical potential. An example of this is using a salt gradient across a membrane, such as an amphiphilic

layer. A salt gradient is disclosed in Holden et al., J Am Chem Soc. 2007 Jul. 11; 129 (27): 8650-5. In some instances, the current passing through the pore as a polynucleotide moves with respect to the pore is used to estimate or determine the sequence of the polynucleotide. This is strand sequencing.

[0482] The method may involve measuring the current passing through the pore as the polynucleotide moves with respect to the pore. Therefore the apparatus used in the method may also comprise an electrical circuit capable of applying a potential and measuring an electrical signal across the membrane and pore. The methods may be carried out using a patch clamp or a voltage clamp. The methods preferably involve the use of a voltage clamp.

[0483] The method of the invention may involve the measuring of a current passing through the pore as the polynucleotide moves with respect to the pore. Suitable conditions for measuring ionic currents through transmembrane protein pores are known in the art and disclosed in the Example. The method is typically carried out with a voltage applied across the membrane and pore. The voltage used is typically from +5 V to -5 V, such as from +4 V to -4 V, +3 V to -3 V or +2 V to -2 V. The voltage used is typically from -600 mV to +600 mV or -400 mV to +400 mV. The voltage used is preferably in a range having a lower limit selected from -400 mV, -300 mV, -200 mV, -150 mV, -100 mV, -50 mV, -20 mV and 0 mV and an upper limit independently selected from +10 mV, +20 mV, +50 mV, +100 mV, +150 mV, +200 mV, +300 mV and +400 mV. The voltage used is more preferably in the range 100 mV to 240 mV and most preferably in the range of 120 mV to 220 mV. It is possible to increase discrimination between different nucleotides by a pore by using an increased applied potential.

[0484] The method is typically carried out in the presence of any charge carriers, such as metal salts, for example alkali metal salt, halide salts, for example chloride salts, such as alkali metal chloride salt. Charge carriers may include ionic liquids or organic salts, for example tetramethyl ammonium chloride, trimethylphenyl ammonium chloride, phenyltrimethyl ammonium chloride, or 1-ethyl-3-methyl imidazolium chloride. In the exemplary apparatus discussed above, the salt is present in the aqueous solution in the chamber. Potassium chloride (KCl), sodium chloride (NaCl), caesium chloride (CsCl) or a mixture of potassium ferrocyanide and potassium ferricyanide is typically used. KCl, NaCl and a mixture of potassium ferrocyanide and potassium ferricyanide are preferred. The charge carriers may be asymmetric across the membrane. For instance, the type and/or concentration of the charge carriers may be different on each side of the membrane.

[0485] The salt concentration may be at saturation. The salt concentration may be 3 M or lower and is typically from 0.1 to 2.5 M, from 0.3 to 1.9 M, from 0.5 to 1.8 M, from 0.7 to 1.7 M, from 0.9 to 1.6 M or from 1 M to 1.4 M. The salt concentration is preferably from 150 mM to 1 M. The method is preferably carried out using a salt concentration of at least 0.3 M, such as at least 0.4 M, at least 0.5 M, at least 0.6 M, at least 0.8 M, at least 1.0 M, at least 1.5 M, at least 2.0 M, at least 2.5 M or at least 3.0 M. High salt concentrations provide a high signal to noise ratio and allow for currents indicative of the presence of a nucleotide to be identified against the background of normal current fluctuations.

[0486] The method is typically carried out in the presence of a buffer. In the exemplary apparatus discussed above, the buffer is present in the aqueous solution in the chamber. Any buffer may be used in the method of the invention. Typically, the buffer is phosphate buffer. Other suitable buffers are HEPES and Tris-HCl buffer. The methods are typically carried out at a pH of from 4.0 to 12.0, from 4.5 to 10.0, from 5.0 to 9.0, from 5.5 to 8.8, from 6.0 to 8.7 or from 7.0 to 8.8 or 7.5 to 8.5. The pH used is preferably about 7.5.

[0487] The method may be carried out at from 0° C. to 100° C., from 15° C. to 95° C., from 16° C. to 90° C., from 17° C. to 85° C., from 18° C. to 80° C., 19° C. to 70° C., or from 20° C. to 60° C. The methods are typically carried out at room temperature. The methods are optionally carried out at a temperature that supports enzyme function, such as about 37° C.

Polynucleotide Binding Protein

[0488] The strand characterisation method preferably comprises contacting the polynucleotide with a polynucleotide binding protein such that the protein controls the movement of the polynucleotide with respect to, such as through, the pore.

[0489] More preferably, the method comprises (a) contacting the polynucleotide with a pore of the invention and a polynucleotide binding protein such that the protein controls the movement of the polynucleotide with respect to, such as through, the pore and (b) taking one or more measurements as the polynucleotide moves with respect to the pore, wherein the measurements are indicative of one or more characteristics of the polynucleotide, and thereby characterising the polynucleotide.

[0490] More preferably, the method comprises (a) contacting the polynucleotide with a pore of the invention and a polynucleotide binding protein such that the protein controls the movement of the polynucleotide with respect to, such as through, the pore and (b) measuring the current through the pore as the polynucleotide moves with respect to the pore, wherein the current is indicative of one or more characteristics of the polynucleotide, and thereby characterising the polynucleotide.

[0491] The polynucleotide binding protein may be any protein that is capable of binding to the polynucleotide and controlling its movement through the pore. It is straightforward in the art to determine whether or not a protein binds to a polynucleotide. The protein typically interacts with and modifies at least one property of the polynucleotide. The protein may modify the polynucleotide by cleaving it to form individual nucleotides or shorter chains of nucleotides, such as di- or trinucleotides. The protein may modify the polynucleotide by orienting it or moving it to a specific position, i.e. controlling its movement.

[0492] The polynucleotide binding protein is preferably derived from a polynucleotide handling enzyme. A polynucleotide handling enzyme is a polypeptide that is capable of interacting with and modifying at least one property of a polynucleotide. The enzyme may modify the polynucleotide by cleaving it to form individual nucleotides or shorter chains of nucleotides, such as di- or trinucleotides. The enzyme may modify the polynucleotide by orienting it or moving it to a specific position. The polynucleotide handling enzyme does not need to display enzymatic activity as long as it is capable of binding the polynucleotide and controlling its movement through the pore. For instance, the enzyme may be modified to remove its enzymatic activity or may be used under conditions which prevent it from acting as an enzyme. Such conditions are discussed in more detail below.

[0493] The polynucleotide handling enzyme is preferably derived from a nucleolytic enzyme. The polynucleotide handling enzyme used in the construct of the enzyme is more preferably derived from a member of any of the Enzyme Classification (EC) groups 3.1.11, 3.1.13, 3.1.14, 3.1.15, 3.1.16, 3.1.21, 3.1.22, 3.1.25, 3.1.26, 3.1.27, 3.1.30 and 3.1.31. The enzyme may be any of those disclosed in International Application No. PCT/GB10/000133 (published as WO 2010/086603).

[0494] Preferred enzymes are polymerases, exonucleases, helicases and topoisomerases, such as gyrases. Suitable enzymes include, but are not limited to, exonuclease I from *E. coli* (SEQ ID NO: 11), exonuclease III enzyme from *E. coli* (SEQ ID NO: 13), RecJ from *T. thermophilus* (SEQ ID NO: 15) and bacteriophage lambda exonuclease (SEQ ID NO: 17), TatD exonuclease and variants thereof. Three subunits comprising the sequence shown in SEQ ID NO: 15 or a variant thereof interact to form a trimer exonuclease. These exonucleases can also be used in the exonuclease method of the invention. The polymerase may be PyroPhage® 3173 DNA Polymerase (which is commercially available from Lucigen® Corporation), SD Polymerase (commercially available from Bioron®) or variants thereof. The enzyme is preferably Phi29 DNA polymerase (SEQ ID NO: 9) or a variant thereof. The topoisomerase is preferably a member of any of the Moiety Classification (EC) groups 5.99.1.2 and 5.99.1.3.

[0495] The enzyme is most preferably derived from a helicase, such as Hel308 Mbu (SEQ ID NO: 18), Hel308 Csy (SEQ ID NO: 19), Hel308 Tga (SEQ ID NO: 20), Hel308 Mhu (SEQ ID NO: 21), TraI Eco (SEQ ID NO: 22), XPD Mbu (SEQ ID NO: 23) or a variant thereof. Any helicase may be

used in the invention. The helicase may be or be derived from a Hel308 helicase, a RecD helicase, such as TraI helicase or a TrwC helicase, a XPD helicase or a Dda helicase. The helicase may be any of the helicases, modified helicases or helicase constructs disclosed in International Application Nos. PCT/GB2012/052579 (published as WO 2013/057495); PCT/GB2012/053274 (published as WO 2013/098562); PCT/GB2012/053273 (published as WO2013098561); PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736. [0496] The helicase preferably comprises the sequence shown in SEQ ID NO: 25 (TrwC Cba) or as variant thereof, the sequence shown in SEQ ID NO: 18 (Hel308 Mbu) or a variant thereof or the sequence shown in SEQ ID NO: 24 (Dda) or a variant thereof. Variants may differ from the native sequences in any of the ways discussed below for transmembrane pores. A preferred variant of SEQ ID NO: 24 comprises (a) E94C and A360C or (b) E94C, A360C, C109A and C136A and then optionally (Δ M1) GIG2 (i.e. deletion of M1 and then addition G1 and G2).

[0497] Any number of helicases may be used in accordance with the invention. For instance, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more helicases may be used. In some embodiments, different numbers of helicases may be used.

[0498] The method of the invention preferably comprises contacting the polynucleotide with two or more helicases. The two or more helicases are typically the same helicase. The two or more helicases may be different helicases.

[0499] The two or more helicases may be any combination of the helicases mentioned above. The two or more helicases may be two or more Dda helicases. The two or more helicases may be one or more Dda helicases and one or more TrwC helicases. The two or more helicases may be different variants of the same helicase.

[0500] The two or more helicases are preferably attached to one another. The two or more helicases are more preferably covalently attached to one another. The helicases may be attached in any order and using any method. Preferred helicase constructs for use in the invention are described in International Application Nos. PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

[0501] A variant of SEQ ID NOs: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25 is an enzyme that has an amino acid sequence which varies from that of SEQ ID NO: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25 and which retains polynucleotide binding ability. This can be measured using any method known in the art. For instance, the variant can be contacted with a polynucleotide and its ability to bind to and move along the polynucleotide can be measured. The variant may include modifications that facilitate binding of the polynucleotide and/or facilitate its activity at high salt concentrations and/or room temperature. Variants may be modified such that they bind polynucleotides (i.e. retain polynucleotide binding ability) but do not function as a helicase (i.e. do not move along polynucleotides when provided with all the necessary components to facilitate movement, e.g. ATP and Mg.sup.2+). Such modifications are known in the art. For instance, modification of the Mg.sup.2+ binding domain in helicases typically results in variants which do not function as helicases. These types of variants may act as molecular brakes (see below).

[0502] Over the entire length of the amino acid sequence of SEQ ID NO: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25, a variant will preferably be at least 50% homologous to that sequence based on amino acid identity. More preferably, the variant polypeptide may be at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% and more preferably at least 95%, 97% or 99% homologous based on amino acid identity to the amino acid sequence of SEQ ID NO: 9, 11, 13, 15, 17, 18, 19, 20, 21, 22, 23, 24 or 25 over the entire sequence. There may be at least 80%, for example at least 85%, 90% or 95%, amino acid identity over a stretch of 200 or more, for example 230, 250, 270, 280, 300, 400, 500, 600, 700, 800, 900 or 1000 or more, contiguous amino acids ("hard homology"). Homology is determined as described above.

The variant may differ from the wild-type sequence in any of the ways discussed above with reference to SEQ ID NO: 2 above. The enzyme may be covalently attached to the pore. Any method may be used to covalently attach the enzyme to the pore.

[0503] A preferred molecular brake is TrwC Cba-Q594A (SEQ ID NO: 25 with the mutation Q594A). This variant does not function as a helicase (i.e. binds polynucleotides but does not move along them when provided with all the necessary components to facilitate movement, e.g. ATP and Mg.sup.2+).

[0504] In strand sequencing, the polynucleotide is translocated through the pore either with or against an applied potential. Exonucleases that act progressively or processively on double stranded polynucleotides can be used on the cis side of the pore to feed the remaining single strand through under an applied potential or the trans side under a reverse potential. Likewise, a helicase that unwinds the double stranded DNA can also be used in a similar manner. A polymerase may also be used. There are also possibilities for sequencing applications that require strand translocation against an applied potential, but the DNA must be first “caught” by the enzyme under a reverse or no potential. With the potential then switched back following binding the strand will pass cis to trans through the pore and be held in an extended conformation by the current flow. The single strand DNA exonucleases or single strand DNA dependent polymerases can act as molecular motors to pull the recently translocated single strand back through the pore in a controlled stepwise manner, trans to cis, against the applied potential.

[0505] Any helicase may be used in the method. Helicases may work in two modes with respect to the pore. First, the method is preferably carried out using a helicase such that it moves the polynucleotide through the pore with the field resulting from the applied voltage. In this mode the 5' end of the polynucleotide is first captured in the pore, and the helicase moves the polynucleotide into the pore such that it is passed through the pore with the field until it finally translocates through to the trans side of the membrane. Alternatively, the method is preferably carried out such that a helicase moves the polynucleotide through the pore against the field resulting from the applied voltage. In this mode 3' end of the polynucleotide is first captured in the pore, and the helicase moves the polynucleotide through the pore such that it is pulled out of the pore against the applied field until finally ejected back to the cis side of the membrane.

[0506] The method may also be carried out in the opposite direction. The 3' end of the polynucleotide may be first captured in the pore and the helicase may move the polynucleotide into the pore such that it is passed through the pore with the field until it finally translocates through to the trans side of the membrane.

[0507] When the helicase is not provided with the necessary components to facilitate movement or is modified to hinder or prevent its movement, it can bind to the polynucleotide and act as a brake slowing the movement of the polynucleotide when it is pulled into the pore by the applied field. In the inactive mode, it does not matter whether the polynucleotide is captured either 3' or 5' down, it is the applied field which pulls the polynucleotide into the pore towards the trans side with the enzyme acting as a brake. When in the inactive mode, the movement control of the polynucleotide by the helicase can be described in a number of ways including ratcheting, sliding and braking. Helicase variants which lack helicase activity can also be used in this way.

[0508] The polynucleotide may be contacted with the polynucleotide binding protein and the pore in any order. It is preferred that, when the polynucleotide is contacted with the polynucleotide binding protein, such as a helicase, and the pore, the polynucleotide firstly forms a complex with the protein. When the voltage is applied across the pore, the polynucleotide/protein complex then forms a complex with the pore and controls the movement of the polynucleotide through the pore.

[0509] Any steps in the method using a polynucleotide binding protein are typically carried out in the presence of free nucleotides or free nucleotide analogues and an enzyme cofactor that facilitates the action of the polynucleotide binding protein. The free nucleotides may be one or more of any of the individual nucleotides discussed above. The free nucleotides include, but are not limited to,

adenosine monophosphate (AMP), adenosine diphosphate (ADP), adenosine triphosphate (ATP), guanosine monophosphate (GMP), guanosine diphosphate (GDP), guanosine triphosphate (GTP), thymidine monophosphate (TMP), thymidine diphosphate (TDP), thymidine triphosphate (TTP), uridine monophosphate (UMP), uridine diphosphate (UDP), uridine triphosphate (UTP), cytidine monophosphate (CMP), cytidine diphosphate (CDP), cytidine triphosphate (CTP), cyclic adenosine monophosphate (cAMP), cyclic guanosine monophosphate (cGMP), deoxyadenosine monophosphate (dAMP), deoxyadenosine diphosphate (dADP), deoxyadenosine triphosphate (dATP), deoxyguanosine monophosphate (dGMP), deoxyguanosine diphosphate (dGDP), deoxyguanosine triphosphate (dGTP), deoxythymidine monophosphate (dTMP), deoxythymidine diphosphate (dTDP), deoxythymidine triphosphate (dTTP), deoxyuridine monophosphate (dUMP), deoxyuridine diphosphate (dUDP), deoxyuridine triphosphate (dUTP), deoxycytidine monophosphate (dCMP), deoxycytidine diphosphate (dCDP) and deoxycytidine triphosphate (dCTP). The free nucleotides are preferably selected from AMP, TMP, GMP, CMP, UMP, dAMP, dTMP, dGMP or dCMP. The free nucleotides are preferably adenosine triphosphate (ATP). The enzyme cofactor is a factor that allows the construct to function. The enzyme cofactor is preferably a divalent metal cation. The divalent metal cation is preferably Mg.sup.2+, Mn.sup.2+, Ca.sup.2+ or Co.sup.2+. The enzyme cofactor is most preferably Mg.sup.2+.

Helicase(s) and Molecular Brake(s)

[0510] In a preferred embodiment, the method comprises: [0511] (a) providing the polynucleotide with one or more helicases and one or more molecular brakes attached to the polynucleotide; [0512] (b) contacting the polynucleotide with a pore of the invention and applying a potential across the pore such that the one or more helicases and the one or more molecular brakes are brought together and both control the movement of the polynucleotide with respect to, such as through, the pore;

[0513] (c) taking one or more measurements as the polynucleotide moves with respect to the pore wherein the measurements are indicative of one or more characteristics of the polynucleotide and thereby characterising the polynucleotide.

[0514] This type of method is discussed in detail in the International Application PCT/GB2014/052737

[0515] The one or more helicases may be any of those discussed above. The one or more molecular brakes may be any compound or molecule which binds to the polynucleotide and slows the movement of the polynucleotide through the pore. The one or more molecular brakes preferably comprise one or more compounds which bind to the polynucleotide. The one or more compounds are preferably one or more macrocycles. Suitable macrocycles include, but are not limited to, cyclodextrins, calixarenes, cyclic peptides, crown ethers, cucurbiturils, pillararenes, derivatives thereof or a combination thereof. The cyclodextrin or derivative thereof may be any of those disclosed in Eliseev, A. V., and Schneider, H-J. (1994) *J. Am. Chem. Soc.* 116, 6081-6088. The agent is more preferably heptakis-6-amino- β -cyclodextrin (am.sub.7- β CD), 6-monodeoxy-6-monoamino- β -cyclodextrin (am.sub.1- β CD) or heptakis-(6-deoxy-6-guanidino)-cyclodextrin (gu.sub.7- β CD).

[0516] The one or more molecular brakes are preferably one or more single stranded binding proteins (SSB). The one or more molecular brakes are more preferably a single-stranded binding protein (SSB) comprising a carboxy-terminal (C-terminal) region which does not have a net negative charge or (ii) a modified SSB comprising one or more modifications in its C-terminal region which decreases the net negative charge of the C-terminal region. The one or more molecular brakes are most preferably one of the SSBs disclosed in International Application No. PCT/GB2013/051924 (published as WO 2014/013259).

[0517] The one or more molecular brakes are preferably one or more polynucleotide binding proteins. The polynucleotide binding protein may be any protein that is capable of binding to the polynucleotide and controlling its movement through the pore. It is straightforward in the art to

determine whether or not a protein binds to a polynucleotide. The protein typically interacts with and modifies at least one property of the polynucleotide. The protein may modify the polynucleotide by cleaving it to form individual nucleotides or shorter chains of nucleotides, such as di- or trinucleotides. The moiety may modify the polynucleotide by orienting it or moving it to a specific position, i.e. controlling its movement.

[0518] The polynucleotide binding protein is preferably derived from a polynucleotide handling enzyme. The one or more molecular brakes may be derived from any of the polynucleotide handling enzymes discussed above. Modified versions of Phi29 polymerase (SEQ ID NO: 8) which act as molecular brakes are disclosed in U.S. Pat. No. 5,576,204. The one or more molecular brakes are preferably derived from a helicase.

[0519] Any number of molecular brakes derived from a helicase may be used. For instance, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more helicases may be used as molecular brakes. If two or more helicases are be used as molecular brakes, the two or more helicases are typically the same helicase. The two or more helicases may be different helicases.

[0520] The two or more helicases may be any combination of the helicases mentioned above. The two or more helicases may be two or more Dda helicases. The two or more helicases may be one or more Dda helicases and one or more TrwC helicases. The two or more helicases may be different variants of the same helicase.

[0521] The two or more helicases are preferably attached to one another. The two or more helicases are more preferably covalently attached to one another. The helicases may be attached in any order and using any method. The one or more molecular brakes derived from helicases are preferably modified to reduce the size of an opening in the polynucleotide binding domain through which in at least one conformational state the polynucleotide can unbind from the helicase. This is disclosed in WO 2014/013260.

[0522] Preferred helicase constructs for use in the invention are described in International Application Nos. PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

[0523] If the one or more helicases are used in the active mode (i.e. when the one or more helicases are provided with all the necessary components to facilitate movement, e.g. ATP and Mg.sup.2+), the one or more molecular brakes are preferably (a) used in an inactive mode (i.e. are used in the absence of the necessary components to facilitate movement or are incapable of active movement), (b) used in an active mode where the one or more molecular brakes move in the opposite direction to the one or more helicases or (c) used in an active mode where the one or more molecular brakes move in the same direction as the one or more helicases and more slowly than the one or more helicases.

[0524] If the one or more helicases are used in the inactive mode (i.e. when the one or more helicases are not provided with all the necessary components to facilitate movement, e.g. ATP and Mg.sup.2+ or are incapable of active movement), the one or more molecular brakes are preferably (a) used in an inactive mode (i.e. are used in the absence of the necessary components to facilitate movement or are incapable of active movement) or (b) used in an active mode where the one or more molecular brakes move along the polynucleotide in the same direction as the polynucleotide through the pore.

[0525] The one or more helicases and one or more molecular brakes may be attached to the polynucleotide at any positions so that they are brought together and both control the movement of the polynucleotide through the pore. The one or more helicases and one or more molecular brakes are at least one nucleotide apart, such as at least 5, at least 10, at least 50, at least 100, at least 500, at least 1000, at least 5000, at least 10,000, at least 50,000 nucleotides or more apart. If the method concerns characterising a double stranded polynucleotide provided with a Y adaptor at one end and a hairpin loop adaptor at the other end, the one or more helicases are preferably attached to the Y

adaptor and the one or more molecular brakes are preferably attached to the hairpin loop adaptor. In this embodiment, the one or more molecular brakes are preferably one or more helicases that are modified such that they bind the polynucleotide but do not function as a helicase. The one or more helicases attached to the Y adaptor are preferably stalled at a spacer as discussed in more detail below. The one or more molecular brakes attach to the hairpin loop adaptor are preferably not stalled at a spacer. The one or more helicases and the one or more molecular brakes are preferably brought together when the one or more helicases reach the hairpin loop. The one or more helicases may be attached to the Y adaptor before the Y adaptor is attached to the polynucleotide or after the Y adaptor is attached to the polynucleotide. The one or more molecular brakes may be attached to the hairpin loop adaptor before the hairpin loop adaptor is attached to the polynucleotide or after the hairpin loop adaptor is attached to the polynucleotide.

[0526] The one or more helicases and the one or more molecular brakes are preferably not attached to one another. The one or more helicases and the one or more molecular brakes are more preferably not covalently attached to one another. The one or more helicases and the one or more molecular brakes are preferably not attached as described in International Application Nos. PCT/GB2013/051925 (published as WO 2014/013260); PCT/GB2013/051924 (published as WO 2014/013259); PCT/GB2013/051928 (published as WO 2014/013262) and PCT/GB2014/052736.

Spacers

[0527] The one or more helicases may be stalled at one or more spacers as discussed in International Application No. PCT/GB2014/050175. Any configuration of one or more helicases and one or more spacers disclosed in the International Application may be used in this invention.

[0528] When a part of the polynucleotide enters the pore and moves through the pore along the field resulting from the applied potential, the one or more helicases are moved past the spacer by the pore as the polynucleotide moves through the pore. This is because the polynucleotide (including the one or more spacers) moves through the pore and the one or more helicases remain on top of the pore.

[0529] The one or more spacers are preferably part of the polynucleotide, for instance they interrupt(s) the polynucleotide sequence. The one or more spacers are preferably not part of one or more blocking molecules, such as speed bumps, hybridised to the polynucleotide.

[0530] There may be any number of spacers in the polynucleotide, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more spacers. There are preferably two, four or six spacers in the polynucleotide. There may be one or more spacers in different regions of the polynucleotide, such as one or more spacers in the Y adaptor and/or hairpin loop adaptor.

[0531] The one or more spacers each provides an energy barrier which the one or more helicases cannot overcome even in the active mode. The one or more spacers may stall the one or more helicases by reducing the traction of the helicase (for instance by removing the bases from the nucleotides in the polynucleotide) or physically blocking movement of the one or more helicases (for instance using a bulky chemical group).

[0532] The one or more spacers may comprise any molecule or combination of molecules that stalls the one or more helicases. The one or more spacers may comprise any molecule or combination of molecules that prevents the one or more helicases from moving along the polynucleotide. It is straightforward to determine whether or not the one or more helicases are stalled at one or more spacers in the absence of a transmembrane pore and an applied potential. For instance, the ability of a helicase to move past a spacer and displace a complementary strand of DNA can be measured by PAGE.

[0533] The one or more spacers typically comprise a linear molecule, such as a polymer. The one or more spacers typically have a different structure from the polynucleotide. For instance, if the polynucleotide is DNA, the one or more spacers are typically not DNA. In particular, if the polynucleotide is deoxyribonucleic acid (DNA) or ribonucleic acid (RNA), the one or more spacers preferably comprise peptide nucleic acid (PNA), glycerol nucleic acid (GNA), threose nucleic acid

(TNA), locked nucleic acid (LNA) or a synthetic polymer with nucleotide side chains. The one or more spacers may comprise one or more nucleotides in the opposite direction from the polynucleotide. For instance, the one or more spacers may comprise one or more nucleotides in 3' to 5' direction when the polynucleotide is in 5' to 3' direction. The nucleotides may be any of those discussed above.

[0534] The one or more spacers preferably comprises one or more nitroindoles, such as one of more 5-nitroindoles, one or more inosines, one or more acridines, one or more 2-aminopurines, one or more 2-6-diaminopurines, one or more 5-bromo-deoxyuridines, one or more inverted thymidines (inverted dTs), one or more inverted dideoxy-thymidines (ddTs), one or more dideoxy-cytidines (ddCs), one or more 5-methylcytidines, one or more 5-hydroxymethylcytidines, one or more 2'-O-Methyl RNA bases, one or more Iso-deoxycytidines (Iso-dCs), one or more Iso-deoxyguanosines (Iso-dGs), one or more iSpC3 groups (i.e. nucleotides which lack sugar and a base), one or more photo-cleavable (PC) groups, one or more hexandiol groups, one or more spacer 9 (iSp9) groups, one or more spacer 18 (iSp18) groups, a polymer or one or more thiol connections. The one or more spacers may comprise any combination of these groups. Many of these groups are commercially available from IDT® (Integrated DNA Technologies®).

[0535] The one or more spacers may contain any number of these groups. For instance, for 2-aminopurines, 2-6-diaminopurines, 5-bromo-deoxyuridines, inverted dTs, ddTs, ddCs, 5-methylcytidines, 5-hydroxymethylcytidines, 2'-O-Methyl RNA bases, Iso-dCs, Iso-dGs, iSpC3 groups, PC groups, hexandiol groups and thiol connections, the one or more spacers preferably comprise 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more. The one or more spacers preferably comprise 2, 3, 4, 5, 6, 7, 8 or more iSp9 groups. The one or more spacers preferably comprise 2, 3, 4, 5 or 6 or more iSp18 groups. The most preferred spacer is four iSp18 groups.

[0536] The polymer is preferably a polypeptide or a polyethylene glycol (PEG). The polypeptide preferably comprises 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more amino acids. The PEG preferably comprises 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more monomer units.

[0537] The one or more spacers preferably comprise one or more abasic nucleotides (i.e. nucleotides lacking a nucleobase), such as 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more abasic nucleotides. The nucleobase can be replaced by —H (idSp) or —OH in the abasic nucleotide. Abasic spacers can be inserted into polynucleotides by removing the nucleobases from one or more adjacent nucleotides. For instance, polynucleotides may be modified to include 3-methyladenine, 7-methylguanine, 1,N6-ethenoadenine inosine or hypoxanthine and the nucleobases may be removed from these nucleotides using Human Alkyladenine DNA Glycosylase (hAAG). Alternatively, polynucleotides may be modified to include uracil and the nucleobases removed with Uracil-DNA Glycosylase (UDG). In one embodiment, the one or more spacers do not comprise any abasic nucleotides.

[0538] The one or more helicases may be stalled by (i.e. before) or on each linear molecule spacers. If linear molecule spacers are used, the polynucleotide is preferably provided with a double stranded region of polynucleotide adjacent to the end of each spacer past which the one or more helicases are to be moved. The double stranded region typically helps to stall the one or more helicases on the adjacent spacer. The presence of the double stranded region(s) is particularly preferred if the method is carried out at a salt concentration of about 100 mM or lower. Each double stranded region is typically at least 10, such as at least 12, nucleotides in length. If the polynucleotide used in the invention is single stranded, a double stranded region may be formed by hybridising a shorter polynucleotide to a region adjacent to a spacer. The shorter polynucleotide is typically formed from the same nucleotides as the polynucleotide, but may be formed from different nucleotides. For instance, the shorter polynucleotide may be formed from LNA.

[0539] If linear molecule spacers are used, the polynucleotide is preferably provided with a blocking molecule at the end of each spacer opposite to the end past which the one or more helicases are to be moved. This can help to ensure that the one or more helicases remain stalled on

each spacer. It may also help retain the one or more helicases on the polynucleotide in the case that it/they diffuse(s) off in solution. The blocking molecule may be any of the chemical groups discussed below which physically cause the one or more helicases to stall. The blocking molecule may be a double stranded region of polynucleotide.

[0540] The one or more spacers preferably comprise one or more chemical groups which physically cause the one or more helicases to stall. The one or more chemical groups are preferably one or more pendant chemical groups. The one or more chemical groups may be attached to one or more nucleobases in the polynucleotide. The one or more chemical groups may be attached to the polynucleotide backbone. Any number of these chemical groups may be present, such as 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more. Suitable groups include, but are not limited to, fluorophores, streptavidin and/or biotin, cholesterol, methylene blue, dinitrophenols (DNPs), digoxigenin and/or anti-digoxigenin and dibenzylcyclooctyne groups.

[0541] Different spacers in the polynucleotide may comprise different stalling molecules. For instance, one spacer may comprise one of the linear molecules discussed above and another spacer may comprise one or more chemical groups which physically cause the one or more helicases to stall. A spacer may comprise any of the linear molecules discussed above and one or more chemical groups which physically cause the one or more helicases to stall, such as one or more abasics and a fluorophore.

[0542] Suitable spacers can be designed depending on the type of polynucleotide and the conditions under which the method of the invention is carried out. Most helicases bind and move along DNA and so may be stalled using anything that is not DNA. Suitable molecules are discussed above.

[0543] The method of the invention is preferably carried out in the presence of free nucleotides and/or the presence of a helicase cofactor. This is discussed in more detail below. In the absence of the transmembrane pore and an applied potential, the one or more spacers are preferably capable of stalling the one or more helicases in the presence of free nucleotides and/or the presence of a helicase cofactor.

[0544] If the method of the invention is carried out in the presence of free nucleotides and a helicase cofactor as discussed below (such that the one or more helicases are in the active mode), one or more longer spacers are typically used to ensure that the one or more helicases are stalled on the polynucleotide before they are contacted with the transmembrane pore and a potential is applied. One or more shorter spacers may be used in the absence of free nucleotides and a helicase cofactor (such that the one or more helicases are in the inactive mode).

[0545] The salt concentration also affects the ability of the one or more spacers to stall the one or more helicases. In the absence of the transmembrane pore and an applied potential, the one or more spacers are preferably capable of stalling the one or more helicases at a salt concentration of about 100 mM or lower. The higher the salt concentration used in the method of the invention, the shorter the one or more spacers that are typically used and vice versa.

[0546] Preferred combinations of features are shown in Table 4 below.

TABLE-US-00003 TABLE 4 Spacer Spacer length Salt Free Helicase Polynucleotide composition* (i.e. number of *) [] nucleotides? cofactor? DNA iSpC3 4 1M Yes Yes DNA iSp18 4 100-1000 mM Yes Yes DNA iSp18 6 <100-1000 mM Yes Yes DNA iSp18 2 1M Yes Yes DNA iSpC3 12 <100-1000 mM Yes Yes DNA iSpC3 20 <100-1000 mM Yes Yes DNA iSp9 6 100-1000 mM Yes Yes DNA idSp 4 1M Yes Yes

[0547] The method may concern moving two or more helicases past a spacer. In such instances, the length of the spacer is typically increased to prevent the trailing helicase from pushing the leading helicase past the spacer in the absence of the pore and applied potential. If the method concerns moving two or more helicases past one or more spacers, the spacer lengths discussed above may be increased at least 1.5 fold, such 2 fold, 2.5 fold or 3 fold. For instance, if the method concerns moving two or more helicases past one or more spacers, the spacer lengths in the third column of

Table 4 above may be increased 1.5 fold, 2 fold, 2.5 fold or 3 fold.

Membrane

[0548] The pore of the invention may be present in a membrane. In the methods of the invention, the polynucleotide is typically contacted with the pore of the invention in a membrane. Any membrane may be used in accordance with the invention. Suitable membranes are well-known in the art. The membrane is preferably an amphiphilic layer. An amphiphilic layer is a layer formed from amphiphilic molecules, such as phospholipids, which have both hydrophilic and lipophilic properties. The amphiphilic molecules may be synthetic or naturally occurring. Non-naturally occurring amphiphiles and amphiphiles which form a monolayer are known in the art and include, for example, block copolymers (Gonzalez-Perez et al., Langmuir, 2009, 25, 10447-10450). Block copolymers are polymeric materials in which two or more monomer sub-units that are polymerized together to create a single polymer chain, Block copolymers typically have properties that are contributed by each monomer sub-unit. However, a block copolymer may have unique properties that polymers formed from the individual sub-units do not possess. Block copolymers can be engineered such that one of the monomer sub-units is hydrophobic (i.e. lipophilic), whilst the other sub-unit(s) are hydrophilic whilst in aqueous media. In this case, the block copolymer may possess amphiphilic properties and may form a structure that mimics a biological membrane. The block copolymer may be a diblock (consisting of two monomer sub-units), but may also be constructed from more than two monomer sub-units to form more complex arrangements that behave as amphiphiles. The copolymer may be a triblock, tetrablock or pentablock copolymer. The membrane is preferably a triblock copolymer membrane.

[0549] Archaeobacterial bipolar tetraether lipids are naturally occurring lipids that are constructed such that the lipid forms a monolayer membrane. These lipids are generally found in extremophiles that survive in harsh biological environments, thermophiles, halophiles and acidophiles. Their stability is believed to derive from the fused nature of the final bilayer. It is straightforward to construct block copolymer materials that mimic these biological entities by creating a triblock polymer that has the general motif hydrophilic-hydrophobic-hydrophilic. This material may form monomeric membranes that behave similarly to lipid bilayers and encompass a range of phase behaviours from vesicles through to laminar membranes. Membranes formed from these triblock copolymers hold several advantages over biological lipid membranes. Because the triblock copolymer is synthesised, the exact construction can be carefully controlled to provide the correct chain lengths and properties required to form membranes and to interact with pores and other proteins.

[0550] Block copolymers may also be constructed from sub-units that are not classed as lipid sub-materials; for example a hydrophobic polymer may be made from siloxane or other non-hydrocarbon based monomers. The hydrophilic sub-section of block copolymer can also possess low protein binding properties, which allows the creation of a membrane that is highly resistant when exposed to raw biological samples. This head group unit may also be derived from non-classical lipid head-groups.

[0551] Triblock copolymer membranes also have increased mechanical and environmental stability compared with biological lipid membranes, for example a much higher operational temperature or pH range. The synthetic nature of the block copolymers provides a platform to customise polymer based membranes for a wide range of applications.

[0552] The membrane is most preferably one of the membranes disclosed in International Application No. PCT/GB2013/052766 or PCT/GB2013/052767.

[0553] The amphiphilic molecules may be chemically-modified or functionalised to facilitate coupling of the polynucleotide.

[0554] The amphiphilic layer may be a monolayer or a bilayer. The amphiphilic layer is typically planar. The amphiphilic layer may be curved. The amphiphilic layer may be supported.

[0555] Amphiphilic membranes are typically naturally mobile, essentially acting as two

dimensional fluids with lipid diffusion rates of approximately 10^{-8} cm s⁻¹. This means that the pore and coupled polynucleotide can typically move within an amphiphilic membrane.

[0556] The membrane may be a lipid bilayer. Lipid bilayers are models of cell membranes and serve as excellent platforms for a range of experimental studies. For example, lipid bilayers can be used for in vitro investigation of membrane proteins by single-channel recording. Alternatively, lipid bilayers can be used as biosensors to detect the presence of a range of substances. The lipid bilayer may be any lipid bilayer. Suitable lipid bilayers include, but are not limited to, a planar lipid bilayer, a supported bilayer or a liposome. The lipid bilayer is preferably a planar lipid bilayer. Suitable lipid bilayers are disclosed in International Application No. PCT/GB08/000563 (published as WO 2008/102121), International Application No. PCT/GB08/004127 (published as WO 2009/077734) and International Application No. PCT/GB2006/001057 (published as WO 2006/100484).

[0557] Methods for forming lipid bilayers are known in the art. Lipid bilayers are commonly formed by the method of Montal and Mueller (Proc. Natl. Acad. Sci. USA., 1972; 69:3561-3566), in which a lipid monolayer is carried on aqueous solution/air interface past either side of an aperture which is perpendicular to that interface. The lipid is normally added to the surface of an aqueous electrolyte solution by first dissolving it in an organic solvent and then allowing a drop of the solvent to evaporate on the surface of the aqueous solution on either side of the aperture. Once the organic solvent has evaporated, the solution/air interfaces on either side of the aperture are physically moved up and down past the aperture until a bilayer is formed. Planar lipid bilayers may be formed across an aperture in a membrane or across an opening into a recess.

[0558] The method of Montal & Mueller is popular because it is a cost-effective and relatively straightforward method of forming good quality lipid bilayers that are suitable for protein pore insertion. Other common methods of bilayer formation include tip-dipping, painting bilayers and patch-clamping of liposome bilayers.

[0559] Tip-dipping bilayer formation entails touching the aperture surface (for example, a pipette tip) onto the surface of a test solution that is carrying a monolayer of lipid. Again, the lipid monolayer is first generated at the solution/air interface by allowing a drop of lipid dissolved in organic solvent to evaporate at the solution surface. The bilayer is then formed by the *Langmuir-Schaefer* process and requires mechanical automation to move the aperture relative to the solution surface.

[0560] For painted bilayers, a drop of lipid dissolved in organic solvent is applied directly to the aperture, which is submerged in an aqueous test solution. The lipid solution is spread thinly over the aperture using a paintbrush or an equivalent. Thinning of the solvent results in formation of a lipid bilayer. However, complete removal of the solvent from the bilayer is difficult and consequently the bilayer formed by this method is less stable and more prone to noise during electrochemical measurement.

[0561] Patch-clamping is commonly used in the study of biological cell membranes. The cell membrane is clamped to the end of a pipette by suction and a patch of the membrane becomes attached over the aperture. The method has been adapted for producing lipid bilayers by clamping liposomes which then burst to leave a lipid bilayer sealing over the aperture of the pipette. The method requires stable, giant and unilamellar liposomes and the fabrication of small apertures in materials having a glass surface.

[0562] Liposomes can be formed by sonication, extrusion or the Mozafari method (Colas et al. (2007) Micron 38:841-847).

[0563] In a preferred embodiment, the lipid bilayer is formed as described in International Application No. PCT/GB08/004127 (published as WO 2009/077734). Advantageously in this method, the lipid bilayer is formed from dried lipids. In a most preferred embodiment, the lipid bilayer is formed across an opening as described in WO2009/077734 (PCT/GB08/004127).

[0564] A lipid bilayer is formed from two opposing layers of lipids. The two layers of lipids are

arranged such that their hydrophobic tail groups face towards each other to form a hydrophobic interior. The hydrophilic head groups of the lipids face outwards towards the aqueous environment on each side of the bilayer. The bilayer may be present in a number of lipid phases including, but not limited to, the liquid disordered phase (fluid lamellar), liquid ordered phase, solid ordered phase (lamellar gel phase, interdigitated gel phase) and planar bilayer crystals (lamellar sub-gel phase, lamellar crystalline phase).

[0565] Any lipid composition that forms a lipid bilayer may be used. The lipid composition is chosen such that a lipid bilayer having the required properties, such surface charge, ability to support membrane proteins, packing density or mechanical properties, is formed. The lipid composition can comprise one or more different lipids. For instance, the lipid composition can contain up to 100 lipids. The lipid composition preferably contains 1 to 10 lipids. The lipid composition may comprise naturally-occurring lipids and/or artificial lipids.

[0566] The lipids typically comprise a head group, an interfacial moiety and two hydrophobic tail groups which may be the same or different. Suitable head groups include, but are not limited to, neutral head groups, such as diacylglycerides (DG) and ceramides (CM); zwitterionic head groups, such as phosphatidylcholine (PC), phosphatidylethanolamine (PE) and sphingomyelin (SM); negatively charged head groups, such as phosphatidylglycerol (PG); phosphatidylserine (PS), phosphatidylinositol (PI), phosphatic acid (PA) and cardiolipin (CA); and positively charged headgroups, such as trimethylammonium-Propane (TAP). Suitable interfacial moieties include, but are not limited to, naturally-occurring interfacial moieties, such as glycerol-based or ceramide-based moieties. Suitable hydrophobic tail groups include, but are not limited to, saturated hydrocarbon chains, such as lauric acid (n-Dodecanoic acid), myristic acid (n-Tetradecanoic acid), palmitic acid (n-Hexadecanoic acid), stearic acid (n-Octadecanoic) and arachidic (n-Eicosanoic); unsaturated hydrocarbon chains, such as oleic acid (cis-9-Octadecanoic); and branched hydrocarbon chains, such as phytanoyl. The length of the chain and the position and number of the double bonds in the unsaturated hydrocarbon chains can vary. The length of the chains and the position and number of the branches, such as methyl groups, in the branched hydrocarbon chains can vary. The hydrophobic tail groups can be linked to the interfacial moiety as an ether or an ester. The lipids may be mycolic acid.

[0567] The lipids can also be chemically-modified. The head group or the tail group of the lipids may be chemically-modified. Suitable lipids whose head groups have been chemically-modified include, but are not limited to, PEG-modified lipids, such as 1,2-Diacyl-sn-Glycero-3-Phosphoethanolamine-N-[Methoxy (Polyethylene glycol)-2000]; functionalised PEG Lipids, such as 1,2-Distearoyl-sn-Glycero-3 Phosphoethanolamine-N-[Biotinyl(Polyethylene Glycol) 2000]; and lipids modified for conjugation, such as 1,2-Dioleoyl-sn-Glycero-3-Phosphoethanolamine-N-(succinyl) and 1,2-Dipalmitoyl-sn-Glycero-3-Phosphoethanolamine-N-(Biotinyl). Suitable lipids whose tail groups have been chemically-modified include, but are not limited to, polymerisable lipids, such as 1,2-bis(10,12-tricosadiynoyl)-sn-Glycero-3-Phosphocholine; fluorinated lipids, such as 1-Palmitoyl-2-(16-Fluoropalmitoyl)-sn-Glycero-3-Phosphocholine; deuterated lipids, such as 1,2-Dipalmitoyl-D62-sn-Glycero-3-Phosphocholine; and ether linked lipids, such as 1,2-Di-O-phytanoyl-sn-Glycero-3-Phosphocholine. The lipids may be chemically-modified or functionalised to facilitate coupling of the polynucleotide.

[0568] The amphiphilic layer, for example the lipid composition, typically comprises one or more additives that will affect the properties of the layer. Suitable additives include, but are not limited to, fatty acids, such as palmitic acid, myristic acid and oleic acid; fatty alcohols, such as palmitic alcohol, myristic alcohol and oleic alcohol; sterols, such as cholesterol, ergosterol, lanosterol, sitosterol and stigmasterol; lysophospholipids, such as 1-Acyl-2-Hydroxy-sn-Glycero-3-Phosphocholine; and ceramides.

[0569] In another preferred embodiment, the membrane comprises a solid state layer. Solid state layers can be formed from both organic and inorganic materials including, but not limited to,

microelectronic materials, insulating materials such as Si.sub.3N.sub.4, Al.sub.2O.sub.3, and SiO₂, organic and inorganic polymers such as polyamide, plastics such as Teflon® or elastomers such as two-component addition-cure silicone rubber, and glasses. The solid state layer may be formed from graphene. Suitable graphene layers are disclosed in International Application No. PCT/US2008/010637 (published as WO 2009/035647). If the membrane comprises a solid state layer, the pore is typically present in an amphiphilic membrane or layer contained within the solid state layer, for instance within a hole, well, gap, channel, trench or slit within the solid state layer. The skilled person can prepare suitable solid state/amphiphilic hybrid systems. Suitable systems are disclosed in WO 2009/020682 and WO 2012/005857. Any of the amphiphilic membranes or layers discussed above may be used.

[0570] The method is typically carried out using (i) an artificial amphiphilic layer comprising a pore, (ii) an isolated, naturally-occurring lipid bilayer comprising a pore, or (iii) a cell having a pore inserted therein. The method is typically carried out using an artificial amphiphilic layer, such as an artificial triblock copolymer layer. The layer may comprise other transmembrane and/or intramembrane proteins as well as other molecules in addition to the pore. Suitable apparatus and conditions are discussed below. The method of the invention is typically carried out in vitro.

Coupling

[0571] The polynucleotide is preferably coupled to the membrane comprising the pore. The method may comprise coupling the polynucleotide to the membrane comprising the pore. The polynucleotide is preferably coupled to the membrane using one or more anchors. The polynucleotide may be coupled to the membrane using any known method.

[0572] Each anchor comprises a group which couples (or binds) to the polynucleotide and a group which couples (or binds) to the membrane. Each anchor may covalently couple (or bind) to the polynucleotide and/or the membrane. If a Y adaptor and/or a hairpin loop adaptors are used, the polynucleotide is preferably coupled to the membrane using the adaptor(s).

[0573] The polynucleotide may be coupled to the membrane using any number of anchors, such as 2, 3, 4 or more anchors. For instance, a polynucleotide may be coupled to the membrane using two anchors each of which separately couples (or binds) to both the polynucleotide and membrane.

[0574] The one or more anchors may comprise the one or more helicases and/or the one or more molecular brakes.

[0575] If the membrane is an amphiphilic layer, such as a copolymer membrane or a lipid bilayer, the one or more anchors preferably comprise a polypeptide anchor present in the membrane and/or a hydrophobic anchor present in the membrane. The hydrophobic anchor is preferably a lipid, fatty acid, sterol, carbon nanotube, polypeptide, protein or amino acid, for example cholesterol, palmitate or tocopherol. In preferred embodiments, the one or more anchors are not the pore.

[0576] The components of the membrane, such as the amphiphilic molecules, copolymer or lipids, may be chemically-modified or functionalised to form the one or more anchors. Examples of suitable chemical modifications and suitable ways of functionalising the components of the membrane are discussed in more detail below. Any proportion of the membrane components may be functionalised, for example at least 0.01%, at least 0.1%, at least 1%, at least 10%, at least 25%, at least 50% or 100%.

[0577] The polynucleotide may be coupled directly to the membrane. The one or more anchors used to couple the polynucleotide to the membrane preferably comprise a linker. The one or more anchors may comprise one or more, such as 2, 3, 4 or more, linkers. One linker may be used couple more than one, such as 2, 3, 4 or more, polynucleotides to the membrane.

[0578] Preferred linkers include, but are not limited to, polymers, such as polynucleotides, polyethylene glycols (PEGs), polysaccharides and polypeptides. These linkers may be linear, branched or circular. For instance, the linker may be a circular polynucleotide. The polynucleotide may hybridise to a complementary sequence on the circular polynucleotide linker.

[0579] The one or more anchors or one or more linkers may comprise a component that can be cut

to broken down, such as a restriction site or a photolabile group.

[0580] Functionalised linkers and the ways in which they can couple molecules are known in the art. For instance, linkers functionalised with maleimide groups will react with and attach to cysteine residues in proteins. In the context of this invention, the protein may be present in the membrane or may be used to couple (or bind) to the polynucleotide. This is discussed in more detail below.

[0581] Crosslinkage of polynucleotides can be avoided using a “lock and key” arrangement. Only one end of each linker may react together to form a longer linker and the other ends of the linker each react with the polynucleotide or membrane respectively. Such linkers are described in International Application No. PCT/GB10/000132 (published as WO 2010/086602).

[0582] The use of a linker is preferred in the sequencing embodiments discussed below. If a polynucleotide is permanently coupled directly to the membrane in the sense that it does not uncouple when interacting with the pore (i.e. does not uncouple in step (b) or (e)), then some sequence data will be lost as the sequencing run cannot continue to the end of the polynucleotide due to the distance between the membrane and the pore. If a linker is used, then the polynucleotide can be processed to completion.

[0583] The coupling may be permanent or stable. In other words, the coupling may be such that the polynucleotide remains coupled to the membrane when interacting with the pore.

[0584] The coupling may be transient. In other words, the coupling may be such that the polynucleotide may decouple from the membrane when interacting with the pore.

[0585] For certain applications, such as aptamer detection, the transient nature of the coupling is preferred. If a permanent or stable linker is attached directly to either 5' or 3' end of a polynucleotide and the linker is shorter than the distance between the membrane and the transmembrane pore's channel, then some sequence data will be lost as the sequencing run cannot continue to the end of the polynucleotide. If the coupling is transient, then when the coupled end randomly becomes free of the membrane, then the polynucleotide can be processed to completion. Chemical groups that form permanent/stable or transient links are discussed in more detail below. The polynucleotide may be transiently coupled to an amphiphilic layer or triblock copolymer membrane using cholesterol or a fatty acyl chain. Any fatty acyl chain having a length of from 6 to 30 carbon atom, such as hexadecanoic acid, may be used.

[0586] In preferred embodiments, a polynucleotide, such as a nucleic acid, is coupled to an amphiphilic layer such as a triblock copolymer membrane or lipid bilayer. Coupling of nucleic acids to synthetic lipid bilayers has been carried out previously with various different tethering strategies. These are summarised in Table 5 below.

TABLE-US-00004 TABLE 5 Anchor Type of comprising coupling Reference Thiol Stable Yoshina-Ishii, C. and S. G. Boxer (2003). “Arrays of mobile tethered vesicles on supported lipid bilayers.” *J Am Chem Soc* 125(13): 3696-7. Biotin Stable Nikolov, V., R. Lipowsky, et al. (2007). “Behavior of giant vesicles with anchored DNA molecules.” *Biophys J* 92(12): 4356-68 Cholesterol Transient Pfeiffer, I. and F. Hook (2004). “Bivalent cholesterol-based coupling of oligonucleotides to lipid membrane assemblies.” *J Am Chem Soc* 126(33): 10224-5 Surfactant Stable van Lengerich, B., R. J. Rawle, et al. (e.g. Lipid, “Covalent attachment of lipid vesicles to a Palmitate, fluid-supported bilayer allows observation of etc) DNA-mediated vesicle interactions.” *Langmuir* 26(11): 8666-72

[0587] Synthetic polynucleotides and/or linkers may be functionalised using a modified phosphoramidite in the synthesis reaction, which is easily compatible for the direct addition of suitable anchoring groups, such as cholesterol, tocopherol, palmitate, thiol, lipid and biotin groups. These different attachment chemistries give a suite of options for attachment to polynucleotides. Each different modification group couples the polynucleotide in a slightly different way and coupling is not always permanent so giving different dwell times for the polynucleotide to the membrane. The advantages of transient coupling are discussed above.

[0588] Coupling of polynucleotides to a linker or to a functionalised membrane can also be

achieved by a number of other means provided that a complementary reactive group or an anchoring group can be added to the polynucleotide. The addition of reactive groups to either end of a polynucleotide has been reported previously. A thiol group can be added to the 5' of ssDNA or dsDNA using T4 polynucleotide kinase and ATP γ S (Grant, G. P. and P. Z. Qin (2007). "A facile method for attaching nitroxide spin labels at the 5' terminus of nucleic acids." *Nucleic Acids Res* 35 (10): e77). An azide group can be added to 5'-phosphate of ssDNA or dsDNA using T4 polynucleotide kinase and γ -[2-Azidoethyl]-ATP or γ -[6-Azidohexyl]-ATP. Using thiol or Click chemistry a tether, containing either a thiol, iodoacetamide OPSS or maleimide group (reactive to thiols) or a DIBO (dibenzocyclooctyne) or alkyne group (reactive to azides), can be covalently attached to the polynucleotide. A more diverse selection of chemical groups, such as biotin, thiols and fluorophores, can be added using terminal transferase to incorporate modified oligonucleotides to 3' of ssDNA (Kumar, A., P. Tchen, et al. (1988). "Nonradioactive labeling of synthetic oligonucleotide probes with terminal deoxynucleotidyl transferase." *Anal Biochem* 169 (2): 376-82). Streptavidin/biotin and/or streptavidin/desthiobiotin coupling may be used for any other polynucleotide. The Examples below describes how a polynucleotide can be coupled to a membrane using streptavidin/biotin and streptavidin/desthiobiotin. It may also be possible that anchors may be directly added to polynucleotides using terminal transferase with suitably modified nucleotides (e.g. cholesterol or palmitate).

[0589] The one or more anchors preferably couple the polynucleotide to the membrane via hybridisation. Hybridisation in the one or more anchors allows coupling in a transient manner as discussed above. The hybridisation may be present in any part of the one or more anchors, such as between the one or more anchors and the polynucleotide, within the one or more anchors or between the one or more anchors and the membrane. For instance, a linker may comprise two or more polynucleotides, such as 3, 4 or 5 polynucleotides, hybridised together. The one or more anchors may hybridise to the polynucleotide. The one or more anchors may hybridise directly to the polynucleotide or directly to a Y adaptor and/or leader sequence attached to the polynucleotide or directly to a hairpin loop adaptor attached to the polynucleotide (as discussed below). Alternatively, the one or more anchors may be hybridised to one or more, such as 2 or 3, intermediate polynucleotides (or "splints") which are hybridised to the polynucleotide, to a Y adaptor and/or leader sequence attached to the polynucleotide or to a hairpin loop adaptor attached to the polynucleotide (as discussed below).

[0590] The one or more anchors may comprise a single stranded or double stranded polynucleotide. One part of the anchor may be ligated to a single stranded or double stranded polynucleotide. Ligation of short pieces of ssDNA have been reported using T4 RNA ligase I (Troutt, A. B., M. G. McHeyzer-Williams, et al. (1992). "Ligation-anchored PCR: a simple amplification technique with single-sided specificity." *Proc Natl Acad Sci USA* 89 (20): 9823-5). Alternatively, either a single stranded or double stranded polynucleotide can be ligated to a double stranded polynucleotide and then the two strands separated by thermal or chemical denaturation. To a double stranded polynucleotide, it is possible to add either a piece of single stranded polynucleotide to one or both of the ends of the duplex, or a double stranded polynucleotide to one or both ends. For addition of single stranded polynucleotides to the a double stranded polynucleotide, this can be achieved using T4 RNA ligase I as for ligation to other regions of single stranded polynucleotides. For addition of double stranded polynucleotides to a double stranded polynucleotide then ligation can be "blunt-ended", with complementary 3' dA/dT tails on the polynucleotide and added polynucleotide respectively (as is routinely done for many sample prep applications to prevent concatemer or dimer formation) or using "sticky-ends" generated by restriction digestion of the polynucleotide and ligation of compatible adapters. Then, when the duplex is melted, each single strand will have either a 5' or 3' modification if a single stranded polynucleotide was used for ligation or a modification at the 5' end, 3' end or both if a double stranded polynucleotide was used for ligation.

[0591] If the polynucleotide is a synthetic strand, the one or more anchors can be incorporated

during the chemical synthesis of the polynucleotide. For instance, the polynucleotide can be synthesised using a primer having a reactive group attached to it.

[0592] Adenylated polynucleotides are intermediates in ligation reactions, where an adenosine-monophosphate is attached to 5'-phosphate of the polynucleotide. Various kits are available for generation of this intermediate, such as 5' DNA Adenylation Kit from NEB. By substituting ATP in the reaction for a modified nucleotide triphosphate, then addition of reactive groups (such as thiols, amines, biotin, azides, etc) to 5' of a polynucleotide can be possible. It may also be possible that anchors could be directly added to polynucleotides using a 5' DNA adenylation kit with suitably modified nucleotides (e.g. cholesterol or palmitate).

[0593] A common technique for the amplification of sections of genomic DNA is using polymerase chain reaction (PCR). Here, using two synthetic oligonucleotide primers, a number of copies of the same section of DNA can be generated, where for each copy the 5' of each strand in the duplex will be a synthetic polynucleotide. Single or multiple nucleotides can be added to 3' end of single or double stranded DNA by employing a polymerase. Examples of polymerases which could be used include, but are not limited to, Terminal Transferase, Klenow and *E. coli* Poly(A) polymerase). By substituting ATP in the reaction for a modified nucleotide triphosphate then anchors, such as a cholesterol, thiol, amine, azide, biotin or lipid, can be incorporated into double stranded polynucleotides. Therefore, each copy of the amplified polynucleotide will contain an anchor.

[0594] Ideally, the polynucleotide is coupled to the membrane without having to functionalise the polynucleotide. This can be achieved by coupling the one or more anchors, such as a polynucleotide binding protein or a chemical group, to the membrane and allowing the one or more anchors to interact with the polynucleotide or by functionalising the membrane. The one or more anchors may be coupled to the membrane by any of the methods described herein. In particular, the one or more anchors may comprise one or more linkers, such as maleimide functionalised linkers.

[0595] In this embodiment, the polynucleotide is typically RNA, DNA, PNA, TNA or LNA and may be double or single stranded. This embodiment is particularly suited to genomic DNA polynucleotides.

[0596] The one or more anchors can comprise any group that couples to, binds to or interacts with single or double stranded polynucleotides, specific nucleotide sequences within the polynucleotide or patterns of modified nucleotides within the polynucleotide, or any other ligand that is present on the polynucleotide.

[0597] Suitable binding proteins for use in anchors include, but are not limited to, *E. coli* single stranded binding protein, P5 single stranded binding protein, T4 gp32 single stranded binding protein, the TOPO V dsDNA binding region, human histone proteins, *E. coli* HU DNA binding protein and other archaeal, prokaryotic or eukaryotic single stranded or double stranded polynucleotide (or nucleic acid) binding proteins, including those listed below.

[0598] The specific nucleotide sequences could be sequences recognised by transcription factors, ribosomes, endonucleases, topoisomerases or replication initiation factors. The patterns of modified nucleotides could be patterns of methylation or damage.

[0599] The one or more anchors can comprise any group which couples to, binds to, intercalates with or interacts with a polynucleotide. The group may intercalate or interact with the polynucleotide via electrostatic, hydrogen bonding or Van der Waals interactions. Such groups include a lysine monomer, poly-lysine (which will interact with ssDNA or dsDNA), ethidium bromide (which will intercalate with dsDNA), universal bases or universal nucleotides (which can hybridise with any polynucleotide) and osmium complexes (which can react to methylated bases). A polynucleotide may therefore be coupled to the membrane using one or more universal nucleotides attached to the membrane. Each universal nucleotide may be coupled to the membrane using one or more linkers. The universal nucleotide preferably comprises one of the following nucleobases: hypoxanthine, 4-nitroindole, 5-nitroindole, 6-nitroindole, formylindole, 3-nitropyrrole, nitroimidazole, 4-nitropyrazole, 4-nitrobenzimidazole, 5-nitroindazole, 4-

aminobenzimidazole or phenyl (C6-aromatic ring). The universal nucleotide more preferably comprises one of the following nucleosides: 2'-deoxyinosine, inosine, 7-deaza-2'-deoxyinosine, 7-deaza-inosine, 2-aza-deoxyinosine, 2-aza-inosine, 2-O'-methylinosine, 4-nitroindole 2'-deoxyribonucleoside, 4-nitroindole ribonucleoside, 5-nitroindole 2'-deoxyribonucleoside, 5-nitroindole ribonucleoside, 6-nitroindole 2'-deoxyribonucleoside, 6-nitroindole ribonucleoside, 3-nitropyrrole 2'-deoxyribonucleoside, 3-nitropyrrole ribonucleoside, an acyclic sugar analogue of hypoxanthine, nitroimidazole 2'-deoxyribonucleoside, nitroimidazole ribonucleoside, 4-nitropyrazole 2'-deoxyribonucleoside, 4-nitropyrazole ribonucleoside, 4-nitrobenzimidazole 2'-deoxyribonucleoside, 4-nitrobenzimidazole ribonucleoside, 5-nitroindazole 2'-deoxyribonucleoside, 5-nitroindazole ribonucleoside, 4-aminobenzimidazole 2'-deoxyribonucleoside, 4-aminobenzimidazole ribonucleoside, phenyl C-ribonucleoside, phenyl C-2'-deoxyribosyl nucleoside, 2'-deoxynebularine, 2'-deoxyisoguanosine, K-2'-deoxyribose, P-2'-deoxyribose and pyrrolidine. The universal nucleotide more preferably comprises 2'-deoxyinosine. The universal nucleotide is more preferably IMP or dIMP. The universal nucleotide is most preferably dPMP (2'-Deoxy-P-nucleoside monophosphate) or dKMP (N6-methoxy-2, 6-diaminopurine monophosphate).

[0600] The one or more anchors may couple to (or bind to) the polynucleotide via Hoogsteen hydrogen bonds (where two nucleobases are held together by hydrogen bonds) or reversed Hoogsteen hydrogen bonds (where one nucleobase is rotated through 180° with respect to the other nucleobase). For instance, the one or more anchors may comprise one or more nucleotides, one or more oligonucleotides or one or more polynucleotides which form Hoogsteen hydrogen bonds or reversed Hoogsteen hydrogen bonds with the polynucleotide. These types of hydrogen bonds allow a third polynucleotide strand to wind around a double stranded helix and form a triplex. The one or more anchors may couple to (or bind to) a double stranded polynucleotide by forming a triplex with the double stranded duplex.

[0601] In this embodiment at least 1%, at least 10%, at least 25%, at least 50% or 100% of the membrane components may be functionalised.

[0602] Where the one or more anchors comprise a protein, they may be able to anchor directly into the membrane without further functionalisation, for example if it already has an external hydrophobic region which is compatible with the membrane. Examples of such proteins include, but are not limited to, transmembrane proteins, intramembrane proteins and membrane proteins. Alternatively the protein may be expressed with a genetically fused hydrophobic region which is compatible with the membrane. Such hydrophobic protein regions are known in the art.

[0603] The one or more anchors are preferably mixed with the polynucleotide before contacting with the membrane, but the one or more anchors may be contacted with the membrane and subsequently contacted with the polynucleotide.

[0604] In another aspect the polynucleotide may be functionalised, using methods described above, so that it can be recognised by a specific binding group. Specifically the polynucleotide may be functionalised with a ligand such as biotin (for binding to streptavidin), amylose (for binding to maltose binding protein or a fusion protein), Ni-NTA (for binding to poly-histidine or poly-histidine tagged proteins) or a peptides (such as an antigen).

[0605] According to a preferred embodiment, the one or more anchors may be used to couple a polynucleotide to the membrane when the polynucleotide is attached to a leader sequence which preferentially threads into the pore. Leader sequences are discussed in more detail below. Preferably, the polynucleotide is attached (such as ligated) to a leader sequence which preferentially threads into the pore. Such a leader sequence may comprise a homopolymeric polynucleotide or an abasic region. The leader sequence is typically designed to hybridise to the one or more anchors either directly or via one or more intermediate polynucleotides (or splints). In such instances, the one or more anchors typically comprise a polynucleotide sequence which is complementary to a sequence in the leader sequence or a sequence in the one or more intermediate

polynucleotides (or splints). In such instances, the one or more splints typically comprise a polynucleotide sequence which is complementary to a sequence in the leader sequence.

[0606] An example of a molecule used in chemical attachment is EDC (1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride). Reactive groups can also be added to the 5' of polynucleotides using commercially available kits (Thermo Pierce, Part No. 22980). Suitable methods include, but are not limited to, transient affinity attachment using histidine residues and Ni-NTA, as well as more robust covalent attachment by reactive cysteines, lysines or non natural amino acids.

Double Stranded Polynucleotide

[0607] The polynucleotide may be double stranded. If the polynucleotide is double stranded, the method preferably further comprises before the contacting step ligating a bridging moiety, such as a hairpin loop, to one end of the polynucleotide. The two strands of the polynucleotide may then be separated as or before the polynucleotide is contacted with the pore in accordance with the invention. The two strands may be separated as the polynucleotide movement through the pore is controlled by a polynucleotide binding protein, such as a helicase, or molecular brake.

[0608] Linking and interrogating both strands on a double stranded construct in this way increases the efficiency and accuracy of characterisation.

[0609] The bridging moiety is capable of linking the two strands of the target polynucleotide. The bridging moiety typically covalently links the two strands of the target polynucleotide. The bridging moiety can be anything that is capable of linking the two strands of the target polynucleotide, provided that the bridging moiety does not interfere with movement of the single stranded polynucleotide through the transmembrane pore.

[0610] The bridging moiety may be linked to the target polynucleotide by any suitable means known in the art. The bridging moiety may be synthesised separately and chemically attached or enzymatically ligated to the target polynucleotide. Alternatively, the bridging moiety may be generated in the processing of the target polynucleotide.

[0611] The bridging moiety is linked to the target polynucleotide at or near one end of the target polynucleotide. The bridging moiety is preferably linked to the target polynucleotide within 10 nucleotides of the end of the target polynucleotide.

[0612] Suitable bridging moieties include, but are not limited to a polymeric linker, a chemical linker, a polynucleotide or a polypeptide. Preferably, the bridging moiety comprises DNA, RNA, modified DNA (such as abasic DNA), RNA, PNA, LNA or PEG. The bridging moiety is more preferably DNA or RNA.

[0613] The bridging moiety is most preferably a hairpin loop or a hairpin loop adaptor. Suitable hairpin adaptors can be designed using methods known in the art. The hairpin loop may be any length. The hairpin loop is typically 110 or fewer nucleotides, such as 100 or fewer nucleotides, 90 or fewer nucleotides, 80 or fewer nucleotides, 70 or fewer nucleotides, 60 or fewer nucleotides, 50 or fewer nucleotides, 40 or fewer nucleotides, 30 or fewer nucleotides, 20 or fewer nucleotides or 10 or fewer nucleotides, in length. The hairpin loop is preferably from about 1 to 110, from 2 to 100, from 5 to 80 or from 6 to 50 nucleotides in length. Longer lengths of the hairpin loop, such as from 50 to 110 nucleotides, are preferred if the loop is involved in the differential selectability of the adaptor. Similarly, shorter lengths of the hairpin loop, such as from 1 to 5 nucleotides, are preferred if the loop is not involved in the selectable binding as discussed below.

[0614] The hairpin adaptor may be ligated to either end of the first and/or second polynucleotide, i.e. the 5' or 3' end. The hairpin adaptor may be ligated to the first and/or second polynucleotide using any method known in the art. The hairpin adaptor may be ligated using a ligase, such as T4 DNA ligase, *E. coli* DNA ligase, Taq DNA ligase, Tma DNA ligase and 9°N DNA ligase.

[0615] The two strands of the polynucleotide may be separated using any method known in the art. For instance, they may be separated by a polynucleotide binding protein or using conditions which favour dehybridisation (examples of conditions which favour dehybridisation include, but are not

limited to, high temperature, high pH and the addition of agents that can disrupt hydrogen bonding or base pairing, such as formamide and urea).

[0616] The hairpin adaptor preferably comprises a selectable binding moiety. This allows the first and/or second polynucleotide to be purified or isolated. A selectable binding moiety is a moiety that can be selected on the basis of its binding properties. Hence, a selectable binding moiety is preferably a moiety that specifically binds to a surface. A selectable binding moiety specifically binds to a surface if it binds to the surface to a much greater degree than any other moiety used in the invention. In preferred embodiments, the moiety binds to a surface to which no other moiety used in the invention binds.

[0617] Suitable selective binding moieties are known in the art. Preferred selective binding moieties include, but are not limited to, biotin, a polynucleotide sequence, antibodies, antibody fragments, such as Fab and ScSv, antigens, polynucleotide binding proteins, poly histidine tails and GST tags. The most preferred selective binding moieties are biotin and a selectable polynucleotide sequence. Biotin specifically binds to a surface coated with avidins. Selectable polynucleotide sequences specifically bind (i.e. hybridise) to a surface coated with homologous sequences. Alternatively, selectable polynucleotide sequences specifically bind to a surface coated with polynucleotide binding proteins.

[0618] The hairpin adaptor and/or the selectable binding moiety may comprise a region that can be cut, nicked, cleaved or hydrolysed. Such a region can be designed to allow the first and/or second polynucleotide to be removed from the surface to which it is bound following purification or isolation. Suitable regions are known in the art. Suitable regions include, but are not limited to, an RNA region, a region comprising desthiobiotin and streptavidin, a disulphide bond and a photocleavable region.

[0619] The double stranded target polynucleotide preferably comprises a leader sequence at the opposite end of the bridging moiety, such as a hairpin loop or hairpin loop adaptor. Leader sequences are discussed in more detail below.

Round the Corner Sequencing

[0620] In a preferred embodiment, a target double stranded polynucleotide is provided with a bridging moiety, such as a hairpin loop or hairpin loop adaptor, at one end and the method comprises contacting the polynucleotide with the pore such that both strands of the polynucleotide move through the pore and taking one or more measurements as the both strands of the polynucleotide move with respect to the pore wherein the measurements are indicative of one or more characteristics of the strands of the polynucleotide and thereby characterising the target double stranded polynucleotide. In another preferred embodiment, a target double stranded polynucleotide is provided with a bridging moiety, such as a hairpin loop or hairpin loop adaptor, at one end and the method comprises contacting the polynucleotide with the pore and exonuclease such that both strands of the polynucleotide are digested to form individual nucleotides. Any of the embodiments discussed above equally apply to this embodiment.

Leader Sequence

[0621] Before the contacting step in the strand characterisation/sequencing method, the method preferably comprises attaching to the polynucleotide a leader sequence which preferentially threads into the pore. The leader sequence facilitates the method of the invention. The leader sequence is designed to preferentially thread into the pore and thereby facilitate the movement of polynucleotide through the pore. The leader sequence can also be used to link the polynucleotide to the one or more anchors as discussed above.

[0622] The leader sequence typically comprises a polymer. The polymer is preferably negatively charged. The polymer is preferably a polynucleotide, such as DNA or RNA, a modified polynucleotide (such as abasic DNA), PNA, LNA, polyethylene glycol (PEG) or a polypeptide. The leader preferably comprises a polynucleotide and more preferably comprises a single stranded polynucleotide. The leader sequence can comprise any of the polynucleotides discussed above. The

single stranded leader sequence most preferably comprises a single strand of DNA, such as a poly dT section. The leader sequence preferably comprises the one or more spacers.

[0623] The leader sequence can be any length, but is typically 10 to 150 nucleotides in length, such as from 20 to 150 nucleotides in length. The length of the leader typically depends on the transmembrane pore used in the method.

[0624] The leader sequence is preferably part of a Y adaptor as defined below.

Double Coupling

[0625] The method of the invention may involve double coupling of a double stranded polynucleotide. In a preferred embodiment, the method of the invention comprises: [0626] (a) providing the double stranded polynucleotide with a Y adaptor at one end and a bridging moiety adaptor, such as a hairpin loop adaptor, at the other end, wherein the Y adaptor comprises one or more first anchors for coupling the polynucleotide to the membrane, wherein the bridging moiety adaptor comprises one or more second anchors for coupling the polynucleotide to the membrane and wherein the strength of coupling of the bridging moiety adaptor to the membrane is greater than the strength of coupling of the Y adaptor to the membrane; [0627] (b) contacting the polynucleotide provided in step (a) with a pore the invention such that the polynucleotide moves with respect to, such as through, the pore; and [0628] (c) taking one or more measurements as the polynucleotide moves with respect to the pore, wherein the measurements are indicative of one or more characteristics of the polynucleotide, and thereby characterising the target polynucleotide.

[0629] This type of method is discussed in detail in the UK Application No. 1406147.7.

[0630] The double stranded polynucleotide is provided with a Y adaptor at one end and a bridging moiety adaptor at the other end. The Y adaptor and/or the bridging moiety adaptor are typically polynucleotide adaptors. They may be formed from any of the polynucleotides discussed above.

[0631] The Y adaptor typically comprises (a) a double stranded region and (b) a single stranded region or a region that is not complementary at the other end. The Y adaptor may be described as having an overhang if it comprises a single stranded region. The presence of a non-complementary region in the Y adaptor gives the adaptor its Y shape since the two strands typically do not hybridise to each other unlike the double stranded portion. The Y adaptor comprises the one or more first anchors. Anchors are discussed in more detail above.

[0632] The Y adaptor preferably comprises a leader sequence which preferentially threads into the pore. This is discussed above.

[0633] The bridging moiety adaptor preferably comprises a selectable binding moiety as discussed above. The bridging moiety adaptor and/or the selectable binding moiety may comprise a region that can be cut, nicked, cleaved or hydrolysed as discussed above.

[0634] If one or more helicases and one or more molecular brakes are used as discussed above, the Y adaptor preferably comprises the one or more helicases and the bridging moiety adaptor preferably comprises the one or more molecular brakes.

[0635] The Y adaptor and/or the bridging moiety adaptor may be ligated to the polynucleotide using any method known in the art. One or both of the adaptors may be ligated using a ligase, such as T4 DNA ligase, *E. coli* DNA ligase, Taq DNA ligase, Tma DNA ligase and 9°N DNA ligase. Alternatively, the adaptors may be added to the polynucleotide using the methods of the invention discussed below.

[0636] In a preferred embodiment, step a) of the method comprises modifying the double stranded polynucleotide so that it comprises the Y adaptor at one end and the bridging moiety adaptor at the other end. Any manner of modification can be used. The method preferably comprises modifying the double stranded polynucleotide in accordance with the invention. This is discussed in more detail below. The methods of modification and characterisation may be combined in any way.

[0637] The strength of coupling (or binding) of the bridging moiety adaptor to the membrane is greater than the strength of coupling (or binding) of the Y adaptor to the membrane. This can be measured in any way. A suitable method for measuring the strength of coupling (or binding) is

disclosed in the Examples of the UK Application No. 1406147.7.

[0638] The strength of coupling (or binding) of the bridging moiety adaptor is preferably at least 1.5 times the strength of coupling (or binding) of the Y adaptor, such as at least twice, at least three times, at least four times, at least five or at least ten times the strength of coupling (or binding) of the anchor adaptor. The affinity constant (K_d) of the bridging moiety adaptor for the membrane is preferably at least 1.5 times the affinity constant of the Y adaptor, such as at least twice, at least three times, at least four times, at least five or at least ten times the strength of coupling of the Y adaptor.

[0639] There are several ways in which the bridging moiety adaptor couples (or binds) more strongly to the membrane than the Y adaptor. For instance, the bridging moiety adaptor may comprise more anchors than the Y adaptor. For instance, the bridging moiety adaptor may comprise 2, 3 or more second anchors whereas the Y adaptor may comprise one first anchor.

[0640] The strength of coupling (or binding) of the one or more second anchors to the membrane may be greater than the strength of coupling (or binding) of the one or more first anchors to the membrane. The strength of coupling (or binding) of the one or more second anchors to the bridging moiety adaptor may be greater than the strength of coupling (or binding) of the one or more first anchors to the Y adaptor. The one or more first anchors and the one or more second anchors may be attached to their respective adaptors via hybridisation and the strength of hybridisation is greater in the one or more second anchors than in the one or more first anchors. Any combination of these embodiments may also be used in the invention. Strength of coupling (or binding) may be measure using known techniques in the art.

[0641] The one or more second anchors preferably comprise one or more groups which couple(s) (or bind(s)) to the membrane with a greater strength than the one or more groups in the one or more first anchors which couple(s) (or bind(s)) to the membrane. In preferred embodiments, the bridging moiety adaptor/one or more second anchors couple (or bind) to the membrane using cholesterol and the Y adaptor/one or more first anchors couple (or bind) to the membrane using palmitate. Cholesterol binds to triblock copolymer membranes and lipid membranes more strongly than palmitate. In an alternative embodiment, the bridging moiety adaptor/one or more second anchors couple (or bind) to the membrane using a mono-acyl species, such as palmitate, and the Y adaptor/one or more first anchors couple (or bind) to the membrane using a diacyl species, such as dipalmitoylphosphatidylcholine.

Adding Hairpin Loops and Leader Sequences

[0642] Before provision, a double stranded polynucleotide may be contacted with a MuA transposase and a population of double stranded MuA substrates, wherein a proportion of the substrates in the population are Y adaptors comprising the leader sequence and wherein a proportion of the substrates in the population are hairpin loop adaptors. The transposase fragments the double stranded polynucleotide analyte and ligates MuA substrates to one or both ends of the fragments. This produces a plurality of modified double stranded polynucleotides comprising the leader sequence at one end and the hairpin loop at the other. The modified double stranded polynucleotides may then be investigated using the method of the invention.

[0643] Each substrate in the population preferably comprises at least one overhang of universal nucleotides such that the transposase fragments the template polynucleotide and ligates a substrate to one or both ends of the double stranded fragments and thereby produces a plurality of fragment/substrate constructs and wherein the method further comprises ligating the overhangs to the fragments in the constructs and thereby producing a plurality of modified double stranded polynucleotides. Suitable universal nucleotides are discussed above. The overhang is preferably five nucleotides in length.

[0644] Alternatively, each substrate in population preferably comprises (i) at least one overhang and (ii) at least one nucleotide in the same strand as the at least one overhang which comprises a nucleoside that is not present in the template polynucleotide such that the transposase fragments the

template polynucleotide and ligates a substrate to one or both ends of the double stranded fragments and thereby produces a plurality of fragment/substrate constructs, and wherein the method further comprises (a) removing the overhangs from the constructs by selectively removing the at least one nucleotide and thereby producing a plurality of double stranded constructs comprising single stranded gaps and (b) repairing the single stranded gaps in the constructs and thereby producing a plurality of modified double stranded polynucleotides. The polynucleotide typically comprises the nucleosides deoxyadenosine (dA), deoxyuridine (dU) and/or thymidine (dT), deoxyguanosine (dG) and deoxycytidine (dC). The nucleoside that is not present in the polynucleotide is preferably abasic, adenosine (A), uridine (U), 5-methyluridine (m5U), cytidine (C) or guanosine (G) or comprises urea, 5, 6 dihydroxythymine, thymine glycol, 5-hydroxy-5 methylhydantoin, uracil glycol, 6-hydroxy-5, 6-dihydrothymine, methyltartronylurea, 7, 8-dihydro-8-oxoguanine (8-oxoguanine), 8-oxoadenine, fapy-guanine, methy-fapy-guanine, fapy-adenine, aflatoxin BI-fapy-guanine, 5-hydroxy-cytosine, 5-hydroxy-uracil, 3-methyladenine, 7-methylguanine, 1,N6-ethenoadenine, hypoxanthine, 5-hydroxyuracil, 5-hydroxymethyluracil, 5-formyluracil or a cis-syn-cyclobutane pyrimidine dimer. The at least one nucleotide preferably is 10 nucleotides or fewer from the overhang. The at least one nucleotide is the first nucleotide in the overhang. All of the nucleotides in the overhang preferably comprise a nucleoside that is not present in the template polynucleotide.

[0645] These MuA based methods are disclosed in International Application No.

PCT/GB2014/052505. They are also discussed in detail in the UK Application No. 1406147.7.

[0646] One or more helicases may be attached to the MuA substrate Y adaptors before they are contacted with the double stranded polynucleotide and MuA transposase. Alternatively, one or more helicases may be attached to the MuA substrate Y adaptors before they are contacted with the double stranded polynucleotide and MuA transposase.

[0647] One or more molecular brakes may be attached to the MuA substrate hairpin loop adaptors before they are contacted with the double stranded polynucleotide and MuA transposase.

Alternatively, one or more molecular brakes may be attached to the MuA substrate hairpin loop adaptors before they are contacted with the double stranded polynucleotide and MuA transposase, Uncoupling

[0648] The method of the invention may involve characterising multiple target polynucleotides and uncoupling of the at least the first target polynucleotide.

[0649] In a preferred embodiment, the invention involves characterising two or more target polynucleotides. The method comprises: [0650] (a) providing a first polynucleotide in a first sample; [0651] (b) providing a second polynucleotide in a second sample; [0652] (c) coupling the first polynucleotide in the first sample to a membrane using one or more anchors; [0653] (d) contacting the first polynucleotide with a pore of the invention such that the polynucleotide moves with respect to, such as through, the pore; [0654] (e) taking one or more measurements as the first polynucleotide moves with respect to the pore wherein the measurements are indicative of one or more characteristics of the first polynucleotide and thereby characterising the first polynucleotide; [0655] (f) uncoupling the first polynucleotide from the membrane; [0656] (g) coupling the second polynucleotide in the second sample to the membrane using one or more anchors; [0657] (h) contacting the second polynucleotide with the pore of the invention such that the second polynucleotide moves with respect to, such as through, the pore; and [0658] (i) taking one or more measurements as the second polynucleotide moves with respect to the pore wherein the measurements are indicative of one or more characteristics of the second polynucleotide and thereby characterising the second polynucleotide.

[0659] This type of method is discussed in detail in the UK Application No. 1406155.0.

[0660] Step (f) (i.e. uncoupling of the first polynucleotide) may be performed before step (g) (i.e. before coupling the second polynucleotide to the membrane). Step (g) may be performed before step (f). If the second polynucleotide is coupled to the membrane before the first polynucleotide is

uncoupled, step (f) preferably comprises selectively uncoupling the first polynucleotide from the membrane (i.e. uncoupling the first polynucleotide but not the second polynucleotide from the membrane). A skilled person can design a system in which selective uncoupling is achieved. Steps (f) and (g) may be performed at the same time. This is discussed in more detail below.

[0661] In step (f), at least 10% of the first polynucleotide is preferably uncoupled from the membrane. For instance, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% or at least 95% of the first polynucleotide may be uncoupled from the membrane. Preferably, all of the first polynucleotide is uncoupled from the membrane. The amount of the first polynucleotide uncoupled from the membrane can be determined using the pore. This is disclosed in the Examples.

[0662] The first polynucleotide and second polynucleotide may be different from one another. Alternatively, the first and second polynucleotides may be different polynucleotides. In such instances, there may be no need to remove at least part of the first sample before adding the second polynucleotide. This is discussed in more detail below. If the method concerns investigating three or more polynucleotides, they may all be different from one another or some of them may be different from one another.

[0663] The first polynucleotide and the second polynucleotide may be two instances of the same polynucleotide. The first polynucleotide may be identical to the second polynucleotide. This allows proof reading. If the method concerns investigating three or more polynucleotides, they may all be three or more instances of the same polynucleotide or some of them may be separate instances of the same polynucleotide.

[0664] The first sample and second sample may be different from one another. For instance, the first sample may be derived from a human and the second sample may be derived from a virus. If the first and second samples are different from one another, they may contain or be suspected of containing the same first and second polynucleotides. If the method concerns investigating three or more samples, they may all be different from one another or some of them may be different from one another.

[0665] The first sample and the second sample are preferably two instances of the same sample. The first sample is preferably identical to the second sample. This allows proof reading. If the method concerns investigating three or more samples, they may all be three or more instances of the same sample or some of them may be separate instances of the same sample.

[0666] Any number of polynucleotides can be investigated. For instance, the method of the invention may concern characterising 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 50, 100 or more polynucleotides. If three or more polynucleotides are investigated using the method of the invention, the second polynucleotide is also uncoupled from the membrane and the requisite number of steps are added for the third polynucleotide. The same is true for four or more polynucleotides.

[0667] The method of the invention involves uncoupling the first polynucleotide from the membrane. The method of the invention may involve uncoupling the second polynucleotide from the membrane if three or more polynucleotides are being investigated.

[0668] The first polynucleotide can be uncoupled from the membrane using any known method. The first polynucleotide is preferably not uncoupled from the membrane in step (f) using the transmembrane pore. The first polynucleotide is preferably not uncoupled from the membrane using a voltage or an applied potential.

[0669] Step (f) preferably comprises uncoupling the first polynucleotide from the membrane by removing the one or more anchors from the membrane. If the anchors are removed, the second polynucleotide is coupled to the membrane using other (or separate) anchors. The anchors used to couple the second polynucleotide may be the same type of anchors used to couple the first polynucleotide or different type of anchors.

[0670] Step (f) more preferably comprises contacting the one or more anchors with an agent which

has a higher affinity for the one or more anchors than the anchors have for the membrane. A variety of protocols for competitive binding or immunoradiometric assays to determine the specific binding capability of molecules are well known in the art (see for example Maddox et al, *J. Exp. Med.* 158, 1211-1226, 1993). The agent removes the anchor(s) from the membrane and thereby uncouples the first polynucleotide. The agent is preferably a sugar Any sugar which binds to the one or more anchors with a higher affinity than the one or more anchors have for the membrane may be used. The sugar may be a cyclodextrin or derivative thereof as discussed below.

[0671] If one or more anchors comprise a hydrophobic anchor, such as cholesterol, the agent is preferably a cyclodextrin or a derivative thereof or a lipid. The cyclodextrin or derivative thereof may be any of those disclosed in Eliseev, A. V., and Schneider, H-J. (1994) *J. Am. Chem. Soc.* 116, 6081-6088. The agent is more preferably heptakis-6-amino- β -cyclodextrin (am.sub.7- β CD), 6-monodeoxy-6-monoamino- β -cyclodextrin (am.sub.1- β CD) or heptakis-(6-deoxy-6-guanidino)-cyclodextrin (gu.sub.7- β CD). Any of the lipids disclosed herein may be used.

[0672] If an anchor comprise(s) streptavidin, biotin or desthiobiotin, the agent is preferably biotin, desthiobiotin or streptavidin. Both biotin and desthiobiotin bind to streptavidin with a higher affinity than streptavidin binds to the membrane and vice versa. Biotin has a stronger affinity for streptavidin than desthiobiotin. An anchor comprising streptavidin may therefore be removed from the membrane using biotin or streptavidin and vice versa.

[0673] If an anchor comprises a protein, the agent is preferably an antibody or fragment thereof which specifically binds to the protein. An antibody specifically binds to a protein if it binds to the protein with preferential or high affinity, but does not bind or binds with only low affinity to other or different proteins. An antibody binds with preferential or high affinity if it binds with a K_d of 1×10^{-6} M or less, more preferably 1×10^{-7} M or less, 5×10^{-8} M or less, more preferably 1×10^{-8} M or less or more preferably 5×10^{-9} M or less. An antibody binds with low affinity if it binds with a K_d of 1×10^{-6} M or more, more preferably 1×10^{-5} M or more, more preferably 1×10^{-4} M or more, more preferably 1×10^{-3} M or more, even more preferably 1×10^{-2} M or more. Any method may be used to detect binding or specific binding. Methods of quantitatively measuring the binding of an antibody to a protein are well known in the art. The antibody may be a monoclonal antibody or a polyclonal antibody. Suitable fragments of antibodies include, but are not limited to, Fv, F(ab') and F(ab')₂ fragments, as well as single chain antibodies. Furthermore, the antibody or fragment thereof may be a chimeric antibody or fragment thereof, a CDR-grafted antibody or fragment thereof or a humanised antibody or fragment thereof.

[0674] Step (f) preferably comprises contacting the one or more anchors with an agent which reduces ability of the one or more anchors to couple to the membrane. For instance, the agent could interfere with the structure and/or hydrophobicity of the one or more anchors and thereby reduce their ability to couple to the membrane. If an anchor comprises cholesterol, the agent is preferably cholesterol dehydrogenase. If an anchor comprises a lipid, the agent is preferably a phospholipase. If an anchor comprises a protein, the agent is preferably a proteinase or urea. Other combination of suitable anchors and agents will be clear to a person skilled in the art.

[0675] Step (f) preferably comprises uncoupling the first polynucleotide from the membrane by separating the first polynucleotide from the one or more anchors. This can be done in any manner. For instance, the linker could be cut in an anchor comprising a linker. This embodiment is particularly applicable to anchors which involve linkage via hybridisation. Such anchors are discussed above.

[0676] Step (f) more preferably comprises uncoupling the first polynucleotide from the membrane by contacting the first polynucleotide and the one or more anchors with an agent which competes with the first polynucleotide for binding to one or more anchors Methods for determining and measuring competitive binding are known in the art. The agent is preferably a polynucleotide which competes with the first polynucleotide for hybridisation to the one or more anchors. For

instance, if the first polynucleotide is coupled to the membrane using one or more anchors which involve hybridisation, the polynucleotide can be uncoupled by contacting the one or more anchors with a polynucleotide which also hybridises to the site of hybridisation. The polynucleotide agent is typically added at a concentration that is higher than the concentration of the first polynucleotide and one or more anchors. Alternatively, the polynucleotide agent may hybridise more strongly to the one or more anchors than the first polynucleotide.

[0677] Step (f) more preferably comprises (i) contacting the first polynucleotide and the one or more anchors with urea, tris(2-carboxyethyl) phosphine (TCEP), dithiothreitol (DTT), streptavidin or biotin, UV light, an enzyme or a binding agent; (ii) beating the first polynucleotide and the one or more anchors; or (iii) altering the pH. Urea, tris(2-carboxyethyl) phosphine (TCEP) or dithiothreitol (DTT) are capable of disrupting anchors and separating the first polynucleotide from the membrane. If an anchor comprises a streptavidin-biotin link, then a streptavidin agent will compete for binding to the biotin. If an anchor comprises a streptavidin-desthiobiotin link, then a biotin agent will compete for binding to the streptavidin. UV light can be used to breakdown photolabile groups. Enzymes and binding agents can be used to cut, breakdown or unravel the anchor. Preferred enzymes include, but are not limited to, an exonuclease, an endonuclease or a helicase. Preferred binding agents include, but are not limited to, an enzyme, an antibody or a fragment thereof or a single-stranded binding protein (SSB). Any of the enzymes discussed below or antibodies discussed above may be used. Heat and pH can be used to disrupt hybridisation and other linkages.

[0678] If the first polynucleotide is uncoupled from the membrane by separating the first polynucleotide from the one or more anchors, the one or more anchors will remain in the membrane. Step (g) preferably comprises coupling the second polynucleotide to the membrane using the one or more anchors that was separated from the first polynucleotide. For instance, the second polynucleotide may also be provided with one or more polynucleotides which hybridise(s) to the one or more anchors that remain in the membrane. Alternatively, step (g) preferably comprises coupling the second polynucleotide to the membrane using one or more separate anchors from the ones separated from the first polynucleotide (i.e. one or more other anchors). The one or more separate anchors may be the same type of anchors used to couple the first polynucleotide to the membrane or may be different types of anchors. Step (g) preferably comprises coupling the second polynucleotide to the membrane using one or more different anchors from the one or more anchors separated from the first polynucleotide.

[0679] In a preferred embodiment, steps (f) and (g) comprise uncoupling the first polynucleotide from the membrane by contacting the membrane with the second polynucleotide such that the second polynucleotide competes with the first polynucleotide for binding to the one or more anchors and replaces the first polynucleotide. For instance, if the first polynucleotide is coupled to the membrane using one or more anchors which involve hybridisation, the first polynucleotide can be uncoupled by contacting the anchors with the second polynucleotide attached to polynucleotides which also hybridise to the sites of hybridisation in the one or more anchors. The second polynucleotide is typically added at a concentration that is higher than the concentration of the first polynucleotide and the one or more anchors. Alternatively, the second polynucleotide may hybridise more strongly to the one or more anchors than the first polynucleotide.

Removal or Washing

[0680] Although the first polynucleotide is uncoupled from the membrane in step (f), it is not necessarily removed or washed away. If the second polynucleotide can be easily distinguished from the first polynucleotide, there is no need to remove the first polynucleotide.

[0681] Between steps (f) and (g), the method preferably further comprises removing at least some of the first sample from the membrane. At least 10% of the first sample may be removed, such as at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80% or at least 90% of the first sample may be removed.

[0682] The method more preferably further comprises removing all of the first sample from the membrane. This can be done in any way. For instance, the membrane can be washed with a buffer after the first polynucleotide has been uncoupled. Suitable buffers are discussed below.

Modified Polynucleotides

[0683] Before characterisation, a target polynucleotide may be modified by contacting the polynucleotide with a polymerase and a population of free nucleotides under conditions in which the polymerase forms a modified polynucleotide using the target polynucleotide as a template, wherein the polymerase replaces one or more of the nucleotide species in the target polynucleotide with a different nucleotide species when forming the modified polynucleotide. The modified polynucleotide may then be provided with one or more helicases attached to the polynucleotide and one or more molecular brakes attached to the polynucleotide. This type of modification is described in UK Application No. 1403096.9. Any of the polymerases discussed above may be used. The polymerase is preferably Klenow or 90 North.

[0684] The template polynucleotide is contacted with the polymerase under conditions in which the polymerase forms a modified polynucleotide using the template polynucleotide as a template. Such conditions are known in the art. For instance, the polynucleotide is typically contacted with the polymerase in commercially available polymerase buffer, such as buffer from New England Biolabs®. The temperature is preferably from 20 to 37° C. for Klenow or from 60 to 75° C. for 90 North. A primer or a 3' hairpin is typically used as the nucleation point for polymerase extension.

[0685] Characterisation, such as sequencing, of a polynucleotide using a transmembrane pore typically involves analyzing polymer units made up of k nucleotides where k is a positive integer (i.e. 'k-mers'). This is discussed in International Application No. PCT/GB2012/052343 (published as WO 2013/041878). While it is desirable to have clear separation between current measurements for different k-mers, it is common for some of these measurements to overlap. Especially with high numbers of polymer units in the k-mer, i.e. high values of k, it can become difficult to resolve the measurements produced by different k-mers, to the detriment of deriving information about the polynucleotide, for example an estimate of the underlying sequence of the polynucleotide.

[0686] By replacing one or more nucleotide species in the target polynucleotide with different nucleotide species in the modified polynucleotide, the modified polynucleotide contains k-mers which differ from those in the target polynucleotide. The different k-mers in the modified polynucleotide are capable of producing different current measurements from the k-mers in the target polynucleotide and so the modified polynucleotide provides different information from the target polynucleotide. The additional information from the modified polynucleotide can make it easier to characterise the target polynucleotide. In some instances, the modified polynucleotide itself may be easier to characterise. For instance, the modified polynucleotide may be designed to include k-mers with an increased separation or a clear separation between their current measurements or k-mers which have a decreased noise.

[0687] The polymerase preferably replaces two or more of the nucleotide species in the target polynucleotide with different nucleotide species when forming the modified polynucleotide. The polymerase may replace each of the two or more nucleotide species in the target polynucleotide with a distinct nucleotide species. The polymerase may replace each of the two or more nucleotide species in the target polynucleotide with the same nucleotide species.

[0688] If the target polynucleotide is DNA, the different nucleotide species in the modified typically comprises a nucleobase which differs from adenine, guanine, thymine, cytosine or methylcytosine and/or comprises a nucleoside which differs from deoxyadenosine, deoxyguanosine, thymidine, deoxycytidine or deoxymethylcytidine. If the target polynucleotide is RNA, the different nucleotide species in the modified polynucleotide typically comprises a nucleobase which differs from adenine, guanine, uracil, cytosine or methylcytosine and/or comprises a nucleoside which differs from adenosine, guanosine, uridine, cytidine or methylcytidine. The different nucleotide species may be any of the universal nucleotides discussed

above.

[0689] The polymerase may replace the one or more nucleotide species with a different nucleotide species which comprises a chemical group or atom absent from the one or more nucleotide species. The chemical group may be a propynyl group, a thio group, an oxo group, a methyl group, a hydroxymethyl group, a formyl group, a carboxy group, a carbonyl group, a benzyl group, a propargyl group or a propargylamine group.

[0690] The polymerase may replace the one or more nucleotide species with a different nucleotide species which lacks a chemical group or atom present in the one or more nucleotide species. The polymerase may replace the one or more of the nucleotide species with a different nucleotide species having an altered electronegativity. The different nucleotide species having an altered electronegativity preferably comprises a halogen atom.

[0691] The method preferably further comprises selectively removing the nucleobases from the one or more different nucleotides species in the modified polynucleotide.

Analyte Delivery

[0692] The target analyte is preferably attached to a microparticle which delivers the analyte towards the membrane. This type of delivery is disclosed in UK Application No. 1418469.1. Any type of microparticle and attachment method may be used.

Other Characterisation Method

[0693] In another embodiment, a polynucleotide is characterised by detecting labelled species that are added to the target polynucleotide by a polymerase and then released. The polymerase uses the polynucleotide as a template. Each labelled species is specific for each nucleotide. The polynucleotide is contacted with a pore of the invention, such as a pore of the invention, a polymerase and labelled nucleotides such that phosphate labelled species are sequentially added to the polynucleotide by the polymerase, wherein the phosphate species contain a label specific for each nucleotide. The labelled species may be detected using the pore before they are released from the nucleotides (i.e. as they are added to the target polynucleotide) or after they are released from the nucleotides.

[0694] The polymerase may be any of those discussed above. The phosphate labelled species are detected using the pore and thereby characterising the polynucleotide. This type of method is disclosed in European Application No. 13187149.3 (published as EP 2682460). Any of the embodiments discussed above equally apply to this method.

[0695] Examples of labelled species include, but are not limited to, polymers, polyethylene glycols, sugars, cyclodextrins, fluorophores, drugs, metabolites, peptides. A non-limiting example of such tags can be found in the work of Kumar et al. Sci Rep. 2012; 2:684. Epub 2012 Sep. 21.

Methods of Forming Sensors

[0696] The invention also provides a method of forming a sensor for characterising a target polynucleotide. The method comprises forming a complex between a pore of the invention and a polynucleotide binding protein, such as a helicase or an exonuclease. The complex may be formed by contacting the pore and the protein in the presence of the target polynucleotide and then applying a potential across the pore. The applied potential may be a chemical potential or a voltage potential as described above. Alternatively, the complex may be formed by covalently attaching the pore to the protein. Methods for covalent attachment are known in the art and disclosed, for example, in International Application Nos. PCT/GB09/001679 (published as WO 2010/004265) and PCT/GB 10/000133 (published as WO 2010/086603). The complex is a sensor for characterising the target polynucleotide. The method preferably comprises forming a complex between a pore of the invention and a helicase. Any of the embodiments discussed above equally apply to this method.

[0697] The invention also provides a sensor for characterising a target polynucleotide. The sensor comprises a complex between a pore of the invention and a polynucleotide binding protein. Any of the embodiments discussed above equally apply to the sensor of the invention.

Kits

[0698] The present invention also provides a kit for characterising a target polynucleotide. The kit comprises a pore of the invention and the components of a membrane. The membrane is preferably formed from the components. The pore is preferably present in the membrane. The kit may comprise components of any of the membranes disclosed above, such as an amphiphilic layer or a triblock copolymer membrane.

[0699] The kit may further comprise a polynucleotide binding protein. Any of the polynucleotide binding proteins discussed above may be used.

[0700] The kit may further comprise one or more anchors for coupling the polynucleotide to the membrane.

[0701] The kit is preferably for characterising a double stranded polynucleotide and preferably comprises a Y adaptor and a hairpin loop adaptor. The Y adaptor preferably has one or more helicases attached and the hairpin loop adaptor preferably has one or more molecular brakes attached. The Y adaptor preferably comprises one or more first anchors for coupling the polynucleotide to the membrane, the hairpin loop adaptor preferably comprises one or more second anchors for coupling the polynucleotide to the membrane and the strength of coupling of the hairpin loop adaptor to the membrane is preferably greater than the strength of coupling of the Y adaptor to the membrane.

[0702] The kit of the invention may additionally comprise one or more other reagents or instruments which enable any of the embodiments mentioned above to be carried out. Such reagents or instruments include one or more of the following: suitable buffer(s) (aqueous solutions), means to obtain a sample from a subject (such as a vessel or an instrument comprising a needle), means to amplify and/or express polynucleotides or voltage or patch clamp apparatus. Reagents may be present in the kit in a dry state such that a fluid sample resuspends the reagents. The kit may also, optionally, comprise instructions to enable the kit to be used in the method of the invention or details regarding for which organism the method may be used.

Apparatus

[0703] The invention also provides an apparatus for characterising a target analyte, such as a target polynucleotide. The apparatus comprises a plurality of pores of the invention and a plurality of membranes. The plurality of pores are preferably present in the plurality of membranes. The number of pores and membranes is preferably equal. Preferably, a single pore is present in each membrane.

[0704] The apparatus preferably further comprises instructions for carrying out the method of the invention. The apparatus may be any conventional apparatus for analyte analysis, such as an array or a chip. Any of the embodiments discussed above with reference to the methods of the invention are equally applicable to the apparatus of the invention. The apparatus may further comprise any of the features present in the kit of the invention.

[0705] The apparatus is preferably set up to carry out the method of the invention.

[0706] The apparatus preferably comprises: [0707] a sensor device that is capable of supporting the plurality of pores and membranes and being operable to perform analyte characterisation using the pores and membranes; and [0708] at least one port for delivery of the material for performing the characterisation.

[0709] Alternatively, the apparatus preferably comprises: [0710] a sensor device that is capable of supporting the plurality of pores and membranes being operable to perform analyte characterisation using the pores and membranes; and [0711] at least one reservoir for holding material for performing the characterisation.

[0712] The apparatus more preferably comprises: [0713] a sensor device that is capable of supporting the membrane and plurality of pores and membranes and being operable to perform analyte characterising using the pores and membranes; [0714] at least one reservoir for holding material for performing the characterising; [0715] a fluidics system configured to controllably

supply material from the at least one reservoir to the sensor device; and [0716] one or more containers for receiving respective samples, the fluidics system being configured to supply the samples selectively from one or more containers to the sensor device.

[0717] The apparatus may be any of those described in International Application No. No. PCT/GB08/004127 (published as WO 2009/077734), PCT/GB10/000789 (published as WO 2010/122293), International Application No. PCT/GB10/002206 (published as WO 2011/067559) or International Application No. PCT/US99/25679 (published as WO 00/28312).

[0718] The following Example illustrates the invention.

Example 1

[0719] This Example describes the simulations which were run to investigate DNA behaviour within CsgG.

Materials and Methods

[0720] Steered molecular dynamics simulations were performed to investigate the magnitude of the energetic barrier of CsgG-Eco and various mutants to DNA translocation. Simulations were performed using the GROMACS package version 4.0.5, with the GROMOS 53a6 forcefield and the SPC water model. The structure of CsgG-Eco (SEQ ID NO: 2) was taken from the protein data bank, accession code 4UV3. In order to make models of the CsgG-Eco mutants, the wild-type protein structure was mutated using PyMOL. The mutants studied were CsgG-Eco-(F56A) (SEQ ID NO: 2 with mutation F56A), CsgG-Eco-(F56A-N55S) (SEQ ID NO: 2 with mutations F56A/N55S) and CsgG-Eco-(F56A-N55S-Y51A) (SEQ ID NO: 2 with mutations F56A/N55S/Y51A).

[0721] DNA was then placed into the pores. Two different systems were set up: [0722] i. A single guanine nucleotide was placed into the pore, just above the constriction region (approximately 5-10 Angstroms above the residue 56 ring) [0723] ii. A single strand of DNA (ssDNA) was placed along the pore axis, with 5' end towards the beta-barrel side of the pore. In this set up, the ssDNA was pre-threaded through the entire length of the pore.

[0724] The simulation box was then solvated and then energy minimised using the steepest descents algorithm.

[0725] Each system was simulated in the NPT ensemble, using the Berendsen thermostat and Berendsen barostat to 300 K. Throughout the simulation, restraints were applied to the backbone of the pore.

[0726] In order to pull the DNA through the pore, a pulling force was applied to the phosphorus atom in the single guanine simulations. In the ssDNA simulations the pulling force was applied to the phosphorus atom at the 5' end of the strand. The pulling force was applied at a constant velocity by connecting a spring between the DNA phosphorus atom mentioned above and an imaginary point travelling at a constant velocity parallel to the pore axis. Note that the spring does not have any shape nor does it undergo any hydrodynamic drag. The spring constant was equal to 5 KJmol.^{sup.}-1 Å.^{sup.}-2.

Results

Single G Translocation

[0727] As shown in FIG. 3, a plot of the pulling force versus time shows that there is a large barrier for nucleotide entry into the ring of phenylalanine residues F56 in the wild type CsgG-Eco pore. There was no significant barrier to guanine translocation observed for the CsgG-Eco mutants studied.

ssDNA Translocation

[0728] For ssDNA translocation, two simulations were run per pore with each run having a different applied pulling velocity (100 Å/ns and 10 Å/ns). As shown in FIG. 4, which illustrates the faster pulling velocity simulations, the CagG wild-type pore required the largest pulling force to enable ssDNA translocation. As shown in FIG. 5, which illustrates the slower pulling velocity simulations, both the CsgG-Eco (wild-type, SEQ ID NO: 2) and CsgG-Eco-(F56A) pores required

the largest applied force to enable ssDNA translocation. Comparisons between the pulling force required for ssDNA translocation through CsgG and MspA baseline pore, suggest that mutation of the CsgG pore is required to allow a similar level of ssDNA translocation.

Example 2

[0729] This Example describes the characterisation of several CsgG mutants.

Materials and Methods

[0730] Prior to setting up the experiment, DNA construct X (final concentration 0.1 nM, see FIG. 13 for cartoon representation of construct X and description) was pre-incubated at room temperature for five minutes with T4 Dda-E94C/C109A/C136A/A360C (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C, final concentration added to the nanopore system 10 nM, which was provided in buffer (151.5 mM KCl, 25 mM potassium phosphate, 5% glycerol, pH 7.0, 1 mM EDTA)). After five minutes, TMAD (100 μ M) was added to the pre-mix and the mixture incubated for a further 5 minutes. Finally, MgCl₂ (1.5 mM final concentration added to the nanopore system), ATP (1.5 mM final concentration added to the nanopore system), KCl (500 mM final concentration added to the nanopore system) and potassium phosphate buffer (25 mM final concentration added to the nanopore system) were added to the pre-mix.

[0731] Electrical measurements were acquired from a variety of single CsgG nanopores inserted in block co-polymer in buffer (25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0). After achieving a single pore inserted in the block co-polymer, then buffer (2 mL, 25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0) was flowed through the system to remove any excess CsgG nanopores. 150 μ L of 500 mM KCl, 25 mM K Phosphate, 1.5 mM MgCl₂, 1.5 mM ATP, pH8.0 was then flowed through the system. After 10 minutes a 150 μ L of 500 mM KCl, 25 mM potassium phosphate, 1.5 mM MgCl₂, 1.5 mM ATP, pH8.0 was flowed through the system and then the enzyme (T4 Dda-E94C/C109A/C136A/A360C, 10 nM final concentration), DNA construct X (0.1 nM final concentration), fuel (MgCl₂ 1.5 mM final concentration, ATP 1.5 mM final concentration) pre-mix (150 μ L total) was then flowed into the single nanopore experimental system. The experiment was run at -120 mV and helicase-controlled DNA movement monitored.

Results

Pores Showing Increased Range (FIGS. 6 to 8, and 18 to 30)

CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) has a range of ~10 pA (see FIG. 6 (a)) whereas the CsgG-Eco pore mutants below exhibited an increased current range- [0732] 1-CsgG-Eco-(Y51N-F56A-D149N-E185R-E201N-E203N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51N/F56A/D149N/E185R/E201N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (See FIG. 6 (b)). [0733] 2-CsgG-Eco-(N55A-StrepII(C))9 (SEQ ID NO: 2 with mutation N55A where StrepII(C) is has SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 6 (c)). [0734] 3-CsgG-Eco-(N55S-StrepII(C))9 (SEQ ID NO: 2 with mutations N55S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~40 pA (see FIG. 7 (a)). [0735] 4-CsgG-Eco-(Y51N-StrepII(C))9 (SEQ ID NO: 2 with mutation Y51N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~40 pA (see FIG. 7 (b)). [0736] 5-CsgG-Eco-(Y51A-F56A-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 7 (c)). [0737] 6-CsgG-Eco-(Y51A-F56N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~20 pA (see FIG. 8 (a)). [0738] 7-CsgG-Eco-(Y51A-N55S-F56A-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/N55S/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 8 (b)). [0739] 8-CsgG-Eco-(Y51A-N55S-F56N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/N55S/F56N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see

FIG. 8 (c)). [0740] 13-CsgG-Eco-(F56H-StrepII(C))9 (SEQ ID NO: 2 with mutation F56H where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 18). [0741] 14-CsgG-Eco-(F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutation F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~40 pA (see FIG. 19). [0742] 15-CsgG-Eco-(F56T-StrepII(C))9 (SEQ ID NO: 2 with mutation F56T where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 20). [0743] 16-CsgG-Eco-(S54P/F56A-StrepII(C))9 (SEQ ID NO: 2 with mutation S54P/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 21). [0744] 17-CsgG-Eco-(Y51T/F56A-StrepII(C))9 (SEQ ID NO: 2 with mutation Y51T/F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 22). [0745] 18-CsgG-Eco-(F56P-StrepII(C))9 (SEQ ID NO: 2 with mutation F56P where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 23). [0746] 19-CsgG-Eco-(F56A-StrepII(C))9 (SEQ ID NO: 2 with mutation F56A where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~40 pA (see FIG. 24). [0747] 20-CsgG-Eco-(Y51T/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 25). [0748] 21-CsgG-Eco-(N55S/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations N55S/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 26). [0749] 22-CsgG-Eco-(Y51T/N55S/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/N55S/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 27). [0750] 23-CsgG-Eco-(F56Q/N102R-StrepII(C))9 (SEQ ID NO: 2 with mutations F56Q/N102R where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~30 pA (see FIG. 28). [0751] 24-CsgG-Eco-(Y51Q/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51Q/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~40 pA (see FIG. 29). [0752] 25-CsgG-Eco-(Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited a range of ~35 pA (see FIG. 30).

Pores Showing Increased Throughput (FIGS. 9 and 10)

[0753] As can be seen from FIGS. 9 and 10, the following mutant pores (9-12 below) exhibited multiple helicase controlled DNA movements (Labelled as X in FIGS. 9 and 10) per channel in 4 hours, whereas CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) shown in FIG. 9 (a) frequently exhibited only 1 or 2 helicase controlled DNA movements (labelled as X in FIG. 9 (a)) per channel in 4 hours and instead exhibited prolonged block regions (labelled as Y in FIG. 9 (a)). [0754] 9-CsgG-Eco-(D149N-E185N-E203N-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 9 (b)) [0755] 10-CsgG-Eco-(D149N-E185N-E201N-E203N-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185N/E201N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 9 (c)) [0756] 11-CsgG-Eco-(D149N-E185R-D195N-E201N-E203N)-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185R/D195N/E201N/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 10 (a)) [0757] 12-CsgG-Eco-(D149N-E185R-D195N-E201R-E203N)-StrepII(C))9 (SEQ ID NO: 2 with mutations D149N/E185R/D195N/E201R/E203N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (FIG. 10 (b))

Pore Showing Increased Insertion (FIGS. 11 and 12)

[0758] As can be seen by comparing FIGS. 11 and 12, the mutant pore CagG-Eco-(T150I-StrepII(C))9 (SEQ ID NO: 2 with mutations T150I where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) shown in FIG. 12. was present in the membrane in increased pore numbers (~4-5 fold) compared with the CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) pore (shown in FIG. 11). Arrows in FIGS. 11

and **12** illustrated the number of CsgG-Eco nanopores which inserted into the block co-polymer in a 4 hour experiment (130-140 in FIGS. **11** and **1-11** in FIG. **12** each corresponded to a separate nanopore experiment). For CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) three experiments showed insertion of one nanopore, whereas for the mutant pore (CsgG-Eco-(T150I-StrepII(C))₉) each experiment showed insertion of at least one nanopore and several experiments showed multiple pore insertions.

Example 3

[0759] This example described an *E. Coli* purification method developed to purify the CsgG pore. Materials and Methods

[0760] DNA encoding the polypeptide Pro-CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and where Pro is SEQ ID NO: 48 and is attached at the N-terminus) was synthesised in GenScript USA Inc. and cloned into a pT7 vector containing ampicillin resistance gene. Protein expression of the pT7 vector was induced by Isopropyl β -D-1-thiogalactopyranoside (IPTG). The concentration of the DNA solution was adjusted to 400 ng/ μ L. DNA (1 μ L) was used to transform Lemo21 (DE3) competent *E. coli* cells (50 μ L, NEB, catalogue number C2528H). Prior to transformation, the CsgG gene was knocked out from Lemo21 (DE3) cells (Gene Bridges GmbH, Germany). The cells were then plated out on LB agar containing ampicillin (0.1 mg/mL) and incubated for approx 16 hours at 37° C.

[0761] Bacterial colonies grown on LB plates, containing ampicillin, incorporated the CsgG plasmid. One such colony was used to inoculate a starter culture of LB media (100 mL) containing carbenicillin (0.1 mg/mL). The starter culture was grown at 37° C. with agitation until OD₆₀₀ was reached to 1.0-1.2. The starter culture was used to inoculate a fresh 500 mL of LB media containing carbenicillin (0.1 mg/mL) and Rhamnose (500 μ M) to an O.D. 600 of 0.1. The culture was grown at 37° C. with agitation until OD₆₀₀ reached 0.6. The temperature of the culture was then adjusted to 18° C. and induction was initiated by the addition of IPTG (0.2 mM final concentration).

Induction was carried out for approximately 18 hours with agitation at 18° C.

[0762] Following induction, the culture was pelleted by centrifugation at 6,000 g for 30 minutes. The pellet was resuspended in 50 mM Tris, 300 mM NaCl, containing protease inhibitors (Merck Millipore 539138), benzonase nuclease (Sigma E1014) and 1 $\times$ bugbuster (Merck Millipore 70921) pH8.0 (approximately 10 mL of buffer per gram of pellet). Suspension was mixed well until it was fully homogeneous, the sample was then transferred to roller mixer at 4° C. for approx 5 hours. Lysate was pelleted by centrifugation at 20,000 g for 45 minutes and the supernatant was filtered through 0.22 μ m PES syringe filter. Supernatant which contained CsgG (known as sample 1) was taken forward for purification by column chromatography.

[0763] Sample 1 was applied to a 5 mL Strep Trap column (GE Healthcare). The column was washed with 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM pH8 until a stable baseline of 10 column volumes was maintained. The column was then washed with 25 mM Tris, 2M NaCl, 2 mM EDTA, 0.01% DDM pH8 before being returned to the 150 mM buffer. Elution was carried out with 10 mM desthiobiotin. An example of a chromatography trace of Strep trap (GE Healthcare) purification of a CsgG protein is shown in FIG. **14**. The elution peak is labelled E1. FIG. **15** shows an example of a typical SDS-PAGE visualization of CsgG-Eco protein after the initial Strep purification. Lanes 1-3 shows the main elution peak (labelled E1 in FIG. **14**) which contained CsgG protein as indicated by the arrow. Lanes 4-6 corresponded to elution fractions of the tail of the main elution peak (labelled E1 in FIG. **14**) which contained contaminants.

[0764] The elution peak was pooled and heated to 65° C. for 15 minutes to remove heat unstable contaminated proteins. The heated solution was subjected to centrifugation at 20,000 g for 10 minutes and the pellet was discarded. The supernatant was subjected to gel filtration on a 120 mL Sephadex S200 column (GE Healthcare) in 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS pH8. Monitoring was carried out at 220 nm due to low Tryptophan component of protein. The sample was eluted at approximately 55 mL volume (FIG. **16** shows the size exclusion

column trace with the 55 mL sample peak labelled with a star). The elution peak was run on a 4-20% TGX (see FIG. 17, Bio Rad) to confirm the presence of the pore of interest CsgG-Eco-(StrepII(C)) (SEQ ID NO: 2 where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus). Identified fractions were pooled and concentrated by 50 kD Amicon spin column.

Example 4

[0765] This example describes the simulations which were run to investigate the interaction between CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20) with T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (Δ M1) G1G2).

Simulation Methods

[0766] Simulations were performed using the GROMACS package version 4.0.5, with the GROMOS 53a6 forcefield and the SPC water model.

[0767] The CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20) model was based on the crystal structures of CsgG found in the protein data bank, accession codes 4UV3 and 4Q79. The relevant mutations were made using PyMOL. The resultant pore model was then energy minimised using the steepest descents algorithm. The T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (Δ M1) G1G2) model was based on the Dda1993 structure found in the protein data bank, accession code 3UPU. Again, relevant mutations were made using PyMOL, and the model was energy minimised using the steepest descents algorithm.

[0768] The T4 Dda-(E94C/C109A/C136A/A360C) (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C and then (Δ M1) G1G2) model was then placed above CsgG-Eco-(Y51T/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51T/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 20). Three simulations were performed with a different initial enzyme conformation (Runs 1 to 3 (0 ns), see FIG. 31):

[0769] In all enzyme conformations, the enzyme was oriented such that the 5' end of the DNA was pointing towards the pore, and the enzyme was unrestrained throughout the simulation. The pore backbone was restrained and the simulation box was solvated. The system was simulated in the NPT ensemble for 40 ns, using the Berendsen thermostat and Berendsen barostat to 300 K.

[0770] The contacts between the enzyme and pore were analysed using both GROMACS analysis software and also locally written code. The tables below show the number of contacts observed for both pore and enzyme amino acids. Tables 6-8 shows the amino acid contact points on pore which interact with the amino acid contact points on the enzyme. In two out of the three simulations the enzyme tilts on top of the pore (see run 2 and 3 (20, 30 and 40 ns), FIGS. 31 and 32). Run 1 shows that the enzyme has not tilted and so points that are shown to have high interaction in table 6 can be optimised in order to increase enzyme stability on the pore cap.

TABLE-US-00005 TABLE 6 =run 1 enzyme and pore contact interactions

Pore	Enzyme # contacts
ASN 102	ASP 198 8200
ASN 102	TYR 438 8130
GLN 100	ASP 212 7369
GLU 101	TRP 195 5979
ARG 97	TYR 350 4873
GLU 101	LEU 215 4851
ASN 102	TRP 195 3988
ARG 97	TYR 415 3798
GLU 101	TYR 350 3759
LEU 113	ASP 212 3718
ASN 102	LYS 358 3124
ARG 97	GLY 211 2765
GLU 101	CYS 412 2715
ARG 97	GLY 193 2708
ASN 102	ILE 196 2342
GLU 101	TYR 415 2268
GLU 101	ARG 216 2158
ARG 110	THR 213 2094
ARG 110	ASP 212 2066
GLY 103	ARG 216 1456
GLU 101	TYR 318 1333
ASN 102	GLU 347 1316
GLU 101	LYS 194 1310
ARG 97	PRO 411 1203
GLU 101	LYS 358 1161
ASN 102	ARG 216 1132
ARG 97	TRP 195 888
LYS 94	TYR 415 793
ASN 102	PRO 315 696
ASN 102	LYS 247 541
GLU 101	ALA 214 449
ASN 102	ASP 346 440
ARG 97	ALA 214 366
ARG 97	LYS 194 336
GLU 101	ASP 212 302
ARG 97	VAL 439 267
ARG 110	THR 210 263
ARG 97	THR 210 259
ARG 97	GLN 422 257
GLU 101	TYR 409 228
ALA 98	TRP 195 207
GLU 101	LYS 247 201
ASN 102	GLU 317 179
ARG 110	ARG 216 147

ARG 97 ASP 212 108 ASN 102 VAL 314 87 GLU 101 THR 213 72 ASN 102 LYS 255 70 VAL
105 ARG 216 69 ASN 102 LEU 215 59 ASN 102 THR 210 55 ILE 111 ASP 212 48 ARG 97 HIS
414 48 THR 104 ARG 216 36 ASN 102 TYR 197 32 GLN 100 THR 213 30 ASN 102 GLU 361 28
ARG 97 VAL 418 28 ALA 98 TYR 415 27 GLU 101 LEU 354 17 GLU 101 TYR 197 16 ASN 102
GLY 316 16 ARG 97 GLU 361 16 ARG 97 GLU 347 14 ILE 107 ARG 216 12 ASN 102 GLY 208
12 ARG 97 TYR 409 11 ARG 97 LYS 247 11 GLU 101 LYS 364 8 ARG 97 PHE 209 7 LYS 94
GLU 419 6 GLU 101 PRO 411 5 GLU 101 GLU 317 5 ASN 102 ILE 251 5 ARG 97 LEU 354 5
LYS 94 VAL 418 3 ASN 102 ARG 321 3 ARG 97 LYS 243 3 LYS 94 CYS 412 2 LEU 113 THR
210 2 GLY 103 GLU 317 2 GLU 101 LYS 351 2 ASN 102 TYR 318 2 ASN 102 MET 219 2 ASN
102 LYS 194 2 ARG 97 VAL 314 2 ARG 97 LYS 364 2 THR 104 PRO 315 1 GLY 103 THR 213 1
GLU 101 PRO 315 1

TABLE-US-00006 TABLE 7 =run 2 enzyme and pore contact interactions Pore Enzyme # contacts
GLU 101 THR 210 14155 SER 115 ASP 202 9477 ARG 97 THR 210 9064 ASN 102 VAL 200
5323 THR 104 ASP 202 4476 ASN 102 ASN 221 3422 GLU 101 PHE 437 3171 ARG 97 ASP 217
2698 GLU 101 ARG 216 2198 ARG 97 GLY 208 1730 GLU 101 LYS 199 1710 SER 115 SER 224
1440 ASN 102 LYS 199 1351 ASN 102 ASP 212 1298 ASN 102 ARG 405 1219 GLU 101 ARG
207 1180 ASN 102 SER 224 1150 ASN 102 LYS 255 1114 ARG 97 ASP 198 946 GLU 101 PHE
209 931 ARG 97 THR 213 791 ARG 97 ARG 216 599 ASN 102 THR 210 589 GLN 114 ASP 202
530 ASN 102 ASP 202 492 ARG 97 ASP 212 490 GLY 103 ARG 405 474 THR 104 SER 224 451
GLU 101 LYS 255 429 ASN 102 ASP 198 405 ASN 102 PHE 209 400 ASN 102 ARG 178 316
ARG 110 GLU 258 309 ASN 102 ASN 180 257 GLN 100 PHE 223 256 GLU 101 TYR 197 220
GLN 114 SER 228 212 LEU 113 PHE 223 210 ASN 102 ILE 225 204 GLN 114 LYS 227 194
GLU 101 GLY 211 189 GLU 101 ASP 212 174 LEU 113 SER 224 159 LEU 113 GLY 203 145
ARG 97 VAL 220 134 GLU 101 THR 213 133 THR 104 SER 228 125 ARG 97 TYR 197 123 LYS
94 ASP 212 118 ASN 102 ARG 216 110 ASN 102 ASN 235 108 ASN 102 GLY 211 104 GLU 101
ARG 405 79 GLN 114 SER 224 69 ASN 102 VAL 220 63 LEU 113 LYS 227 49 ASN 102 VAL
201 42 ARG 97 PHE 209 42 GLU 101 ASN 180 40 ARG 97 TYR 438 38 ARG 97 ARG 207 32
ASN 102 PHE 407 28 SER 115 ASN 221 23 ARG 110 HIS 204 22 GLU 101 PHE 223 21 ARG 97
ASP 189 19 ARG 110 PHE 223 16 THR 104 ILE 225 13 GLY 103 ASN 180 11 ARG 97 LYS 194
11 GLU 101 PHE 407 10 ARG 97 MET 219 9 THR 104 ASN 235 8 ARG 110 ARG 405 8 ARG 97
TRP 195 7 ILE 111 PHE 223 6 GLU 101 GLY 208 6 LEU 113 ASP 202 5 GLU 101 ARG 178 5
ASN 102 THR 213 5 ALA 98 ARG 216 5 ASN 102 ASP 217 4 ARG 97 LYS 199 4 THR 104 LEU
229 3 THR 104 ARG 405 3 GLU 101 VAL 201 3 GLU 101 MET 219 3 ARG 110 ASP 202 3 ARG
110 ARG 207 2 THR 104 VAL 201 1 GLY 103 SER 224 1 GLY 103 LYS 255 1 GLY 103 GLU 258
1 GLY 103 ASN 235 1 GLU 101 ASP 198 1 ASN 102 PHE 437 1 ARG 97 PHE 437 1 ARG 110
LYS 227 1

TABLE-US-00007 TABLE 8 =run 3 enzyme and pore contact interactions Pore Enzyme # contacts
ARG 97 THR 174 15557 GLN 100 ASP 5 10353 GLU 101 LYS 177 9238 ARG 97 SER 179 6630
LEU 116 ASP 202 6545 GLU 101 TYR 434 6524 SER 115 ASP 202 5693 GLU 101 HIS 204 5457
ARG 97 GLN 10 5106 ARG 93 ASP 202 4646 ARG 93 GLU 8 4446 SER 115 LYS 11 4342 LEU
113 ASP 5 3871 ASN 102 SER 224 3605 GLU 101 ASN 12 3344 GLU 101 GLN 10 3327 ARG 97
GLU 175 3096 GLU 101 SER 224 3028 LEU 116 GLU 8 2936 LYS 94 ASP 185 2708 ARG 97
ASN 180 2700 GLU 101 PHE 3 2500 THR 104 LYS 11 2352 SER 115 GLU 8 2323 ARG 93 ASN
180 1912 ASN 102 LYS 177 1838 LYS 94 ASP 198 1828 ARG 110 ASP 5 1714 ALA 98 GLY 203
1701 ASN 102 ASN 12 1695 GLU 101 TYR 169 1691 ARG 97 THR 7 1593 ARG 110 ASP 4 1404
ARG 97 ASP 212 1381 ASN 102 HIS 204 1226 ASN 102 ASN 15 1173 ARG 97 VAL 176 1096
ALA 98 HIS 204 998 ARG 97 ASP 202 875 ASN 102 TYR 434 850 ALA 98 ASN 12 716 GLU
101 THR 213 702 GLU 101 ARG 178 642 GLU 101 ASN 221 600 ASN 102 LYS 11 588 ARG 97
ASP 217 585 ARG 97 ARG 207 537 GLU 101 ARG 207 525 ARG 97 PHE 437 511 GLU 101
ARG 216 510 ASN 102 LYS 19 482 ARG 97 HIS 204 473 LEU 113 LYS 11 409 ARG 97 THR

213 358 ARG 93 ASP 212 354 ARG 97 TYR 169 316 ARG 97 GLY 203 308 ARG 97 ASP 435 300
GLN 87 LYS 199 249 THR 104 ASN 15 221 ARG 97 ALA 181 220 ASN 102 LYS 227 198 LYS
94 ARG 178 184 ASN 102 GLU 8 183 LEU 113 LEU 6 182 ARG 93 SER 179 179 LEU 90 ASN
180 172 LEU 90 ASP 202 144 ARG 97 ILE 225 138 GLU 101 ASN 15 135 GLU 101 LYS 19 113
LYS 94 ASN 180 109 LYS 94 GLU 175 105 ARG 93 THR 7 81 LYS 94 ARG 207 77 GLN 100
PHE 3 72 ASN 102 ARG 216 66 ARG 97 LYS 177 62 GLU 101 THR 210 59 ARG 97 ARG 178 56
LYS 94 ASP 212 55 ARG 97 GLU 172 53 GLU 101 VAL 176 51 ALA 98 ARG 207 49 ARG 110
PHE 3 48 ALA 98 ASP 202 47 ARG 97 VAL 200 40 ALA 98 VAL 201 36 LYS 94 THR 210 35 ILE
111 ASP 5 32 ARG 97 ARG 405 27 LEU 90 VAL 200 26 ARG 97 THR 210 26 GLY 103 PHE 3 25
GLU 101 PHE 209 25 ARG 97 ARG 216 22 ASN 102 VAL 220 21 LYS 94 GLY 211 19 ARG 97
PHE 209 17 GLU 101 LYS 227 15 GLN 114 LYS 11 15 GLY 103 LYS 19 13 ARG 97 PHE 3 13
GLU 101 THR 2 12 GLU 101 ILE 225 12 ARG 97 ILE 184 12 ALA 98 GLU 8 12 ALA 98 ARG
178 12 ASN 102 ILE 225 11 LYS 94 LYS 199 10 GLU 101 ARG 433 8 ARG 97 ASN 221 8 LYS
94 VAL 200 7 ASN 102 ASP 202 7 ASN 102 ASN 221 7 ARG 97 LEU 173 7 SER 115 HIS 204 6
ASN 102 GLY 203 6 GLU 101 CYS 171 5 ARG 97 ASN 12 5 ASN 102 PHE 223 4 ASN 102 LYS
166 4 ARG 97 GLY 211 4 ARG 97 GLN 170 4 GLU 101 ARG 405 3 ASN 102 PHE 3 3 GLU 101
GLU 175 2 ARG 97 VAL 220 2 ARG 93 GLY 203 2 LYS 94 THR 174 1 LEU 90 LYS 199 1 LEU
116 ASN 180 1 LEU 113 ASP 212 1 LEU 113 ASP 202 1 GLY 103 ASN 15 1 GLU 101 THR 7 1
GLU 101 PHE 437 1 GLN 114 ASP 202 1 ASN 102 ARG 405 1 ARG 97 TYR 434 1 ARG 97 PRO
182 1 ARG 97 GLY 9 1 ARG 97 GLU 8 1 ALA 99 ASP 202 1

Example 5

[0771] This example describes the simulations which were run to investigate the interaction between a) CsgG-Eco-(Y51A/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 25) with T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) GIG2) and b) CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 26) with T4 Dda (E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) GIG2).

Simulation Methods

[0772] Simulations were performed as described in Example 4.

[0773] The CsgG-Eco-(Y51A/F56Q)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 25) and CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus pore mutant No. 26) models were based on the crystal structures of CsgG found in the protein data bank, accession codes 4UV3 and 4Q79. The relevant mutations were made using PyMOL. The resultant pore model was then energy minimised using the steepest descents algorithm.

[0774] The T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) G1G2) model was based on the Dda1993 structure found in the protein data bank, accession code 3UPU. Again, relevant mutations were made using PyMOL, and the model was energy minimised using the steepest descents algorithm.

[0775] The T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C) (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C and then (Δ M1) G1G2) model was then placed above mutant pores 25 and 26.

[0776] In the simulations the enzyme was oriented such that the 5' end of the DNA was pointing towards the pore, and the enzyme was unrestrained throughout the simulation. For each of the mutant pores investigated two simulations were run-in the first the pore backbone was restrained

and the simulation box was solvated and in the second the pore backbone was restrained except for the cap region and the simulation was box solvated. The system was simulated in the NPT ensemble for 40 ns, using the Berendsen thermostat and Berendsen barostat to 300 K. [0777] The contacts between the enzyme and pore were analysed using both GROMACS analysis software and also locally written code. The tables below show the number of contacts observed for both pore and enzyme amino acids (for mutants 25 and 26 with T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 9 (pore backbone restrained) and 10 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 25 and the number of contacts that they make with the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 11 (pore backbone restrained) and 12 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 25 (CsgG-Eco-(Y51A/F56Q)-StrepII(C)) 9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) which interact with the amino acid contact points on the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 13 (pore backbone restrained) and 14 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 26 and the number of contacts that they make with the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). Tables 15 (pore backbone restrained) and 16 (pore backbone restrained with cap region unrestrained) show the amino acid contact points on pore mutant 26 (CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)) which interact with the amino acid contact points on the enzyme (T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C)). FIG. 33 shows an initial snapshot of pore mutant 26 and T4 Dda-(E94C/F98W/C109A/C136A/K194L/A360C).

TABLE-US-00008 TABLE 9 Amino Acid Position Number of Contacts
 GLU 101 26081 ARG 97 12985 GLN 100 6485 ASN 102 4941 LEU 113 3923 LYS 94 3159 ARG 110 2348 GLN 114 617 ILE 111 195 ALA 98 74 ILE 107 67 THR 104 24 PRO 112 21 SER 115 17 GLY 103 17 GLN 87 13 LEU 90 6 ASN 108 3

TABLE-US-00009 TABLE 10 Amino Acid Position Number of Contacts
 ARG 97 17462 GLU 101 12262 ASN 102 9008 LYS 94 5982 GLN 100 3471 GLY 103 3366 LEU 113 1511 GLN 87 734 THR 104 500 ARG 110 420 ASN 91 273 GLN 114 178 ALA 98 94 ARG 93 44 ILE 111 21 PRO 112 13 LEU 90 13 SER 115 4 VAL 105 2 ALA 99 2 ILE 95 1

TABLE-US-00010 TABLE 11 Pore Enzyme Number of Amino Acid Position Amino Acid Position
 Contacts
 GLU 101 TYR 318 7116 GLU 101 THR 210 6306 GLN 100 ASN 365 4693 GLU 101 TRP 195 4608 LEU 113 ASN 367 3910 LYS 94 ASP 212 2369 ASN 102 LYS 371 2088 GLU 101 LYS 364 1903 ARG 97 SER 224 1883 ARG 97 TYR 438 1764 ARG 110 LYS 368 1727 GLU 101 LYS 358 1443 GLN 100 ASN 367 1390 GLU 101 GLN 422 1356 ARG 97 TRP 195 1318 ARG 97 PHE 209 1095 ARG 97 TYR 318 965 ARG 97 TYR 350 895 ASN 102 LYS 358 833 ARG 97 GLU 361 818 ARG 97 GLY 208 706 GLU 101 GLY 316 697 GLU 101 GLU 317 678 ARG 97 GLU 317 647 GLN 114 ASN 367 617 GLU 101 LYS 371 506 ASN 102 TRP 366 466 ARG 97 THR 210 463 ASN 102 ARG 207 457 LYS 94 ASP 217 423 GLU 101 ARG 207 409 ASN 102 GLU 419 405 GLN 100 LYS 368 402 GLU 101 ARG 321 354 ARG 110 GLY 369 332 ARG 97 GLN 422 324 ASN 102 GLY 313 313 ARG 97 ASN 221 306 ARG 97 VAL 439 301 ARG 110 ASN 365 274 SER 257 GLU 258 267 GLU 101 LYS 310 217 ILE 111 ASN 367 195 ARG 97 ASP 198 154 GLU 101 SER 224 154 ARG 97 VAL 418 145 ARG 97 ALA 157 137 GLU 101 GLY 208 136 ARG 97 GLU 419 135 LYS 94 TYR 415 111 ARG 97 VAL 220 105 ASN 102 PRO 315 104 ASN 102 GLU 93 103 ARG 97 LYS 364 97 LYS 94 TYR 350 94 ARG 97 GLU 154 92 ARG 97 ALA 416 82 GLU 101 PHE 223 76 ILE 107 LYS 368 67 ARG 97 GLU 347 66 ARG 97 TYR 415 60 ASN 102 TYR 197 57 LYS 94 GLY 211 49 LYS 94 THR 213 47 GLU 101 ASN 365 46 ARG 97 LEU 354 41 LYS 94 ARG 216 41 ALA 98 TRP 195 37 ASN 102 TRP 195 34 ALA 98 LYS 358 32 ASN 102 LEU 194 27 THR 104 TRP 366 24 ARG 97 GLY 211 23 PRO 112 ASN 367 21 GLU 101 PHE 209 21

ASN 102 LYS 364 20 LYS 94 ALA 416 19 SER 115 TRP 366 17 GLY 103 LEU 194 17 ARG 97
ASP 212 16 GLU 101 TYR 415 15 ARG 110 GLY 370 15 ARG 97 LYS 351 14 ASN 102 LYS 310
14 ASN 102 GLU 154 14 ARG 97 CYS 360 13 ARG 97 THR 156 12 GLU 101 LEU 194 11 GLN
87 ASP 212 9 ARG 97 PHE 308 8 GLU 101 LEU 354 8 LEU 113 TRP 366 7 ARG 97 ARG 321 7
GLU 101 ALA 157 7 LEU 113 ASN 365 6 GLU 101 GLY 193 6 ARG 97 ASP 217 5 ALA 98 THR
210 5 GLN 87 ARG 216 4 ARG 97 LEU 319 4 ARG 97 ASN 155 4 LYS 94 VAL 220 4 LEU 90
TYR 415 3 LEU 90 ASP 212 3 ARG 97 PHE 223 3 ARG 97 GLY 193 3 ARG 97 ALA 421 3 ASN
108 LYS 368 3 ASN 102 GLY 316 3 GLU 101 VAL 418 3 ARG 97 ILE 159 2 GLU 101 PRO 315
2 SER 257 ASP 260 1 ARG 97 GLY 357 1 ASN 102 THR 210 1 ASN 102 ILE 196 1 ASN 102
GLU 317 1 GLU 101 LYS 227 1 GLU 101 GLU 361 1 GLU 101 ASN 221 1 LYS 94 VAL 418 1
LYS 94 GLU 154 1

TABLE-US-00011 TABLE 12 Pore Enzyme Number of Amino Acid Position Amino Acid Position
Contacts ARG 97 ASN 365 3386 GLU 101 GLY 316 2967 GLY 103 ASP 189 2773 LYS 94 ASP
212 2524 GLU 101 GLU 317 2401 ASN 102 TYR 197 2377 GLN 100 ASN 365 2241 GLU 101
LYS 358 2110 ARG 97 GLU 317 2031 GLU 101 THR 210 2025 ASN 102 ASP 189 1985 LYS 94
THR 213 1689 LEU 113 ASN 367 1509 ARG 97 GLU 361 1421 ARG 97 TRP 195 1361 ARG 97
TYR 438 1315 ARG 97 VAL 439 1145 LYS 94 ASP 217 957 GLU 101 TYR 318 852 GLU 101
ASN 365 797 SER 257 ASP 260 784 ARG 97 GLY 193 744 ARG 97 GLU 419 738 ARG 97 PHE
209 731 ASN 102 ARG 321 719 ASN 102 TRP 195 708 ASN 102 LYS 371 686 ARG 97 GLN 422
637 GLN 87 ASP 212 603 ARG 97 TYR 318 603 ARG 97 ASN 367 575 ARG 97 ASP 198 543
GLN 100 ASN 367 517 LYS 94 TYR 350 470 ARG 97 ALA 157 446 ARG 97 GLY 208 444 SER
257 GLU 258 425 GLU 101 LYS 371 407 GLU 101 LYS 364 400 ASN 102 GLU 93 389 GLN 100
LYS 368 383 ARG 110 LYS 368 357 ARG 97 THR 210 303 ASN 91 ARG 216 273 THR 104 ASN
367 214 GLY 103 TYR 197 204 GLN 114 ASN 367 178 GLU 101 ARG 207 173 ASN 102 TRP
366 171 ASN 102 GLY 369 153 ASN 102 LEU 194 123 ASN 102 GLY 313 120 ARG 97 THR 362
100 THR 104 TRP 366 99 GLY 103 THR 164 97 LYS 94 TRP 195 94 GLN 100 GLY 193 84 GLY
103 LYS 177 81 ASN 102 LYS 166 80 ARG 97 ILE 159 78 LYS 94 ALA 416 76 GLN 87 ARG
216 75 ASN 102 SER 306 73 ASN 102 ARG 207 73 ALA 98 TRP 195 68 ARG 97 TYR 158 65
ASN 102 GLU 347 65 ASN 102 GLN 170 65 SER 257 LYS 261 62 ARG 97 ASN 192 62 GLN
100 GLY 369 61 THR 104 LYS 199 60 ARG 97 LEU 194 57 ASN 102 PHE 163 54 ASN 102 GLU
348 53 GLY 103 GLY 313 53 GLN 100 GLU 419 53 ASN 102 LYS 351 52 GLN 87 THR 213 51
LYS 94 GLY 211 49 GLN 100 GLN 423 49 THR 104 LYS 166 48 GLU 101 ARG 321 45 ARG 97
PHE 308 44 ARG 93 GLU 419 44 ASN 102 GLY 370 41 ARG 110 GLY 369 41 SER 257 THR 259
40 ARG 97 GLY 211 40 ARG 97 THR 156 39 ASN 102 THR 164 39 LYS 94 ARG 216 39 ARG 97
ARG 321 38 GLY 103 LYS 166 35 ASN 102 GLY 208 33 GLU 101 GLY 370 30 LYS 94 GLU 361
28 ASN 102 SER 224 27 GLU 101 ILE 159 25 THR 104 PHE 308 23 THR 104 ASN 365 21 ASN
102 PRO 315 21 LYS 94 GLU 154 21 ILE 111 ASN 367 21 ARG 110 ILE 159 20 GLY 103 ARG
321 20 GLN 100 LEU 194 20 ARG 97 GLU 347 18 ARG 97 ARG 122 16 GLY 103 TRP 366 16
GLY 103 GLY 369 15 PRO 112 ASN 367 13 ALA 98 ARG 216 13 ARG 97 SER 345 12 ASN 102
LYS 364 12 GLN 100 GLY 370 12 GLU 101 PRO 315 11 GLN 100 ILE 159 11 GLN 100 GLU
317 11 LEU 90 VAL 418 10 ASN 102 VAL 220 10 ASN 102 LYS 255 10 ALA 98 LYS 358 10
THR 104 ASN 180 9 GLY 103 LEU 194 9 THR 104 LYS 368 8 LYS 94 TYR 415 8 GLN 100 THR
164 8 ARG 97 VAL 418 7 ARG 97 VAL 220 7 ARG 97 TRP 366 7 ASN 102 LEU 319 7 GLY 103
LYS 255 7 LYS 94 ILE 413 7 GLN 100 PHE 163 7 THR 104 LYS 255 6 THR 104 LYS 177 6 LYS
94 GLU 347 6 GLN 100 ASN 192 6 GLN 87 ALA 416 5 ARG 97 LYS 368 5 ARG 97 ASP 217 5
ASN 102 ARG 216 5 LYS 94 VAL 439 5 ASN 102 VAL 200 4 ASN 102 ASN 365 4 GLU 101 LYS
310 4 GLU 101 GLY 313 4 GLY 103 PHE 223 4 LYS 94 ASN 155 4 GLN 100 LEU 420 4 ARG 97
SER 224 3 ARG 97 ASP 417 3 ASN 102 THR 362 3 SER 115 GLU 419 3 GLU 101 LEU 194 3
GLU 101 GLY 369 3 GLY 103 ASN 192 3 LYS 94 ASP 417 3 VAL 105 LYS 166 2 LEU 113 ASN
365 2 LEU 90 ASP 212 2 ARG 97 TYR 92 2 ARG 97 GLU 154 2 ARG 97 ARG 207 2 THR 104

TYR 197 2 THR 104 ASP 185 2 ASN 102 PHE 209 2 ASN 102 ILE 159 2 GLY 103 PHE 308 2
GLY 103 LEU 319 2 GLY 103 ILE 159 2 ALA 98 THR 210 2 ALA 99 ILE 159 2 GLN 100 THR
210 2 SER 257 LYS 254 1 ILE 95 ARG 216 1 LEU 90 THR 213 1 ARG 97 TYR 304 1 ARG 97
ASP 212 1 THR 104 GLY 369 1 THR 104 ASN 192 1 ASN 102 VAL 314 1 ASN 102 THR 210 1
ASN 102 PHE 223 1 ASN 102 LYS 199 1 SER 115 ASP 212 1 ARG 110 GLU 317 1 ARG 110
ASN 365 1 GLY 103 PRO 315 1 GLY 103 ASN 367 1 ALA 98 ALA 157 1 LYS 94 VAL 418 1
TABLE-US-00012 TABLE 13 Amino Acid Position Number of Contacts GLU 101 18418 TRP 97
12195 ASN 102 5232 GLY 103 212 ALA 98 113 SER 115 89 THR 104 57 LYS 94 52 LEU 113 11
GLN 100 5 ARG 110 5 GLN 114 4 ARG 93 1
TABLE-US-00013 TABLE 14 Amino Acid Position Number of Contacts TRP 97 16770 ASN 102
11609 GLU 101 4947 GLY 103 2211 THR 104 2187 GLN 100 1589 LYS 94 686 ALA 98 289 SER
115 274 ARG 110 251 ARG 93 44 ILE 95 14 LEU 113 5 ASN 91 5 LEU 116 4 VAL 105 1 LEU 90
1
TABLE-US-00014 TABLE 15 Pore Enzyme Number of Amino Acid Position Amino Acid Position
Contacts GLU 101 TRP 195 5230 TRP 97 GLY 211 4360 GLU 101 THR 210 3265 GLU 101 LYS
358 3046 GLU 101 GLN 422 2476 ASN 102 ASP 212 1980 TRP 97 ARG 216 1707 TRP 97 ASN
365 1445 TRP 97 GLU 361 944 GLU 101 LYS 368 937 TRP 97 GLU 419 909 GLU 101 ALA 157
906 ASN 102 LYS 368 842 GLU 101 TYR 318 764 TRP 97 THR 210 720 ASN 102 TRP 366 626
GLU 101 ARG 216 518 TRP 97 ASN 155 487 GLU 101 LYS 364 482 ASN 102 GLU 361 409
ASN 102 THR 210 292 GLU 101 ASN 365 284 GLU 101 GLY 211 261 TRP 97 TRP 366 253 ASN
102 TRP 195 243 ASN 102 ASN 365 239 TRP 97 GLN 422 230 TRP 97 VAL 418 205 ASN 102
LYS 358 204 TRP 97 THR 156 195 GLY 103 GLY 193 152 TRP 97 GLU 154 148 ASN 102 THR
213 129 TRP 97 ASP 212 128 ALA 98 ASP 212 112 TRP 97 THR 213 112 TRP 97 ALA 157 101
SER 115 ASP 212 76 ASN 102 LEU 194 75 GLU 101 PRO 315 61 ASN 102 VAL 314 59 THR
104 TRP 366 56 TRP 97 TYR 350 48 GLU 101 GLY 193 46 TRP 97 LYS 364 45 ASN 102 PRO
315 45 GLU 101 THR 156 38 ASN 102 GLU 317 37 GLU 101 LEU 194 36 TRP 97 PHE 209 32
GLY 103 LEU 194 31 TRP 97 TYR 318 30 LYS 94 GLU 154 30 TRP 97 GLY 193 28 TRP 97 ALA
214 27 GLU 101 GLU 317 25 ASN 102 GLU 154 22 TRP 97 LEU 354 18 GLY 103 TRP 366 14
ALA 115 TRP 366 13 TRP 97 ASP 417 12 LEU 113 TRP 366 11 GLU 101 VAL 418 11 ASN 102
GLY 211 10 GLU 101 ARG 207 10 GLY 103 LYS 368 9 GLU 101 VAL 314 8 LYS 94 VAL 418 8
ASN 102 LYS 166 7 GLU 101 LYS 166 7 GLY 103 ASN 365 6 TRP 97 LYS 358 5 ASN 102 LYS
364 5 ARG 110 ASN 365 5 LYS 94 TYR 350 5 GLN 114 TRP 366 4 LYS 94 ASN 155 4 GLN 100
ASN 365 4 ASN 102 GLY 313 3 GLU 101 ASP 212 3 LYS 94 ALA 416 3 ASN 102 GLY 316 2
ASN 102 ALA 157 2 GLU 101 LYS 126 2 ALA 98 GLY 211 1 TRP 97 VAL 439 1 TRP 97 VAL
314 1 TRP 97 TRP 195 1 TRP 97 GLU 347 1 TRP 97 GLU 317 1 TRP 97 ARG 122 1 THR 104
LYS 368 1 ARG 93 TRP 366 1 ASN 102 THR 362 1 GLU 101 CYS 360 1 GLU 101 ARG 321 1
LYS 94 HIS 414 1 LYS 94 ASP 417 1 GLN 100 LYS 368 1
TABLE-US-00015 TABLE 16 Pore Enzyme Number of Amino Acid Position Amino Acid Position
Contacts TRP 97 GLU 361 3862 TRP 97 GLU 317 3269 TRP 97 GLU 154 2166 ASN 102 GLY
313 2086 ASN 102 THR 210 2009 GLU 101 TRP 195 1896 ASN 102 ASN 365 1656 THR 104
GLU 419 1510 TRP 97 ASN 365 1488 TRP 97 ARG 216 1435 GLN 100 ALA 157 1204 GLY 103
TRP 195 1191 GLU 101 ASN 365 1106 GLU 101 LYS 368 1100 ASN 102 TRP 195 822 SER 257
ASP 260 779 ASN 102 GLU 361 738 TRP 97 TRP 195 719 TRP 97 ASN 155 591 ASN 102 ARG
207 535 ASN 102 TRP 366 524 GLY 103 LYS 364 514 GLU 101 THR 210 402 TRP 97 THR 210
377 ASN 102 GLY 208 371 TRP 97 TYR 318 365 TRP 97 GLN 422 333 TRP 97 PRO 315 316
TRP 97 VAL 220 309 THR 104 LYS 368 307 TRP 97 THR 156 304 ASN 102 ALA 157 274 SER
115 ASP 212 274 TRP 97 GLY 316 266 GLU 101 GLN 422 264 GLY 103 TYR 197 214 ASN 102
ARG 148 203 ASN 102 LYS 368 199 ASN 102 ASP 198 199 ASN 102 LYS 255 192 ASN 102
PHE 308 180 TRP 97 GLU 419 174 LYS 94 GLY 211 174 GLN 100 ASN 155 168 ASN 102 PHE
223 153 TRP 97 LYS 358 152 TRP 97 ASP 198 151 ARG 110 GLY 193 148 GLN 100 GLU 419

144 ASN 102 LYS 364 139 ALA 98 ALA 157 135 ASN 102 ILE 159 122 ALA 98 TRP 195 113
THR 104 TYR 197 113 TRP 97 THR 362 103 GLY 103 GLY 369 103 LYS 94 THR 210 103 LYS
94 ASP 212 89 ASN 102 GLU 154 87 ASN 102 ILE 196 84 LYS 94 TYR 415 84 THR 104 GLU
154 77 TRP 97 GLY 193 75 ASN 102 ARG 321 71 TRP 97 VAL 418 69 GLU 101 TRP 366 69
THR 104 ASP 198 67 THR 104 ALA 157 64 GLY 103 GLU 154 60 LYS 94 GLY 153 60 LYS 94
GLU 154 59 TRP 97 TRP 366 58 GLU 101 ARG 207 56 ARG 110 LEU 194 52 ASN 102 GLY 369
46 LYS 94 GLU 361 46 GLY 103 LYS 368 42 ARG 93 ASP 212 40 ASN 102 GLU 317 39 GLN
100 THR 156 34 TRP 97 VAL 314 32 ALA 98 ASP 212 31 GLY 103 ARG 207 31 LYS 94 PHE
209 29 TRP 97 GLY 211 28 ASN 102 ARG 312 28 ARG 110 GLY 313 25 TRP 97 ARG 321 22
ASN 102 VAL 220 22 TRP 97 ASP 417 18 THR 104 GLY 369 18 ASN 102 THR 164 17 LYS 94
ARG 216 17 TRP 97 GLY 153 16 ASN 102 TYR 158 16 GLY 103 LEU 194 16 GLU 101 ILE 159
15 GLN 100 PRO 315 15 ILE 95 ASP 212 14 GLY 103 TRP 366 14 TRP 97 LEU 194 13 GLU 101
ARG 321 13 GLN 100 LYS 368 13 ASN 102 GLY 153 11 ARG 110 ASN 192 11 GLN 100 TYR
158 10 ASN 102 LYS 145 9 GLU 101 ARG 216 9 ARG 110 PRO 315 8 ALA 98 THR 210 7 ASN
102 GLY 211 7 ASN 102 ARG 216 7 ARG 110 LYS 368 7 LYS 94 HIS 414 7 LYS 94 ALA 214 7
SER 257 GLU 258 6 TRP 97 LYS 255 6 THR 104 ASP 212 6 LYS 94 ASP 417 6 ASN 91 ASP 212
5 TRP 97 THR 213 5 TRP 97 LYS 364 5 THR 104 TRP 195 5 THR 104 PHE 308 5 GLY 103 LYS
255 5 GLY 103 GLU 419 5 GLY 103 ARG 321 5 LEU 116 ASP 212 4 THR 104 TYR 438 4 THR
104 THR 210 4 THR 104 ILE 159 4 ARG 93 ARG 216 4 GLU 101 ALA 157 4 GLY 103 ILE 159 4
GLY 103 ASP 198 4 LYS 94 ASN 155 4 LEU 113 ARG 216 3 ASN 102 LEU 319 3 GLU 101 VAL
314 3 GLU 101 GLY 193 3 LEU 113 TRP 366 2 ASN 102 VAL 314 2 ASN 102 TYR 197 2 ASN
102 PHE 209 2 ASN 102 LEU 194 2 GLU 101 THR 362 2 GLU 101 GLY 313 2 GLY 103 ILE 196
2 ALA 98 PRO 315 1 ALA 98 GLY 211 1 ALA 98 ASN 155 1 VAL 105 LYS 368 1 LEU 90 ASP
212 1 TRP 97 PHE 223 1 TRP 97 PHE 209 1 TRP 97 ASP 217 1 TRP 97 ASP 212 1 THR 104 VAL
200 1 THR 104 GLY 370 1 THR 104 ARG 207 1 ASN 102 PRO 152 1 ASN 102 LYS 310 1 ASN
102 LYS 227 1 ASN 102 GLU 419 1 ASN 102 ARG 122 1 ASN 102 ALA 311 1 GLU 101 VAL
418 1 GLU 101 PHE 308 1 GLU 101 LEU 194 1 GLY 103 PHE 308 1 LYS 94 VAL 439 1 GLN
100 ARG 122 1

Example 6

[0778] This Example describes the characterisation of several CsgG mutants which show improved characterisation accuracy.

Materials and Methods

[0779] The materials and methods that were used in this example are the same as those described above for example 2. The enzyme used to control movement was either Enzyme 1=T4 Dda-E94C/C109A/C136A/A360C (SEQ ID NO: 24 with mutations E94C/C109A/C136A/A360C) or Enzyme 2=T4 Dda-E94C/F98W/C109A/C136A/K194L/A360C (SEQ ID NO: 24 with mutations E94C/F98W/C109A/C136A/K194L/A360C).

[0780] The ID accuracy characterisation measurements were calculated using methods as disclosed in the International Application PCT/GB2012/052343 (published as WO/2013/041878)

Results

[0781] The 1D basecall characterisation accuracy for MspA mutant x=MspA-((Del-L74/G75/D118/L119) D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K) 8 (SEQ ID NO: 50 with mutations D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K and deletion of the amino acids L74/G75/D118/L119) with DNA translocation controlled by T4 Dda-E94C/F98W/C109A/C136A/K194L/A360C was 68.7%. All of the mutants tested (see Table 17 below) showed improved 1D basecall characterisation accuracy in comparison to MspA mutant X.

[0782] 27-CsgG-Eco-(Y51A/F56Q/R97W/R192Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0783] 28-CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)

[0784] 29-CsgG-Eco-(Y51A/F56Q/K135L/T150I/S208V-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135L/T150I/S208V where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0785] 30-CsgG-Eco-(Y51A/F56Q/T150I/S208V-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/T150I/S208V where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0786] 31-CsgG-Eco-(Y51A/F56Q/S208V-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/S208V where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0787] 32-CsgG-Eco-(Y51A/F56Q/T150I-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/T150I where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0788] 33-CsgG-Eco-(Y51A/F56Q/K135V/T150Y-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135V/T150Y where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0789] 34-CsgG-Eco-(Y51A/F56Q/K135L-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135L where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0790] 35-CsgG-Eco-(Y51A/F56Q/R97F/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97F/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0791] 36-CsgG-Eco-(Y51A/F56Q/K135L/T150I-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135L/T150I where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0792] 37-CsgG-Eco-((Del-D195/Y196/Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids D195/Y196/Q197/R198/L199) [0793] 38-CsgG-Eco-((Del-R192/F193/I194/D195/Y196/Q197/R198/L199/L200)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids R192/F193/I194/D195/Y196/Q197/R198/L199/L200) [0794] 39-CsgG-Eco-((Del-Q197/R198/L199/L200)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids Q197/R198/L199/L200) [0795] 40-CsgG-Eco-((Del-I194/D195/Y196/Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids 1194/D195/Y196/Q197/R198/L199) [0796] 41-CsgG-Eco-((Del-V139/G140/D149/T150/V186/Q187/V204/G205)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids V139/G140/D149/T150/V186/Q187/V204/G205) [0797] 42-CsgG-Eco-((Del-D195/Y196/Q197/R198/L199/L200)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids D195/Y196/Q197/R198/L199/L200) [0798] 43-CsgG-Eco-((Del-Y196/Q197/R198/L199/L200/E201)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids Y196/Q197/R198/L199/L200/E201) [0799] 44-CsgG-Eco-((Del-Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids Q197/R198/L199) [0800] 45-CsgG-Eco-((Del-F193/I194/D195/Y196/Q197/R198/L199)-Y51A/F56Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus and deletion of the amino acids F193/I194/D195/Y196/Q197/R198/L199) [0801] 46-CsgG-Eco-(Y51A/F56Q/R192T-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192T where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0802] 47-CsgG-Eco-(Y51A/F56Q/N102S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/N102S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0803] 48-CsgG-Eco-(Y51A/F56Q/Q42R-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/Q42R where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0804] 49-CsgG-Eco-(Y51A/F56Q/R192S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0805]

50-CsgG-Eco-(Y51A/F56Q/G103N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/G103N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0806]
51-CsgG-Eco-(Y51A/F56Q/R97N/N102R-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97N/N102R where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0807]
52-CsgG-Eco-(Y51A/F56Q/R97L-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97L where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0808]
53-CsgG-Eco-(Y51A/F56Q/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0809]
54-CsgG-Eco-(Y51A/F56Q/R97N/N102G-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97N/N102G where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0810]
55-CsgG-Eco-(Y51A/F56Q/F48S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/F48S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0811]
56-CsgG-Eco-(Y51A/F56Q/G103S-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/G103S where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0812]
57-CsgG-Eco-(Y51A/F56Q/E101L-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/E101L where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0813]
58-CsgG-Eco-(Y51A/F56Q/R192Q-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192Q where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0814]
59-CsgG-Eco-(Y51A/F56Q/K135N/R142N/R192N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/K135N/R142N/R192N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0815]
60-CsgG-Eco-(Y51A/F56Q/R97N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0816]
61-CsgG-Eco-(Y51A/F56Q/R192N-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R192N where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0817]
62-CsgG-Eco-(Y51A/F56Q/Y130W-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/Y130W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) [0818]
63-CsgG-Eco-(Y51A/F56Q/E101G-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/E101G where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)
TABLE-US-00016 TABLE 17 1D Basecall Characterisation CsgC Mutant No. Enzyme Accuracy (%) 25 2 72.6 26 2 79.2 27 2 81.3 28 2 80.2 29 2 72.2 30 2 74.7 31 2 74.5 32 2 73.3 33 2 73.8 34 2 74.3 35 2 75.5 36 2 73.5 37 2 73.4 38 2 75.3 39 2 72.0 40 2 74.5 41 2 74.8 42 2 71.0 43 2 70.9 44 2 71.9 45 2 73.0 46 1 75.0 47 1 72.1 48 1 75.0 49 1 74.8 50 1 75.2 51 1 73.1 52 1 75.4 53 1 77.1 54 1 75.9 55 1 73.7 56 1 73.1 57 1 73.2 58 1 76.8 59 1 72.7 60 1 76.1 61 1 75.0 62 1 74.7 63 1 73.1 25 1 74.8 26 1 77.4

Example 7

[0819] This example compares DNA capture of a number of different mutant nanopores.

Materials and Methods

[0820] Electrical measurements were acquired from a variety of single CsgG or MspA nanopores inserted in block co-polymer in buffer (25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0). After achieving a single pore inserted in the block co-polymer, then buffer (2 mL, 25 mM K Phosphate buffer, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III), pH 8.0) was flowed through the system to remove any excess nanopores. 150 µL of 500 mM KCl, 25 mM K Phosphate, 1.5 mM MgCl₂, 1.5 mM ATP, pH8.0 was then flowed through the system. After 10 minutes a 150 µL of DNA (SEQ ID NO: 51, 200 nM) was then flowed into the single nanopore experimental system. The experiment was run at -120 mV and helicase-controlled DNA movement monitored.

Results

[0821] The CsgG mutant CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/E101S/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) (see FIG. 36) shows a higher rate of capture (e.g. captures DNA

polynucleotides more easily) than MspA-((Del-L74/G75/D118/L119) D56F/E59R/L88N/D90N/D91N/Q126R/D134R/E139K) 8 (see FIG. **34**). Each spike in the current traces corresponds to translocation of the DNA polynucleotide (SEQ ID NO: 51) through the nanopore without being controlled by an enzyme. The 10 second current traces in FIG. **34** show fewer DNA translocations than the 10 second current traces for the CsgG nanopore-CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9.

[0822] The mutation of position E101 to E101S resulted in an increase in the capture rate when compared to a CsgG mutant without the E101S mutation. FIG. **35** shows that the 10 second current traces for CsgG mutant CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) exhibited fewer translocations than CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9. The average number of translocations for CsgG-Eco-(Y51A/F56Q/R97W/R192D-StrepII(C))9 was 7.25 per second (n=12) and the average number of translocations for CsgG-Eco-(Y51A/F56Q/R97W/E101S/R192D-StrepII(C))9 was 18 per second (n=14).

Example 8

[0823] This example compares the level of expression of two different CsgG mutant pores.

Materials and Methods

[0824] The materials and methods that were used to make the nanopores (A=CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) and B=CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus)) in this example are the same as those described above for example 3.

Results

[0825] The two nanopores A=CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) and B=CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 (SEQ ID NO: 2 with mutations Y51A/F56Q/R97W/R192D where StrepII(C) is SEQ ID NO: 47 and is attached at the C-terminus) were expressed and purified using exactly the same protocol and the same volumes of each nanopore were analysed using gel filtration chromatograms (120 mL S200 Column, see FIG. **37**) and SDS-PAGE analysis (see FIG. **38**). The absorbance value for B (470.3 mAu) was much higher than A (11.4 mAu) which indicated that CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9 expressed at a much higher level than CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9. The intensity of the bands in FIG. **38** also indicate the expression level of the two pores. Bands A-C (containing CsgG-Eco-(Y51A/F56Q/R97W)-StrepII(C))9) were less intense than bands D-E (containing CsgG-Eco-(Y51A/F56Q/R97W/R192D)-StrepII(C))9). The two methods of analysis both indicated that the addition of R192D mutation greatly increased the observed expression of the CsgG mutant.

Example 9

[0826] This Example describes the characterisation of several CsgG mutants which show improved characterisation accuracy.

Materials and Methods

CsG Pores

[0827] The following 8 CsgG mutant pores were tested. Mutant 28 described in Example X above was used as the baseline pore. The mutations are made in SEQ ID NO: 2 and the purification tag StrepII has the sequence shown in SEQ ID NO: 47. [0828] Baseline pore (mutant 28): CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9 [0829] Mutant A: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-del (D195-L199) [0830] Mutant B: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-del (F193-L199) [0831] Mutant C: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-F191T [0832] Mutant D: CsgG-(WT-Y51A/F56Q/R97W/R192D-del (V105-I107)-StrepII)9 [0833] Mutant E:

CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107) [0834] Mutant F: CsgG-(WT-Y51A/F56Q/R192D-StrepII)9-R93W [0835] Mutant G: CsgG-(WT-Y51A/F56Q/R192D-StrepII)9-R93W-del (D195-L199) [0836] Mutant H: CsgG-(WT-Y51A/F56Q/R192D-StrepII)9-R93Y/R97Y

Nanopore Preparation

[0837] To prepare a nanopore array chip that contain multiple wells of block co-polymer membrane each with a single CsgG mutant nanopore inserted the following method was used. CsgG mutants expressed in *E. coli* were purified and stored in buffer containing 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS, 0.1% Brij 58 at pH8. These mutant CsgG pores were diluted to 1 in 1 million using buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 and added to chips to obtain single pores in each of the wells. After pore insertion, the array chip was washed with 1 mL buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 to remove excess pores. After a few minutes, each chip was flushed twice with 500 mL of sequencing mix containing 470 mM KCl, 25 mM HEPES, 11 mM ATP and 10 mM MgCl₂.

DNA Sample Preparation

[0838] DNA sample was prepared for sequencing using the following method. 1 µg of DNA analyte was incubated with the 40 nM of adapter mix containing a T4 Dda helicase enzyme prebound to the adapter and blunt TA ligase for 10 minutes (available from <https://store.nanoporetech.com/>). The structure of the adapter is shown in FIG. 43 and the sequences contained in the adapter are set out in SEQ ID NOs: 52 to 55. The ligation mixture was then purified to remove unligated free adapter using Spri purification. The final ligated mixture was eluted in 25 µL elution buffer containing 40 mM CAPS at pH10, 40 mM KCl and 400 nM cholesterol tether. For each chip, 12 µL of DNA-adapter ligated mix was mixed with the sequencing mix (final volume of 150 µL) and added to chip for sequencing. The experiment was then run for 6 hours at 160 mV.

[0839] The 1D accuracy characterisation measurements were calculated using methods as disclosed in WO2013/041878.

Measuring Template Speed

[0840] The template speed was measured by the following method. The basecall of each squiggle was aligned to the reference sequence. The number of bases that spanned the alignment (alignment end position subtracted from the alignment start position) was divided by the time between the event that corresponded to the end of the alignment and by the event that corresponded to the start of the alignment.

Results

Basecall Accuracy

[0841] As shown in FIG. 39, all 8 CsgG mutant pores were found to have an improved basecall accuracy compared to the baseline pore, mutant 28 (CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9) described in Example 6. As shown in Table 17, mutant 28 showed the highest basecall accuracy (80.2%) of all the CsgG mutants tested in Example 6, in which mutant 25 (CsgG-(WT-Y51A/F56Q-StrepII)9) described in Example 5 was the baseline mutant. Therefore the deletion of D195-L199, of F193-L199 or V105-I107, or the substitution of F191T results in a further improvement in accuracy in addition to the improvement in accuracy resulting from the R97W and R192D substitutions in mutant 28.

[0842] The deletion of D195-L199, of F193-L199 or V105-I107, or the substitution of F191T, would each also be expected to improve accuracy in the presence of other mutations in the CsgG sequence or in the absence of the R97W, R192D, Y51A and/or F56Q mutations. For example, mutant E, which contains the mutation K94Q in addition to the del (V105-I107) mutation has an improved accuracy compared to mutant 28, as does the del (V105-I107)-containing mutant G which does not contain the R97W substitution, but instead contains a R93W substitution.

[0843] Table 17 in Example 26 shows that mutant 26, which contains the R97W substitution has almost as good a basecall accuracy (79.2%) as mutant 28 (80.2%). This suggests that the R97W mutation underlies the increased basecall accuracy. In this Example, two mutants which do not contain the R97W mutation (and also do not contain the R192D mutation), but instead contain a R93W substitution (mutant F) or both a R93Y substitution and a R97Y substitution (mutant H) were tested and found to have a higher basecall accuracy than mutant 28. This shows that R93W and R93Y/R97Y substitutions can be used to improve the basecall accuracy of CsgG nanopores.

Template Speed and Template Accuracy

[0844] As shown in FIG. 40A, mutant D (CsgG-(WT-Y51A/F56Q/R97W/R192D-del (V105-I107)-StrepII)9) has a tightened distribution of speed population as compared to the baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9.

[0845] FIG. 40B shows that mutant D (CsgG-(WT-Y51A/F56Q/R97W/R192D-del (V105-I107)-StrepII)9) has tightened distribution of template accuracy as compared to the baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9.

[0846] It is an advantage to have a tightened speed and accuracy distribution since there is less variation in the data. Also, the median accuracy of the data will increase by reducing the amount of lower accuracy data generated. Thus, deleting the amino acids from V105 to I107 of a CsgG nanopore can be used to produce a CsgG nanopore with improved properties for characterising polynucleotides.

Example 10

[0847] This Example describes the characterisation of CsgG mutants which show a reduction in noisy pore signal.

Materials and Methods

CsG Pores

[0848] The following CsgG mutant pores were tested. Mutant 28 described in Example X above was used as the baseline pore. The mutations are made in SEQ ID NO: 2 and the purification tag StrepII has the sequence shown in SEQ ID NO: 47. [0849] Baseline pore (mutant 28): CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9 [0850] Mutant I: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-K94N. [0851] Mutant J: CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9-K94Q.

Nanopore Preparation

[0852] Chips containing multiple wells of block co-polymer membrane each with a single CsgG mutant nanopore inserted were prepared as described in Example 9.

DNA Sample Preparation

[0853] DNA sample was prepared for sequencing using the method described in Example 9.

Determining Time Spent in Noisy Pore State

[0854] The percentage of time spent in a noisy pore state was calculated as follows. The event detected signal within each channel was split up into non-overlapping short windows. For each window the mean average of the current levels and the dispersion of the current levels was calculated. The values obtained were then passed to a classifier which returned a label denoting whether the window contained a noisy signal. The classifier was trained to detect noisy signals by providing it with pre-labelled data.

Results

Noisy Pore State

[0855] FIG. 41 displays an example “squiggle” that shows the “noisy” pore error mode exhibited by baseline mutant 28 CsgG-(WT-Y51A/F56Q/R97W/R192D-StrepII)9. The top panel of FIG. 41 shows the difference in flow of current through the pore during the “good” and “noisy” pore states. The bottom panel of FIG. 41 shows an expanded view of the transition from “good” state to “noisy” state.

[0856] FIG. 42 shows the reduction in noisy pore state of mutant pores I and J when compared to baseline mutant 28, averaged over at least 5 runs.

[0857] The percentage of time spent in noise pore state by both Mutant J and Mutant I is significantly reduced compared to the baseline. Mutants I and J differ from the mutant 28 used as a baseline at just one residue. Both mutant I and mutant J contain a substitution of K94. Mutant I contains a K94N mutation and mutant II contains a K94Q mutation. Therefore substitution of K94 in a CsgG nanopore, particularly substitution with N or Q, can be used to produce a CsgG nanopore with improved properties for characterizing polynucleotides.

Example 11

[0858] This Example describes the characterisation of several CsgG mutants which show increased capture activity.

Materials and Methods

CsG Pores

[0859] The following CsgG mutant pores were tested. Mutant E described in Example 9 above was used as the baseline pore. The mutations are made in SEQ ID NO: 2 and the purification tag StrepII has the sequence shown in SEQ ID NO: 47. [0860] Baseline (mutant E): CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107)). [0861] Mutant K: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-Q42K [0862] Mutant L: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E44N [0863] Mutant M: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E44Q [0864] Mutant N: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-L90R [0865] Mutant O: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-N91R [0866] Mutant P: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-195R [0867] Mutant Q: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-A99R [0868] Mutant R: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E101H [0869] Mutant S: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E101K [0870] Mutant T: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E101N [0871] Mutant U: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E101Q [0872] Mutant V: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-E101T [0873] Mutant W: CsgG-(WT-Y51A/F56Q/K94Q/R97W/R192D-del (V105-I107))-Q114K

Nanopore Preparation

[0874] To prepare nanopore array chips that contain multiple wells of block co-polymer membrane each with a single CsgG mutant nanopore inserted the following method was used. CsgG mutants expressed in *E. coli* were purified and stored in buffer containing 25 mM Tris, 150 mM NaCl, 2 mM EDTA, 0.01% DDM, 0.1% SDS, 0.1% Brij 58 at pH8. These mutant CsgG pores were diluted to 1 in 1 million using buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 and added to the chips to obtain single pores in each of the wells. After pore insertion, the chips were washed with 1 mL buffer comprising 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 to remove excess pores. 1 mL of solution containing 240 nM TBA analyte in 25 mM Potassium Phosphate, 150 mM Potassium Ferrocyanide (II), 150 mM Potassium Ferricyanide (III) at pH 8.0 was flushed into the chip.

Determining Capture Ability

[0875] The ability of a mutant pore to capture DNA analyte is then assessed by its ability to capture Thrombin Binding Aptamer (TBA) (SEQ ID NO: 51). The experiment was run at 180 mV. In order to measure TBA capture by a pore, the median time between TBA events was calculated.

Results

[0876] As shown in FIG. 44, the median time between TBA events for 13 mutants was significantly reduced compared to the baseline, indicating that all 13 mutants display increased capture rates of template DNA.

[0877] Each of the 13 mutants had a single amino acid substitution compared to the baseline pore. The particular substitutions were: Q42K, E44N, E44Q, L90R, N91R, 195R, A99R, E101H, E101K,

E101N, E101Q, E101T and Q114K. All of these mutations involve the substitution of a negatively charged amino acid with an uncharged amino acid or a positively charged amino acid, or of an uncharged amino acid with a positively charged amino acid. Therefore, it can be concluded that substitution of the amino acid at one or more of positions Q42, E44, L90, N91, I95, A99, E101 and Q114 with amino acids that remove the negative charge and/or increase the positive charge at these positions results in increased capture of a polynucleotide.

Sequence Alignment of the Various CsgG Homologues

[0878] FIG. 45 shows the sequence alignment between the twenty-one CsgG homologues as detailed above. A multiple sequence alignment was performed on SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30, SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40 and SEQ ID NO: 41.

[0879] Praline software, a multiple sequence alignment toolbox that integrates homology-extended and secondary structure information was used to perform the alignment

<http://www.ibi.vu.nl/programs/pralinewww/>, see also Simossis V A1, Heringa J.; Nucleic Acids Res. 2005 Jul. 1; 33 (Web Server issue): W289-94. The alignment was scored using the BLOSUM62 residue exchange matrix. For details of this method, see for example Henikoff S, Henikoff J G; Proc. Natl. Acad. Sci. USA Vol. 89, pp. 10915-10919, November 1992. Gap opening and extension penalties of 12 and 1 were used respectively. Secondary structure prediction using PSIPRED was used to guide the alignment. For details of this method, see for example Jones D. T.; J Mol Biol. 1999 Sep. 17; 292 (2): 195-202. The above methods used to align the sequences are exemplary and other methods of sequence alignment known in the art may be used.

[0880] With reference to the sequence alignments of FIG. 45, each section of sequence alignment the conservation at each position is indicated by a histogram and a score. Numbers 0-9 on the scale indicate increasing conservation, columns with mutations which result in similar properties of that amino being conserved are marked with a plus ('+') and the star symbol ('*') indicates 100% sequence identity at that position. It can be seen from the conservation values of the sequence alignment that many of the residues show very high or even perfect sequence identity indicating that these 21 homologues are closely related.

[0881] FIG. 46 shows the same relative sequence alignments as FIG. 45 with predicted alpha helical secondary structure regions additionally shaded in grey. FIG. 47 shows the same relative sequence alignments as FIG. 45 with predicted beta sheet secondary structure regions additionally shaded in grey. FIGS. 46 and 47 show that the regions of predicted alpha helices and beta sheets of these homologues, important secondary structures for CsgG nanopores are highly conserved.

[0882] The multiple sequence alignment strongly suggests the sequences are related; not only is there a high degree of conservation along the alignment, but the predicted secondary structural elements are also aligned.

[0883] The sequence alignments in FIGS. 45, 46 and 47 may be used as a reference to show the relative positions that align with each other. Thus amino-acid residues identified with respect to SEQ ID NO 2 and the corresponding amino-acid residues in other CsgG homologues can be identified. For ease of identification, residues R97 and R192 have been located with an asterisk. It can be seen from the table that for example R192 of SEQ ID NO: 2 corresponds to residue R191 of SEQ ID NO: 32 and residue K177 of SEQ ID NO: 37.

[0884] As will be readily appreciated with reference to FIGS. 45 to 47, the CsgG monomers are highly conserved. Furthermore, from knowledge of the mutations in relation to SEQ ID NO: 2 it is possible to determine the equivalent positions for mutations of CsgG monomers other than that of SEQ ID NO: 2.

[0885] Thus reference to a mutant CsgG monomer comprising a variant of the sequence as shown in SEQ ID NO: 2 and specific amino-acid mutations thereof as set out in the claims and elsewhere

in the specification also encompasses a mutant CsgG monomer comprising a variant of the sequence as shown in SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30, SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40 and SEQ ID NO: 41 and corresponding amino-acid mutations thereof. Likewise reference to a construct, pore or method involving the use of a pore relating to a mutant CsgG monomer comprising a variant of the sequence as shown in SEQ ID NO: 2 and specific amino-acid mutations thereof as set out in the claims and elsewhere in the specification also encompasses a construct, pore or method relating to a mutant CsgG monomer comprising a variant of the sequence according the above disclosed SEQ ID NOS and corresponding amino-acid mutations thereof. It will further be appreciated that the invention extends to other variant CsgG monomers not expressly identified in the specification that show highly conserved regions.

Claims

1.-46. (canceled)

47. A kit for characterising a target analyte, the kit comprising: (a) a CsgG pore comprising a mutant monomer which is a variant of the amino acid sequence set forth in SEQ ID NO: 2, and the variant comprises R192D, R192Q, R192F, R192S, or R192T substitutions; and (b) the components of a membrane.

48. The kit of claim 47 wherein the variant comprises R192D substitution.

49. The kit of claim 47, wherein the variant comprises (a) one or more mutations at the following positions: I41, R93, A98, Q100, G103, T104, A106, I107, N108, L113, S115, T117, Y130, K135, E170, S208, D233, D238 and E244 and/or (b) one or more of D43S, E44S, F48S/N/Q/Y/W/I/V/H/R/K, Q87N/R/K, N91K/R, K94R/F/Y/W/L/S/N, R97F/Y/W/V/I/K/S/Q/H, E101I/L/A/H, N102K/Q/L/I/V/S/H, R110F/G/N, Q114R/K, R142Q/S, T150Y/A/V/L/S/Q/N and D248S/N/Q/K/R; (A) deletion of one or more positions F193, I194, D195, Y196, Q197, R198, L199, L200 and E201 and/or (B) deletion of one or more of V139/G140/D149/T150/V186/Q187/V204/G205, G137/G138/Q151/Y152/Y184/E185/Y206/T207 and A141/R142/G147/A148/A188/G189/G202/E203; and/or one or more of the following: (i) one or more mutations at the following positions N40, D43, E44, S54, S57, Q62, R97, E101, E124, E131, R142 and T150; (ii) mutations at Y51/N55, Y51/F56, N55/F56 or Y51/N55/F56; (iii) Q42R or Q42K; (iv) K49R or K49Q; (v) N102R, N102F, N102Y or N102W; (vi) D149N, D149Q or D149R; (vii) E185N, E185Q or E185R; (viii) D195N, D195Q or D195R; (ix) E201N, E201Q or E201R; (x) E203N, E203Q or E203R; (xi) deletion of one or more of the following positions F48, K49, P50, Y51, P52, A53, S54, N55, F56 and S57; (xii) one or more mutations at the following positions L90, N91, I95, A99, Q114; and (xiii) R97W; R93W; R93Y and R97Y; F191T; deletion of V105, A106 and I107; and/or deletion of F193, I194, D195, Y196, Q197, R198 and L199 or deletion of D195, Y196, Q197, R198 and L199.

50. The kit of claim 47, wherein the variant comprises a mutation at Y51 and/or Y56.

51. The kit of claim 50, wherein the mutation at Y51 is Y51A and/or the mutation at Y56 is F56Q.

52. The kit of claim 47, wherein (a) the variant comprises one or more of the following substitutions N40R, N40K, D43N, D43Q, D43R, D43K, E44N, E44Q, E44R, E44K, S54P, S57P, Q62R, Q62K, R97N, R97G, R97L, E101N, E101Q, E101R, E101K, E101F, E101Y, E101W, E101T, E124N, E124Q, E124R, E124K, E124F, E124Y, E124W, E131D, R142E, R142N and T150I; (b) the variant comprises F56N/N55Q, F56N/N55R, F56N/N55K, F56N/N55S, F56N/N55G, F56N/N55A, F56N/N55T, F56Q/N55Q, F56Q/N55R, F56Q/N55K, F56Q/N55S, F56Q/N55G, F56Q/N55A, F56Q/N55T, F56R/N55Q, F56R/N55R, F56R/N55K, F56R/N55S, F56R/N55G, F56R/N55A, F56R/N55T, F56S/N55Q, F56S/N55R, F56S/N55K, F56S/N55S,

[illegible]

F56R/N55A/Y51V, F56R/N55A/Y51N, F56R/N55A/Y51Q, F56R/N55A/Y51S, F56R/N55A/Y51G, F56R/N55T/Y51L, F56R/N55T/Y51V, F56R/N55T/Y51A, F56R/N55T/Y51N, F56R/N55T/Y51Q, F56R/N55T/Y51S, F56R/N55T/Y51G, F56S/N55Q/Y51L, F56S/N55Q/Y51V, F56S/N55Q/Y51A, F56S/N55Q/Y51N, F56S/N55Q/Y51Q, F56S/N55Q/Y51S, F56S/N55Q/Y51G, F56S/N55R/Y51L, F56S/N55R/Y51V, F56S/N55R/Y51A, F56S/N55R/Y51N, F56S/N55R/Y51Q, F56S/N55R/Y51S, F56S/N55R/Y51G, F56S/N55K/Y51L, F56S/N55K/Y51V, F56S/N55K/Y51A, F56S/N55K/Y51N, F56S/N55K/Y51Q, F56S/N55K/Y51S, F56S/N55K/Y51G, F56S/N55S/Y51L, F56S/N55S/Y51V, F56S/N55S/Y51A, F56S/N55S/Y51N, F56S/N55S/Y51Q, F56S/N55S/Y51S, F56S/N55S/Y51G, F56S/N55G/Y51L, F56S/N55G/Y51V, F56S/N55G/Y51A, F56S/N55G/Y51N, F56S/N55G/Y51Q, F56S/N55G/Y51S, F56S/N55G/Y51G, F56S/N55A/Y51L, F56S/N55A/Y51V, F56S/N55A/Y51A, F56S/N55A/Y51N, F56S/N55A/Y51Q, F56S/N55A/Y51S, F56S/N55A/Y51G, F56S/N55T/Y51L, F56S/N55T/Y51V, F56S/N55T/Y51A, F56S/N55T/Y51N, F56S/N55T/Y51Q, F56S/N55T/Y51S, F56S/N55T/Y51G, F56G/N55Q/Y51L, F56G/N55Q/Y51V, F56G/N55Q/Y51A, F56G/N55Q/Y51N, F56G/N55Q/Y51Q, F56G/N55Q/Y51S, F56G/N55Q/Y51G, F56G/N55R/Y51L, F56G/N55R/Y51V, F56G/N55R/Y51A, F56G/N55R/Y51N, F56G/N55R/Y51Q, F56G/N55R/Y51S, F56G/N55R/Y51G, F56G/N55K/Y51L, F56G/N55K/Y51V, F56G/N55K/Y51A, F56G/N55K/Y51N, F56G/N55K/Y51Q, F56G/N55K/Y51S, F56G/N55K/Y51G, F56G/N55S/Y51L, F56G/N55S/Y51V, F56G/N55S/Y51A, F56G/N55S/Y51N, F56G/N55S/Y51Q, F56G/N55S/Y51S, F56G/N55S/Y51G, F56G/N55G/Y51L, F56G/N55G/Y51V, F56G/N55G/Y51A, F56G/N55G/Y51N, F56G/N55G/Y51Q, F56G/N55G/Y51S, F56G/N55G/Y51G, F56G/N55A/Y51L, F56G/N55A/Y51V, F56G/N55A/Y51A, F56G/N55A/Y51N, F56G/N55A/Y51Q, F56G/N55A/Y51S, F56G/N55A/Y51G, F56G/N55T/Y51L, F56G/N55T/Y51V, F56G/N55T/Y51A, F56G/N55T/Y51N, F56G/N55T/Y51Q, F56G/N55T/Y51S, F56G/N55T/Y51G, F56A/N55Q/Y51L, F56A/N55Q/Y51V, F56A/N55Q/Y51A, F56A/N55Q/Y51N, F56A/N55Q/Y51Q, F56A/N55Q/Y51S, F56A/N55Q/Y51G, F56A/N55R/Y51L, F56A/N55R/Y51V, F56A/N55R/Y51A, F56A/N55R/Y51N, F56A/N55R/Y51Q, F56A/N55R/Y51S, F56A/N55R/Y51G, F56A/N55K/Y51L, F56A/N55K/Y51V, F56A/N55K/Y51A, F56A/N55K/Y51N, F56A/N55K/Y51Q, F56A/N55K/Y51S, F56A/N55K/Y51G, F56A/N55S/Y51L, F56A/N55S/Y51V, F56A/N55S/Y51A, F56A/N55S/Y51N, F56A/N55S/Y51Q, F56A/N55S/Y51S, F56A/N55S/Y51G, F56A/N55G/Y51L, F56A/N55G/Y51V, F56A/N55G/Y51A, F56A/N55G/Y51N, F56A/N55G/Y51Q, F56A/N55G/Y51S, F56A/N55G/Y51G, F56A/N55A/Y51L, F56A/N55A/Y51V, F56A/N55A/Y51A, F56A/N55A/Y51N, F56A/N55A/Y51Q, F56A/N55A/Y51S, F56A/N55A/Y51G, F56A/N55T/Y51L, F56A/N55T/Y51V, F56A/N55T/Y51A, F56A/N55T/Y51N, F56A/N55T/Y51Q, F56A/N55T/Y51S, F56A/N55T/Y51G, F56K/N55Q/Y51L, F56K/N55Q/Y51V, F56K/N55Q/Y51A, F56K/N55Q/Y51N, F56K/N55Q/Y51Q, F56K/N55Q/Y51S, F56K/N55Q/Y51G, F56K/N55R/Y51L, F56K/N55R/Y51V, F56K/N55R/Y51A, F56K/N55R/Y51N, F56K/N55R/Y51Q, F56K/N55R/Y51S, F56K/N55R/Y51G, F56K/N55K/Y51L, F56K/N55K/Y51V, F56K/N55K/Y51A, F56K/N55K/Y51N, F56K/N55K/Y51Q, F56K/N55K/Y51S, F56K/N55K/Y51G, F56K/N55S/Y51L, F56K/N55S/Y51V, F56K/N55S/Y51A, F56K/N55S/Y51N, F56K/N55S/Y51Q, F56K/N55S/Y51S, F56K/N55S/Y51G, F56K/N55G/Y51L, F56K/N55G/Y51V, F56K/N55G/Y51A, F56K/N55G/Y51N, F56K/N55G/Y51Q, F56K/N55G/Y51S, F56K/N55G/Y51G, F56K/N55A/Y51L, F56K/N55A/Y51V, F56K/N55A/Y51A, F56K/N55A/Y51N, F56K/N55A/Y51Q, F56K/N55A/Y51S, F56K/N55A/Y51G, F56K/N55T/Y51L, F56K/N55T/Y51V, F56K/N55T/Y51A, F56K/N55T/Y51N, F56K/N55T/Y51Q, F56K/N55T/Y51S, F56K/N55T/Y51G, F56E/N55R, F56E/N55K, F56D/N55R, F56D/N55K, F56R/N55E, F56R/N55D, F56K/N55E or F56K/N55D; and/or (c) the variant comprises deletion of

D149R/E185N/D195N/E201Q/E203N, D149R/E185Q/D195N/E201N/E203Q, D149R/E185Q/D195Q/E201N/E203N, D149R/E185Q/D195N/E201Q/E203N, D149R/E185Q/D195N/E201N/E203Q, D149R/E185N/D195Q/E201Q/E203N, D149R/E185N/D195Q/E201N/E203Q, D149R/E185N/D195N/E201Q/E203Q, D149R/E185Q/D195Q/E201Q/E203N, D149R/E185Q/D195Q/E201N/E203Q, D149R/E185Q/D195N/E201Q/E203Q, D149R/E185N/D195Q/E201Q/E203Q, D149R/E185Q/D195Q/E201Q/E203Q, D149N/E185R/D195N/E201N/E203N, D149Q/E185R/D195N/E201N/E203N, D149N/E185R/D195Q/E201N/E203N, D149N/E185R/D195N/E201Q/E203N, D149N/E185R/D195N/E201N/E203Q, D149Q/E185R/D195Q/E201N/E203N, D149Q/E185R/D195N/E201Q/E203N, D149Q/E185R/D195N/E201N/E203Q, D149N/E185R/D195Q/E201Q/E203N, D149N/E185R/D195N/E201Q/E203Q, D149Q/E185R/D195Q/E201Q/E203N, D149Q/E185R/D195Q/E201N/E203Q, D149Q/E185R/D195N/E201Q/E203Q, D149N/E185R/D195Q/E201Q/E203Q, D149N/E185R/D195N/E201R/E203N, D149Q/E185R/D195N/E201R/E203N, D149N/E185R/D195Q/E201R/E203N, D149N/E185R/D195N/E201R/E203Q, D149Q/E185R/D195Q/E201R/E203N, D149Q/E185R/D195N/E201R/E203Q, D149N/E185R/D195Q/E201R/E203Q, D149Q/E185R/D195Q/E201R/E203Q, E131D/K49R, E101N/N102F, E101N/N102Y, E101N/N102W, E101F/N102F, E101F/N102Y, E101F/N102W, E101Y/N102F, E101Y/N102Y, E101Y/N102W, E101W/N102F, E101W/N102Y, E101W/N102W, E101N/N102R, E101F/N102R, E101Y/N102R or E101W/N102F.

54. The kit of claim 47, wherein the variant comprises a mutation at T150.

55. The kit of claim 47, wherein the CsgG pore comprises nine monomers.

56. The kit of claim 55, wherein the CsgG pore comprises at least one mutant monomer.

56. The kit of claim 55, wherein the CsgG pore comprises two or more mutant monomers.

57. The kit of claim 55, wherein the two or more mutant monomers are genetically fused.

58. The kit of claim 55, wherein the two or more mutant monomers are attached via one or more linkers.

59. The kit of claim 55, wherein the target analyte is a metal ion, an inorganic salt, a polymer, an amino acid, a peptide, a polypeptide, a protein, a nucleotide, an oligonucleotide, a polynucleotide, a dye, a bleach, a pharmaceutical, a diagnostic agent, a recreational drug, an explosive or an environmental pollutant.

60. The kit of claim 59, wherein the target analyte is a polynucleotide.

61. The kit of claim 59, wherein the target analyte is a polypeptide.
